

Claude — compiled path

Compiled from `/home/dnikolic/Documents/Learning/Plan učenja/paths/Claude`. Canonical sources stay in place.

Phase — 01-claude-foundations

Directory: `01-claude-foundations/`

Source: `01-claude-foundations/01-claude-workflow-thinking.md`

Claude workflow — from a question to a problem that produces an outcome

This unit defines your persistent mode: **AI-assisted architect** (the goal is not “get text”; you steer **one concrete work problem**). **No calendar fiction** (“one fake week per topic”) — repeat the rituals as often as needed.

What changes if you internalize this

Old habit	New habit
One question → polished story as the goal	One concrete work problem → Claude workflow → artefact/plan you can check against code or incident process
One giant prompt	Sessions where you decide what goes in next (session ownership)
Everything in my head → “implement now”	Decomposition stated in plain language first, then execution (thinking vs execution)

Procedure (sessions that matter for practice)

- Name the problem in one sentence** — not abstract (“explain DDD”) but grounded (“Our `Order` aggregate — cancel, refund, payment retry — I need a change plan against the code we run”).
- Break into 3-7 sub-steps in your own words** before you send anything to the model — required **thinking** slice.
- Give the model:**
 - what you already know about the domain (short, factual),
 - what is **out of scope** today,
 - exact output shape** (e.g. “steps first, then module boundaries, then risks”).
- Model proposes — you pick** what the next increment is; don’t spiral into vague “keep going”.
- Iterate from a flaw:** round two cites a **specific problem** (“no idempotency point at consumer edge”), not “do better”.

Checklist before Send

- [] **One sentence:** what problem is today?
- [] **3-7 steps** written by **you** without copy/paste from the model
- [] **Out-of-scope** called out explicitly
- [] Desired artefact stated: **plan** | **architecture** | **incident note** | **refactor plan**

Symphony drill

Bad: “build an order service”

Good: “DDD `Order` aggregate. Rules: cancel, refund, payment retry. I need an **implementation plan** (bounds, use-case list, invariants, tests) — **no** full PHP in this round.”

Go drill

Take one worker/queue scenario with **timeout** and **retry**. First ask for **flow sketch / components** plus where **idempotency** is missing — **then** code.

Ops drill (Docker)

Enforce ordering:

1. **What logs say** (minimal, safe extract).
2. **Hypotheses** (list).
3. **Smallest experiment** probing one hypothesis (not “nuke reinstall everything”).

Mini-lab (your cadence — not calendar)

After a substantive session jot in private notes:

Metric	Why
First-answer quality (1-5)	whether framing is precise enough
Iterations to acceptable artefact	whether decomposition needs work
How often you changed direction	session ownership signal

If you split “iteration tuning” later

Keep **guided iteration** over the **same task** for several tightening rounds as **its own subtopic sheet** later if helpful; basics already live above under iterative improvement.

Source: `01-claude-foundations/02-context-ownership.md`

Context ownership — what Claude actually “sees”

Goal: always remember Claude has **no magic access** — it sees **only what you supply**. **You own** context; the model consumes it efficiently when it is truthful and scoped.

No calendar gimmicks — practice until reflexively asking “what is actually on input right now?”

Bad vs good mental model

Bad (illusion)	Good (reality)
“Claude knows my whole repo”	Claude knows messages + attachments + open editor context + Rules + Skills nothing else unless you pasted it
Tiny prompt vs huge unknown system	Big problems deserve layered deliberate prompts (explicit ROLE / CONTEXT / SPEC / CONSTRAINT / OUTPUT FORMAT once you cross beyond one vague sentence)

Exercise: equally hard prompt, richer vs thinner input

Task A — “small” prompt

Example: “Build retry for an HTTP client.”

Watch: guesses about **missing** libs, semantics, infra.

Task B — “fat” prompt (same domain intuition, sharper edges)

Example **Go-ish** briefing:

- RabbitMQ
- who owns retries / queues
- idempotency
- DLQ
- context timeouts
- sync vs async handoff

Watch: surprises shrink when borders are spelled out.

Required comparison note (5-10 bullets after A & B)

- where A was fast but misleading architecturally
- where B slowed reading yet matched real constraint surface

Symphony drill

Run the **same** architecture task twice:

1. **Without** spec (spoken intent only).
2. **With** spec: CQRS/DDD fences you genuinely use (**boundaries, invariants**).

Contrast first vs second response quality (whole chat logs optional).

Lab — repeat freely

Pick one realistic task → **three** context shapes:

1. **Minimal** — one-sentence issue.
2. **Healthy** — problem + fences + tiny relevant excerpt or ASCII sketch.
3. **Huge** — “everything”: whole module bundles, backlog noise, stray files.

Observe where (1)-(2) best trade signal/cost tokens and where (3) adds noise only.

Checklist

- [] I wrote down **everything the model absolutely did NOT see**.
- [] Listed **explicit out-of-scope** guardrails.
- [] There is exactly **one** snippet that earns its bytes — **not** “whole 200-file tree pasted”.

Source: 01-claude-foundations/03-token-ownership.md

Token ownership — never burn context unless it changes a decision

Goal: every message carries **only information that earns** the next architectural or technical decision — not “the whole universe” everywhere.

No calendar fluff — remeasure repeatedly on identical problems until instinctive.

Principle

Bad	Good
Paste whole modules / monstrous inline diffs	Ship precisely enough prose + snippets to conclude exactly one decisive question

Symphony exercise path

Grab a **fat** bounded context/module you honestly know well.

Round 1: ask for critique with **everything** pasted (your definition of overstuffed attachments).

Round 2: same question but **single aggregate (+ immediate interfaces coupling it)** input only.

Round 3: narrower still — **ownership boundary sketch** (“who talks to whom”) **or** one use case plus invariants exclusively.

Measure each round

- guessed context weight (simple 1–3 scale works)
- answer usefulness (self score 1–5)
- wall-clock latency to actionable plan headline

Go mirror run

Triple input sweep over one backlog item:

1. entire `services` subtree dump
2. only enqueue function body + immediate helpers
3. retry policy excerpt + enqueue call snippet alone

Journal which pairing **optimizes payoff vs pasted bytes**.

Ops

Incident triage layering: symptom + **tiny** trustworthy log excerpt first → only then Compose/manifest excerpts when hypotheses demand them → never whole noisy cluster yarns unless hypotheses require them.

Checklist before Send

- [] I can articulate **exactly one** pending architecture decision tied to prompt.
- [] Deleted attachments that fail that decision hypothesis (old refactor notes, orphaned diffs).
- [] Requested deterministic output shape (**decision bullets**, comparative table, phased steps).

Lab

Reuse one business-scoped issue three times altering only context sizing (table comparing **iteration × usefulness × pasted surface**).

Source: `01-claude-foundations/04-prompt-layering.md`

Layered prompting — five blocks instead of “just do X for me”

Goal: non-trivial requests open with **five visible blocks** so the model inherits your org reality intentionally (see context + token hygiene units). Topics ordered by numbering here — never calendar weeks baked into filenames-as-schedule mythology.

The five blocks

1. **ROLE** — persona who adjudicates architectural trade-offs (senior CQRS-owner for subsystem X?).
2. **CONTEXT** — tight slice of truthful system circumstance (no lore dump).
3. **SPEC** — bullet measurable outcomes reviewers can tick blindly.
4. **CONSTRAINT** — bans + stack conventions the model cannot hand-wave past.

5. **OUTPUT FORMAT** — deterministic shape readers expect (**ASCII diagrams, two-column deltas, enumerated steps**, etc.)

“Write me an API” alone forces invented constraints → usually wastes an entire conversational revolution.

Small template (adapt names & stack)

ROLE: Senior Go engineer guarding internal payout webhooks; double application of monetary side-effects forbidden.

CONTEXT: HTTP intake, Rabbit worker fleet, upstream partners replay identical payloads.

SPEC:

- Retry strategy + rationale (fixed vs exponential).
- Exact DLQ hook + semantics for discarded poison messages.
- Field bundle forming idempotent processing key(s).
- Per-consumer SLA budget assumptions.

CONSTRAINT:

- Standard Go error ergonomics – no swallowed errors.
- New dependency → short human-readable trade-off blurb referencing alternatives.

OUTPUT FORMAT:

1. ASCII funnel HTTP -> worker -> retry -> DLQ
2. Two-column viability matrix (minimal vs resilient paths)
3. Invariants resilient to webhook replay bombardment

Cross-stack rehearsal (reuse identical SPEC skeleton)

Reuse one architectural spine (replay-happy webhook). Swap **CONTEXT/CONSTRAINT** only:

Stack	Twist
Symfony	CQRS/module boundaries spelled from your codebase
Go	Packages + concurrency expectations mirror actual modules
Ops	Incident triad: observable symptom → hypothesis → microscopic experiment probing one hypothesis

Practice spin (frequency up to you)

Queue small architecture riddles → **three prompts** each differing **ONLY** by **CONTEXT** granularity or **CONSTRAINT** strength → count conversational turns senior engineers deem acceptable vs noise.

Checklist prior Send

- [] One declarative clause states **precise unlocking decision**.
- [] Every SPEC line is objectively pass/fail without model collusion.
- [] A teammate reading **OUTPUT FORMAT** alone anticipates eventual document skeleton.

Source: 01-claude-foundations/05-rules-and-skills.md

Rules and Skills (Cursor)

Goal: encode recurring **ROLE / CONSTRAINT** fragments as persistent **Rules** (AGENTS.md , .cursor/rules/*.mdc , ...) and **Skills** (SKILL.md where Cursor expects them). Session prompts stay terse because heavy policy already lives centrally.

Numerical prefixes = **concept order**, pretend calendar weeks **not** mandated.

What’s what — current mental model (verify against Cursor docs when paths change)

- **Rules:** short evergreen guardrails (**always-ish** surfaced). Style, layering laws, autonomy limits.
- **Skills:** procedural playbooks the agent may lazily hydrate when recognizing matching tasks — **provided** filenames + SKILL metadata advertise scenarios clearly according to Cursor’s current SKILL format (**double-check Cursor documentation** whenever something feels off).

Treat both as “stuff I historically re-pasted every session”; rewrite when product naming drifts (**flag outdated SKILL paths to the user**, don’t silently trust this sheet forever).

Symfony policy seeds

- Modular DDD-ish boundaries consciously named
- Explicit CQRS write/read separation expectations
- No “utility sprawl burying domain decisions”

Go policy seeds

- Small cohesive interfaces; composition favored
- Repos vs services vs transport handlers spelled per repo precedent
- Dependency adds require justification snippets (e.g. agreed `database/sql`, `sqlx`, etc.)

Ops policy seeds

- Incident docs: evidence → rollback knobs → timelines
- Forbid “everything at once”; align with microscopic observability experiments identical to Ops drill elsewhere

Skills complement rules whenever a playbook would create unreadable RULE walls.

Lightweight rule authoring skeleton

Draft **exactly**:

1. **Scope** directories / languages affected
2. **Must / must-not** ($\leq$ **five** razor statements)
3. **Evidence format** reviewers expect (bullet diff plan, rollback recipe, comparative table...)

Anything longer than a coffee skim will die unread.

Skills mind-map (example profiles — invent real SKILL entries yourself)

Persona-ish profile	Typical CONSTRAINT knobs
Go backend reviewer	concurrency, latency, package graph
Symfony architect	modules, aggregates, CQRS invariants
Ops debugger	hypotheses, blast-radius caps

Author concrete `SKILL.md` bodies strictly per Cursor’s published instructions.

Drill — before vs after codification

1. Pick grounded architecture ticket (replay-harden webhook, SLA bust, risky migration).
2. **Baseline** prompt naked (no RULE/SKILL references).
3. **Replay** attaching relevant Rules / Skills / workspace pointers (**same SPEC body** otherwise).

Measure: minutes-to-usable-draft, regressions/backtracks, subjective prompt char count.

Notebook which noise classes evaporate thanks to layering.

Checklist

- [] RULE length $\leq$ what a newcomer reads between meetings.

- [] Must/must-not items machine-checkably binary (no fuzzy poetry).
- [] Quarterly prune unused RULE noise / merge duplicated cousins.

Source: 01-claude-foundations/06-foundation-integration-lab.md

Foundations integration lab

Purpose: rehearse foundational moves **together:** incident triage triad (**symptom** → **hypothesis** → **micro experiment**); context + token stewardship (no junk pasted per message); layered five-block prompting; Cursor **Rules / Skills** as hardened defaults. Repeated “exam sprint” reinforcing **architecture assistant bounded by workspace rules**, not generic copy-pasta.

Numerical prefixes = study ordering only — invent personal calendar rhythms separately.

System snapshot card template (overwrite with truthful stack)

Treat as **stable appendix** pasted alongside prompts needing shared mental map:

```
Symfony HTTP API boundary -> Go processing worker fleet -> authoritative MySQL
-> RabbitMQ retry / DLQ story -> Compose dev fidelity -> observability /
on-call escalation hooks
```

Second lab iterations may pivot (k8s, alternative broker...) — insist that **diagram stays canonical** unless you consciously fork branches of analysis so contradictions glare.

Model output cascade you insist on (OUTPUT FORMAT clause duplicate every time)

1. **Executive plan** sentence batch (systems changing + why risky).
2. **SPEC** — acceptance criteria cleanly split from tech noise.
3. **Architectural schematic** noting concurrency + duplicate-processing hazards.
4. **Operationalized steps** (owners per step).
5. **Explicit trade-offs** contrasting lean vs hardened paths.

Whenever steps blur into stakeholder theater without falsifiable TECH spec bullets, tighten follow-ups carrying **missing slices only** (token discipline).

Minimum measurement grid

Measure	Insight
Draft senior-accept threshold	qualitative bar
Correction rounds counted	iterative overhead
Ballpark prompt mass	surrogate token budgeting
Wall clock to first sane memo	pacing

If tooling hides token charts, subjective logging still beats blindness.

Post-run journal (whole chat transcripts optional bullets)

- what landed first-shot
- which **CONTEXT / CONSTRAINT** you forgot
- RULE/SKILL debt discovered (hypothesized doc patch — align with `.cursor/rules` expectations; don’t shotgun reorganize without agreement)

Signals when certain problem classes systematically explode rework volume.

One-pass playbook table

Step	Activity
0	Replace diagram card truthfully for your infra.
1	Load relevant RULE/SKILL assets already authored.
2	First message = five layered blocks + card + narrowly scoped systemic question.
3	If blueprint leaks/contradicts, succeeding chat turns append scoped surgical patches referencing prior sections rather than rewriting entire scrollback wholesale.
4	Fill metric grid once first acceptable scaffold exists.
5	Note which RULEs helped damp noise + candidates for retirement/merge sweep

Common pitfalls

- Dumping immutable incident archaeology / PR blobs into CONTEXT without new architectural decisions riding that baggage — wastes tokens viciously.
- MIXED business SLA language + infra checklist inside single unordered SPEC stew — bifurcate lists when ambiguity surfaces.

Maintainability caveat: SKILL paths, MCP availability, Claude model quirks evolve — when product docs diverge from this workbook, refresh both this file **and** the workspace RULE calling out refreshed expectations (`plan-ucenja-ciljevi-i-rad.mdc`).

Phase — 02-prompt-engineering

Directory: 02-prompt-engineering/

Source: 02-prompt-engineering/01-constraint-prompting.md

Constraint prompting

Phase framing (this learning area)

Numbered topics **01 - 07** = topic order.

Decomposition: 3-7 steps in **your own words** before you prompt — same reflex as Phase 1 workflow-thinking. Poor decomposition ⇒ bloated CONTEXT.

#	Focus
01	Constraint prompting (<i>here</i>)
02	Role prompting
03	Architecture prompting
04	Debug prompting
05	Critique + failure prompting
06	Optimization prompting
07	Integration lab

Practice stacks: PHP/Symfony (DDD+CQRS) • Go/ `sqlx` /workers • Ops/incidents • Terraform-style IaC where relevant.

Measure when useful: quality (would a senior merge it?), approximate tokens or turns, time to usable draft.

Stop saying	Start saying
"I know prompts"	"I know workflows for problem type X "

Checkpoint: Shift from chasing one strong prompt alone to owning **pattern ≠ problem-type** workflows (architecture / debug / ops / review / spec / refactor / incident).

Theme: Constraints are how you stop the model from inventing your org's real boundaries.

The usual failure mode

Weak: "Build a payment service."

Strong: Same ask, plus **explicit CONSTRAINTS** the answer must satisfy.

Example constraint set (adjust to your truth)

- **Go** + `sqlx` (or your agreed DB layer)
- **DDD** + **CQRS** boundaries you actually run
- **Retry, timeout, idempotency** spelled as non-negotiables
- **RabbitMQ** (or your broker) in the picture
- **No ORM** if that is your rule

Practice rotations

Track	Exercise idea
PHP / Symfony	One aggregate -level change (rules, invariants, boundaries).
Go	End-to-end payment flow sketch: where idempotency lives, who owns retries.
Ops	One Terraform module (or slice): what it touches, rollback story, blast radius.

Lab — same problem, two passes

1. **Without** listing constraints (observe hallucinated stack / missing invariants).
2. **With** the constraint block (observe fewer surprises, more reviewable output).

Measure: quality (1-5 or pass/fail checklist), **iteration count** to “good enough”.

Checklist

- [] CONSTRAINTS are **testable** (yes/no), not vibes.
 - [] You stated **out of scope** explicitly.
 - [] You named **one decision** this session must unlock.
-

Source: 02-prompt-engineering/02-role-prompting.md

Role prompting

Theme: The model’s “stance” changes the kind of mistakes it makes — pick the role to match the review you need.

Failure mode

Weak: “Explain Redis.”

Strong: Give a **ROLE** that encodes **ownership and stakes**, then a narrow question.

Example

“You are a **senior backend architect** (Symfony ~14y pattern is illustrative — use your real tenure). You own **Go backend** platform decisions. Explain **cache ownership** for Redis in **our** bounded contexts — not a textbook tour.”

Roles to rehearse

Rotate the same technical question through multiple roles and compare outputs:

- Backend / platform **architect**
- **Ops engineer** (deployability, rollback, SLO)
- **Security reviewer** (threats, data paths, blast radius)
- **Staff engineer** (trade-offs, long-term cost)
- **Incident responder** (time pressure, evidence-first)
- **Performance engineer** (latency, queueing, hot paths)

Lab

- **One problem** → **five ROLE variants** (everything else as equal as possible).
- Compare: depth, missed risks, verbosity, actionability.

Measure: quality, iterations, subjective “would I ship this advice?”.

Checklist

- [] ROLE binds to **decisions you can make**, not fictional company lore.
 - [] You still attach **CONTEXT + CONSTRAINTS** when the problem is real-system sized.
-

Architecture prompting

Theme: Architecture work needs a **fixed skeleton** so every answer is comparable and reviewable.

Template (six blocks)

Aligns with layered prompting; **RISKS** is explicit here because architecture without risk is fantasy.

1. **ROLE** — who signs off technically.
2. **CONTEXT** — system truth slice (not a novel).
3. **SPEC** — measurable outcomes / boundaries.
4. **CONSTRAINTS** — stack + organizational rules.
5. **RISKS** — what breaks, what scales badly, unknowns.
6. **OUTPUT FORMAT** — diagram shape, headings, mandatory sections.

Overlap with Phase 1: The foundations track uses five blocks (ROLE / CONTEXT / SPEC / CONSTRAINT / OUTPUT FORMAT). This unit adds **RISKS** as a first-class block for architecture-sized asks — merge them in practice.

Practice rotations

Track	Prompt shape
PHP / Symfony	DDD ecommerce bounded contexts: commands/events, consistency boundaries.
Go	Payment platform: services, workers, idempotency, failure domains.
Ops / IaC	Terraform (or IaC) architecture: environments, modules, state, blast radius.

Lab — quality bar

The model output must visibly include:

- **Trade-offs** (at least two real alternatives).
- **Risks** (technical + operational).
- **Ownership** (who runs it, who changes it).
- **Scaling** story (what breaks first under load or org growth).

Measure: how many **correction rounds** you need before those four show up cleanly.

Checklist

- [] SPEC bullets are falsifiable without the model agreeing.
- [] OUTPUT FORMAT demands **diagram + narrative**, not vibes.

Debug prompting

Theme: An architecture assistant **does not** jump to “try X” without a **hypothesis ladder**.

Required flow

problem (evidence) → hypothesis → validation plan → only then fix

Practice rotations

Track	Scenario idea
Go	Worker slow — queue lag vs CPU vs IO vs retry storms.
DB	Query latency — plan, locks, hot rows, missing index vs bad access pattern.
Ops	Pod restart loop — crash vs probe vs OOM vs config vs dependency.

Lab — forbidden output pattern

Ask the model **not** to output bare “**try X**” without structure.

Require instead:

1. **At least five hypotheses** (ranked or grouped).
2. **Order of checks** with **expected signal** if true/false.
3. **Risk** of each check (blast radius, time cost, false confidence).

Measure: how often the first draft still sneaks in magic fixes — that’s your prompt debt.

Checklist

- [] You pasted **minimal logs / metrics** first, not entire bundles.
- [] You demanded **evidence-linked** reasoning.

Source: 02-prompt-engineering/05-critique-and-failure-prompting.md

Critique + failure prompting

Theme: Use the model to **attack** its own draft before you treat it as a plan.

Workflow

generate → critique → surface risks → repair / rewrite

Practice rotations

Track	Exercise
PHP / Symfony	CQRS refactor — command/query split, event boundaries.
Go	Distributed retry — double delivery, idempotency, poison messages.
Ops	Incident rollback — what to revert, what to keep, comms shape.

Lab — three-pass minimum

1. **Implement / plan** (first answer).
2. **Critique your previous answer** under a reviewer ROLE (security, staff engineer, SRE — pick one).
3. **Produce improved v2** that lists what changed and why.

Measure: count of **caught risks** in step 2 that step 3 actually fixes.

Failure prompting (explicit)

Add CONSTRAINTS like:

- “Assume partial outage of dependency D.”
- “Assume duplicate message delivery.”
- “Assume deploy during incident.”

Then require **failure modes + mitigations**, not happy path only.

Checklist

- [] Critique pass uses a **different ROLE** than generation.
- [] You ask for **diff-style** “what failed in v1” before v2.

Source: 02-prompt-engineering/06-optimization-prompting.md

Optimization prompting

Theme: Same outcome, **fewer iterations** and **less context burned** — this is where Phase 1 token ownership meets Phase 2 pattern choice.

Decomposition first

Optimization is mostly **throwing away noise** once you’ve **decomposed** the real decision tree (same habit as the Phase 1 workflow / decomposition practice). Prompt v3 should still carry every constraint that moves the architectural outcome — shrink everything else.

What “optimized” means here

You want the **same quality bar** with:

- fewer back-and-forth loops, and/or
- smaller prompts (fewer tokens) without losing decision-critical facts.

Practice — same problem, three prompt versions

Pick one scenario per stack (rotate):

Track	Micro-scenario
PHP / Symfony	One aggregate behavior change.
Go	One worker behavior (retry, DLQ, idempotency).
Ops	One incident triage write-up + next steps.

Produce:

- **Prompt v1** — verbose, everything you “might” need.
- **Prompt v2** — cut fat; keep constraints + spec + output format.
- **Prompt v3** — minimal but still passes your quality checklist.

Measure: quality (same rubric), **iterations**, **tokens** (estimate if no meter).

Checklist

- [] You did not delete **constraints** that change the answer.
- [] Each version states the **same SPEC** so comparison is fair.

Source: 02-prompt-engineering/07-integration-lab.md

Integration lab (Phase 2)

Theme: One **Payment platform** scenario end-to-end through every pattern from this phase.

You supply a concise **CONTEXT** card (your real or toy architecture). Claude must traverse:

1. **Architecture** — layered prompt with **ROLE / CONTEXT / SPEC / CONSTRAINT / RISK / OUTPUT**.
2. **Trade-offs** — explicit alternatives you could defend in review.

3. **Implementation plan** — steps, owners, risky migrations.
4. **Debug** — hypothetical failure injected; hypotheses + validation ordering.
5. **Ops** — deploy, rollback, observability gaps.
6. **Review** — critique pass on the whole storyline.
7. **Optimization** — tighten prompts or context for a second lap with measurements.

Pattern inventory (use consciously)

- **Constraint** prompting
- **Role** prompting
- **Architecture** prompting
- **Critique / failure** prompting
- **Optimization** prompting

Measure

Metric	Notes
Quality	Senior-acceptable without rewrite?
Iterations	Count of “fix this section” loops
Tokens	Rough prompt + reply mass
Speed	Time to usable artefact

Document after the lap

Brief note (private notes OK): **which pattern**, at **which step**, bought the biggest quality jump — that’s your personal cheat sheet going forward.

Phase 2 checkpoint

You stop optimizing **isolated prompts** and keep **problem-type workflows**: architecture vs debug vs incident vs refactor each has a recognizable opening pattern.

Outdated-alert habit

Payment platforms, broker semantics, and Cursor capabilities drift — if this lab’s assumed stack doesn’t match yours, rewrite the CONTEXT card and constraints instead of fighting the model.

Phase — 03-prompt-failure-engineering

Directory: 03-prompt-failure-engineering/

Source: 03-prompt-failure-engineering/01-ambiguity-failure.md

Ambiguity failure

Phase framing (Prompt Failure Engineering)

Topics 01 – 05 .

Mindset: bad output → classify **failure class** → repair prompt/context/spec — not only patch prose by hand.

Class	Meaning
Ambiguity	Underspecified wording; model guesses stack.
Missing context	Not enough system truth.
Constraint failure	Forbidden/allowed tech not stated.
Spec drift	Answer leaves your locked SPEC.
Token / over-context	Too much noisy paste.
Role failure	Wrong reviewer stance.
Verification failure	You skipped SPEC/criteria check.

Daily drill (recommended): 1× Symphony + 1× Go + 1× Ops — each with an **intentionally bad prompt** → diagnose → repaired prompt → re-run → one-line lesson.

Checkpoint: Stop at “wrong”; start at “**I know why it failed.**”

Theme: The most common failure — you said a **word**, not a **decision surface**.

Pattern

Weak: “Build a worker.”

The model must guess: **RabbitMQ consumer? cron? goroutine pool? cloud task?** — each is a different architecture.

Diagnosis label: **ambiguity** (noun/verb not tied to your system).

Practice — tighten the same ask

PHP / Symfony

Weak	Stronger
“Build an order flow.”	Name DDD boundaries, CQRS split, payment retry rules, refund ownership — who may initiate what.

Go

Weak	Stronger
“Build a queue.”	Name RabbitMQ (or your broker), retry , DLQ , worker pool sizing assumptions, idempotency expectations.

Lab — daily drill (short)

Write **five intentionally vague prompts** (mix stacks). For each, **classify** the likely failure as **ambiguity** (or note if another class also applies). Then rewrite **one** into a non-ambiguous prompt (add CONTEXT + CONSTRAINT + SPEC).

Measure: count how many guesses disappear in the first reply.

Checklist

- [] Every actor noun has a **single** interpretation in your org (worker = what exactly?).
- [] Verbs like “optimize” / “build” are replaced by **observable outcomes**.

Source: 03-prompt-failure-engineering/02-missing-context-failure.md

Missing context failure (and over-context)

Theme: The model is not psychic — “your project” is not in the prompt unless you put it there. Too much undifferentiated context creates **token pressure** and wrong focus.

Missing context pattern

Weak: “Optimize the query.”
Stronger: Attach minimal **ground truth**: **MySQL, ~120M rows, replica topology, read-heavy, p95 latency ~900ms**, access pattern (OLTP vs reporting), what “good” means.
Diagnosis labels: **missing context**; sometimes **ambiguity** if “optimize” is still undefined.

Over-context / token pressure pattern

Symptom: answer chases the wrong detail, contradicts itself, or stays generic — you pasted whole modules, old tickets, and three unrelated configs.
Diagnosis label: **token pressure / over-context** (same family: wrong *amount* and *relevance* of context).

Practice rotations

Track	Focus
Symfony	CQRS read path vs write path — paste only the slice that owns the decision.
Go	Worker path — only handler + retry + message shape that matters today.
Ops	Terraform (or IaC) — only the module + env + state risk for <i>this</i> change.

Lab — three deliberate sizes

On **one** real question:

1. **Too little context** — observe confident wrong assumptions.
2. **About right** — minimal facts that change the answer.
3. **Too much** — observe noise, drift, or generic advice.

Measure: quality (quick rubric), **iterations** to acceptable output.

Checklist

- [] You can state **one** decision the message must unlock.
- [] Everything in the paste **earns its tokens** for that decision.

Constraint failure

Theme: Without CONSTRAINTS, the model reaches for **familiar** patterns from other ecosystems — not your agreed stack.

Pattern

You asked for “persistence in Go” and got **Laravel-shaped** or **Doctrine-shaped** thinking because you never said: **Go**, `sqlx`, **small interfaces**, **no ORM**.

Diagnosis label: **constraint failure** (rules of construction were absent).

Practice rotations

Track	Exercise
Symfony	Same feature with explicit module / CQRS / framework version constraints vs without — compare wrong imports of patterns.
Go	Persistence or API layer with <code>sqlx</code> , error style, DI rules vs without.
Ops	Module with blast-radius and state constraints vs “just write terraform”.

Lab — A/B same task

1. **Run without** a CONSTRAINT block — capture the “wrong idiom” class.
2. **Run with** CONSTRAINT block — confirm the answer stays in-bounds.

Classify first attempt: almost always **constraint failure** if the stack drifts.

Checklist

- [] CONSTRAINTS are **binary** (allowed / forbidden), not suggestions.
- [] You named **forbidden** patterns (“no ORM”, “no hidden globals”) when they matter.

Specification drift

Theme: The model (or you) loses the thread — **implementation** diverges from the **SPEC** you intended to lock.

Pattern

SPEC said: retry **max 3**.
Answer implements: “retry until success” / unbounded loop.
Diagnosis label: **spec drift** (answer no longer satisfies the stated acceptance criteria).

Why it happens

- SPEC buried inside long prose.
- Follow-up messages introduce new goals without restating SPEC.
- No explicit “diff against SPEC” step in OUTPUT FORMAT.

Practice rotations

Track	Exercise
Go	Payment retry — numeric cap, idempotency key, DLQ behavior: write SPEC bullets first.
PHP / Symfony	CQRS aggregate change — invariants and command outcomes as checkable bullets.

Lab — force verification

1. Ask for **implementation / plan**.
2. **Without new features in the prompt**, ask for a **SPEC compliance table**: each SPEC line → satisfied / violated / unclear → evidence quote from the answer.
3. Mark **drift** rows and rewrite prompt or add CONSTRAINT / OUTPUT FORMAT requiring the table **in the first pass** next time.

Checklist

- [] SPEC lives in **short bullets** that survive copy-paste into follow-ups.
- [] OUTPUT FORMAT demands **traceability** back to SPEC when risk is high.

Source: 03-prompt-failure-engineering/05-failure-classification-integration.md

Failure classification integration

Theme: On a **Payment platform** (or your single big scenario), you **inject** known-bad model behavior and practice **classify** → **repair** → **verify**.

Bad output example (intentional stress test)

A plan that has, e.g.:

- **Unbounded retries**
- **No timeouts**
- **No idempotency** where money moves

Your job is **not** to rage-edit the text — it is to **label** *why* the session produced this.

Classification workflow

failure output → label (one or more classes) → repair plan → re-run → verify

Labels to use (natural catalog by end of Phase 2.5)

Class	Plain meaning
Ambiguity	You never pinned what “worker”, “flow”, or “optimize” meant.
Constraint failure	Stack / style / bans were missing; wrong-idiom answer.
Missing context	Not enough system truth.
Over-context / token pressure	Too much or irrelevant paste; model fixates wrong.
Spec drift	Answer drifted from numeric / logical SPEC.
Role failure	Wrong reviewer mindset (e.g. no SRE when you needed blast-radius).
Verification failure	You never compared output to SPEC/constraints.

Repair plan template (short)

1. **Primary class** (and secondary if any).

2. **Prompt change** — which block you add or tighten: CONTEXT, CONSTRAINT, SPEC, OUTPUT FORMAT, ROLE.
3. **Verification** — one step that proves the next run can't repeat the same drift without you seeing it.

Lab — end checkpoint

Run one **deliberately weak** prompt → classify → repair → re-run. Log **one line**: *which class* bought the biggest quality jump.

Phase 2.5 checkpoint

You move from “**Claude was wrong**” to “**The failure was X; the fix is Y.**”

Phase — 04-specification-guided-development

Directory: 04-specification-guided-development/

Source: 04-specification-guided-development/01-specification-first-thinking.md

Specification-first thinking

Phase framing (Specification Guided Development)

Highest-leverage framing for this track: validated **specification before** substantial implementation churn.
Topics 01 - 08 .

Core workflow

Problem → Specification → Architecture → Implementation → Validation

Validation ownership: you compare each delivered increment against SPEC + acceptance + NFR **before** the next slice merges. Claude assists — **you** keep veto/sign-off.

Mindset shift

Stop	Start
"Build it", then regret in code	Problem → SPEC → bounded plan → code

Exit criteria for this curriculum

Before non-trivial **Symfony**, **Go**, **Terraform**, or **MySQL** churn in these exercises you can produce a SPEC artefact: problem, acceptance, boundaries where needed, constraints/NFR, implementation strategy — **before** IDE-driven delivery.

Practice stacks here

Symfony (PHP / DDD / CQRS) • Go APIs/workers • Terraform • MySQL truth for data+NFR workloads.

Theme checklist across units

Specification first • acceptance criteria • boundaries/ownership • constraints + NFR • implementation specification • validation ownership • spec evolution • drift detection • partition • dependency graph • cross-spec consistency • ownership hierarchy among SPECS.

Checkpoint wording

Prefer **Claude implementing a SPEC** — not prompting straight to unsolicited large codebases.

Unified SPEC template (same spine for Symfony, Go, Ops/IaC, MySQL-backed work)

Use the same headings; omit only what truly does **not** apply. Never skip **Acceptance** nor **validation** when behaviour moves.

Section	Purpose
Problem	Grounded sentence + why now.
Goal	Measurable organisational outcome where possible.
Constraint	Hard bans / compulsory stack rules.
NFR	Throughput, availability, latency, audit, retention, safety baselines — mark hypothesis vs measured.
Boundary / ownership	Which context/module owns decisions (Order , Payment , Inventory , ...).
Acceptance	Short checkable behaviours (often ≥ 5 on feature-sized slices).
Implementation Strategy	Modules, rollout, infra touchpoints — spec fidelity , still not pasted production code dumps.
Tradeoff	≥ 2 plausible options + deliberate pick.
Risk	First-failure modes, mitigations, unknowns.
Implementation spec (when warranted)	Named tech + numeric knobs (e.g. RabbitMQ retries= 3 , DLQ policy, timeouts, worker counts) immediately before coding .

Cross-spec consistency: if several SPECS exist (`api` , `worker` , `db` , `terraform`) keep a terse **dependency/consistency ledger** referencing shared noun definitions so they cannot silently diverge (“must agree column X semantics”).

Ownership hierarchy: parent programme SPEC adjudicates clashes; child bounded-context SPEC cannot quietly violate parent acceptance.

Theme: The largest tactical shift — Claude gets SPEC-shaped answers **before** wall-of-code drive-by.

Bad vs good

Weak: “Build an order system.”

Stronger starting skeleton:

- **Problem:** truthful **order flow** span.
- **Goal:** purchase outcomes you certify.
- **Constraint:** **refund**, **cancel**, **retry**, **audit** inside scope fences.
- **NFR (starter):** e.g. cite **500 rps target** honestly as throughput hypothesis pending measurement.

Produce **readable SPEC** collaborators accept before implementation begins.

Practice

Track	Micro-focus
Symfony	Order aggregate invariants enumerated before method sketches.
Go	Worker / queue contracts articulated before handlers.

Lab — invariant rule

Every sizeable task ships a SPEC Markdown using the **unified headings above** *before* diff-sized implementation output.

Reflect: enumerate ≥ 3 branching unknowns extinguished SPEC-first vs impulse coding.

Checklist

- [] Separate visible blocks exist for **Problem / Goal / Constraint / starter NFR**.
 - [] Model’s heavyweight first replies stay SPEC-shaped whenever change risk is meaningful.
-

Acceptance criteria ownership

Theme: Architect language is not “works” — it is **what “done” means in observable checks**.

Bad vs good

Weak: “Refund works.”

Strong: Refund implies (example bundle — tune to reality):

- **Payment status** transitions per policy.
- **Inventory restoration** succeeds or saga compensates deterministically.
- **Audit** record persisted with immutable trail.
- **Notification** emitted on terminal states.
- **Rollback rule** when inventory step fails mid-flight.

Every line should be binary checkable (**pass/fail**) if you exercised the behaviour.

Practice

Track	Focus
Symfony	Refund flow spanning aggregate + integrations.
Go	Payment retry — success criteria per attempt class.

Lab

Each feature-shaped task: ship **minimum five acceptance criteria**. If merging two features, multiply carefully — duplication is intentional until your taxonomy matures.

Measure: can a teammate sign off via checklist without trusting prose?

Checklist

- [] Criteria read like **verification steps**, not marketing.
- [] Each criterion binds to **ownership** (“who observes / tests this?”).

Boundary ownership

Theme: DDD payoff — SPEC states **who owns what** across contexts.

Bad vs good

Weak: “Order does everything.”

Strong: Explicit separation with ownership:

Context / module	Owns decisions on
Order	...
Payment	...
Inventory	...
Notification	...

Interfaces (events, REST, synchronous calls) cite **upstream/downstream SPEC IDs** once you archive multiple docs.

Spec ownership hierarchy

- **Platform / programme SPEC** owns cross-cutting vocabulary and non-negotiables.
- **Bounded-context SPECS** own local rules; they reference parent invariants explicitly.
- **Child must not mute parent acceptance** silently.

Practice

Track	Focus
Symfony	Bounded context sketch + anti-ball-of-mud guardrails.
Go	Payment platform module boundaries vs generic “utilities”.

Lab

Ask for an **ownership / boundary diagram ASCII** plus a **sentence per edge** citing contract direction — **before** implementation files change.

If edges disagree with prior SPECS → **cross-spec inconsistency**: fix SPEC graph first.

Checklist

- [] Each arrow has naming: **owns data / emits event / invokes command**.
- [] Forbidden “reach across” behaviour is spelled as **explicit integration contract**.

Source: 04-specification-guided-development/04-constraints-and-nfr.md

Constraints + NFR

Theme: Without constraints + measurable **non-functional** truths, Claude drifts toward generic CRUD.

Functional guarantees (subset examples)

Enumerate behaviour-level constraints you mean:

- Retry semantics, cancel windows, refunds, entitlement rules ...

NFR catalogue (iterate per system)

Concrete numbers beat vibes:

Class	Examples
Load	Peak RPS (e.g. 500 sustained vs burst)
Availability	Tier target (e.g. 99.95%) and how measured
Latency	p50 / p95 / p99 thresholds that matter monetarily
Audit	immutable log, correlation IDs, retention
Security posture	Secrets handling baseline, GDPR touchpoints if any

Adjust false precision — cite “target / hypothesis / TBD instrumentation” honestly.

Practice

Track	Focus
Go	Payment API — explicit quotas + timeout floors.
Ops / Terraform	Deploy SPEC blast radius per module; rollback SLA.

Lab

Attach an **NFR block** before implementation on every sizeable task retroactively add if missing).

Checklist

- [] At least **one measurable** NFR or explicit “defer / unknown” rationale.
 - [] Constraints say **cannot** vs **prefer** clearly.
-

Source: `04-specification-guided-development/05-implementation-specification.md`

Implementation specification

Theme: Immediately before coding: an **operations-level spec** naming technologies and numeric knobs — still specification, **not yet** the codebase.

Bad vs good

Weak: “Create a queue.”

****Stronger example knob block:**

- Broker: **RabbitMQ** (or substitute)
- **Retry:** max **3**, exponential policy detail
- **DLQ:** attach policy + poisoning handling
- **Timeouts:** inbound HTTP, ACK budget, outbound calls
- **Worker concurrency:** baseline **5** (hypothesis)

Practice

Track	Focus
Go	Distributed worker fleet behaviour.
Symfony	CQRS flow sequencing + async hand-off spec.

Lab policy

Hard rule for training runs: Claude **does not emit production code blocks** until an **implementation spec** subsection exists naming stack + limits + rollout guard rails.

Implementation spec is glue between architecture narrative and keystrokes — keep it reversible if wrong.

Checklist

- [] Retry / DLQ / idempotency / timeout story has **explicit numbers**.
 - [] MySQL specifics (indexes, partitioning hints) deferred here only if SPEC says “schema phase gate”.
-

Source: `04-specification-guided-development/06-spec-evolution-and-drift.md`

Spec evolution + drift

Theme: Living systems rearrange truths — SPEC must evolve **with rationale** or implementation will silently diverge.

Evolution rehearsal

Historical narrative pattern:

SPEC baseline: synchronous payment acknowledgement.
SYSTEM shift: asynchronous settlement path required.
SPEC duty: changelog entry — drivers, migrating acceptance, deprecated bullets.

Detecting spec drift

- Symptoms:**
- Implemented retry loop violates **bounded retry SPEC** (“infinite-ish”).
 - Timeouts vanished relative to mandated maximum connect time in SPEC.

- Process:**
1. Extract **golden SPEC excerpt** unchanged from prior baseline.
 2. Diff behavioural claims in implementation/design doc.
 3. Mark **violations**.
 4. Either **repair code** toward SPEC or — better when reality changed — **revise SPEC** with traceable history.

Lab

Produce **before/after SPEC delta** prose for each intentional evolution; run a Claude-assisted **SPEC compliance sweep** referencing stored acceptance bullets.

Checklist

- [] SPEC version or date stamp increments when behavioural truth moves.
- [] Violations categorized: **truth changed** vs **engineering mistake**.

Source: 04-specification-guided-development/07-spec-partition-and-dependency-graph.md

Spec partition + dependency graph

Theme: Scale past one monolith SPEC — partitioned docs + clarity on **upstream/downstream** truth.

Partition pattern

Separate living documents instead of cosmic scroll:

Example slice
frontend-spec.md <i>(if applicable)</i>
api-spec.md
worker-spec.md
db-schema-spec.md
terraform-spec.md

Each references **upstream dependencies succinctly**:

worker SPEC	→ consumes queue contract declared in api SPEC revision N
worker SPEC	→ persistence invariants delegated to db SPEC section 3.x

Dependency graph artefacts

Maintain lightweight graph (ASCII or Mermaid-compatible text) tying:
frontend UI contracts → API payloads → worker processing → datastore invariants → IaC rollout ordering
Forbidden training anti-pattern here: Claude attempts **whole system code** simultaneously — partitioning enforces phased verification.

Cross-spec consistency sweeps

When two SPECS cite the same noun (“PaymentIdempotencyKey”), periodic diff must show **matching definitions** — otherwise classification: **SPEC conflict** needing merge session.

Ownership recap

Higher-level SPEC adjudicates clashes between child SPECS if teams disagree until governance steps in.

Lab

Produce **five SPECS stubs** minimal but linked for a hypothetical payment platform skeleton + dependency arrows only — no full implementation.

Checklist

- [] Every partitioned SPEC lists **consumes / provides**.
- [] Graph updates when critical contract fields rename.

Source: 04-specification-guided-development/08-integration-lab.md

Integration lab (Phase 3)

Theme: Payment Platform umbrella — Claude leads you through SPEC spine end-to-end, you keep **ownership of validation**.

Required chain

Problem

- SPEC (fill every heading from the Unified SPEC template – Problem through Implementation spec gate)
- Acceptance
- Architecture notes where needed
- Implementation Strategy + Tradeoff + Risk
- Implementation spec (numbered tech knobs – still pre-code gate)
- Implementation artefacts (only after SPEC gate)
- Validation proof (diff vs Acceptance bullets + flagged NFRs)

Claude may draft intermediate docs — **you** decide merge worthiness vs drift.

Measure deliberately

Metric	Notes
Iterations	Rework loops before acceptable SPEC
Quality	Peer sign-off heuristic
Spec drift incidents	Number of detected mismatches after implementation snapshots

Avoid premature coding windows — reopen SPEC when mismatch count spikes.

Stack rotation within lab

Sweep **Symfony + Go + Terraform + MySQL** surfaces so each SPEC partition faces reality constraints at least once per full exercise cycle when possible.

Phase checkpoint restated

Operational phrase to internalize:

Stop: Claude wrote code spontaneously.

Start: Claude instantiated / refined / aligned implementation to validated specification.

Phase — 05-test-engineering-ownership

Directory: 05-test-engineering-ownership/

Source: 05-test-engineering-ownership/01-unit-testing-ownership.md

Unit testing ownership

Phase framing (Test Engineering Ownership)

Phase numbering: topical track 01 - 06 (topic order).

Working pipeline (closes the loop each increment)

Specification → Implementation → Test → Verification → Repair

- Verification ownership:** you certify evidence **against acceptance + risk appetite** — green CI alone ≠ finished if claims about safety/scalability stay unexamined.
- Regression ownership:** when behaviour changes, owning **who updates / deletes which tests**, and guarding against orphan assertions.
- Test pyramid stance:** biased toward **cheap fast foundational tests** widening toward costlier strata only when seams demand them — Claude should not hallucinate pyramid inversions casually.
- Failure ownership:** surfaced defect → reproducible specimen → characterised root class → guarded test added or explicitly waived debt.

Mindset pivot

Old comfort	Owned comfort
“It runs / looks fine” ⇒ done	SPEC-backed change implemented ⇒ verified ⇒ stressed (where owed) ⇒ repaired knowingly

Exit bar for stacks in this lane

Fluent loop across realistic slices:

- Symfony ≠ persistence expectations
- Go service / worker logic ≠ behavioural truth
- Queues + DB interplay where your architecture demands it

Claude-assisted systems must articulate **why** artefacts look safe-enough plus **scalability caveats**, not shrug “passes smoke”.

Theme inventory carried across numbered units

Unit • Integration • Contract • Property • Chaos • Load modalities plus pyramid + verification/regression/failure ownership consciousness.

Theme: Behaviour-level proof — deterministic, narrowly scoped assertions about domain rules.

Symfony cue

CQRS aggregate kernels: mutations + invariants + rejection paths surfaced as succinct PHPUnit (or successor) suites — avoid testing framework plumbing instead of business law.

Go cue

Worker/business helpers devoid of flaky clock/network — deterministic clocks & DI seams so retry math is honest.

Suggested labs (mix & match)

Refund orchestration branching

Retry backoff edge math

Aggressive validator permutations rejecting illegal domain states sooner than integration stage

Concept tags surfaced now

● **unit test posture** ● **deterministic verification**

Checklist

- [] Each test cites **business rule headline** mirrored from SPEC lineage.
- [] Forbidden randomness sans explicit seed rationale.

Source: 05-test-engineering-ownership/02-integration-testing-ownership.md

Integration testing ownership

Theme: Confidence where **pieces actually meet wiring** — not mocked away.

Bridges to prove in exercises:

Symfony $\rightleftharpoons$ persistence (DB fixture / transaction strategy you trust)

Go $\rightleftharpoons$ RabbitMQ $\rightleftharpoons$ authoritative database path

Lab scaffold

Scenario path:

worker publishes/consumes → queue semantics hold → database reflects expected saga / idempotency

Compose **narrow** slices: one unhappy path plus one sunny path beats ten shallow happy paths.

Concept tags

Integration ownership — you say which dependencies are **in-scope real** versus test doubles today.

Dependency verification — assert observable effects at seams (queued message shape, persisted row mutations, transactional boundaries).

Checklist

- [] Test names state **boundary under proof** (“after ack, ledger row uniqueness holds”).
- [] Fixtures stay **minimal** — avoid copying production dumps wholesale.

Source: 05-test-engineering-ownership/03-contract-testing-ownership.md

Contract testing ownership

Theme: Stability at **service edges** — request/response (or event schema) is a contractual obligation owned by somebody.

Sketch:

producer shape ↔ enforced consumer expectations

Labs rotate:

Side	Artefact
Symfony HTTP API	Contract tests on handler DTO symmetry + rejection codes
Go worker integrations	Message contract vs publisher truth

Boundary ownership recap

Owning **boundary** implies owning **changelog discipline** — breaking edits require deliberate version bump rituals you document.

Concept tags

Contract ownership

Boundary ownership (overlap with SPEC phase vocabulary — tighten language so both specs and tests cite same glossary fields)

Checklist

- [] Fixtures include representative **illegal** payloads, not golden paths only.
- [] Schema drift alerts land before unrelated teams burn days.

Source: 05-test-engineering-ownership/04-property-testing-ownership.md

Property testing ownership

Theme: Assertions over **rules**, not heroic single-shot examples alone.

Rather than chasing one miraculous input snapshot, articulate **predicates**:

Policy surface	Sketch property
Retry counter	retries are never below zero and never surpass configured ceiling
Timeouts configured	durations never negative

Lab intuition — payment-ish ownership

Enumerate invariants owed to treasury / treasury-adjacent systems before coding generators — property tests amplify once invariants crystallise.

Concept tags

Invariant ownership (you must list them before shrinking tools help)

Property testing posture (`testing/quick` Go style / mature PHP equivalents / external harnesses acceptable)

Practical guardrail

Generators must stay **controlled blast radius**: cap complexity so failures shrink to comprehensible replay seeds.

Checklist

- [] Predicates cite **pure** domain kernels first (easier shrinking).
 - [] Logs print **minimal counterexample artefact**, not oceans of stderr.
-

Load + chaos testing ownership

Theme: Produce mindset for **risk under reality’s thumb** — not “unit green” narcissism alone.

Stress scenarios (exemplary ladder — tailor numbers):

Class	Spike
Load	1 k → 5 k sustained-ish coherent requests respecting auth reality
Worker crash	SIGKILL midway batch — queue drain & duplicate semantics survive
Queue delay / backlog	consumer slower than spike — SLA story breathes honestly
Network latency jitter	slow dependency responses — timeouts don’t deadlock platform

Interpretation rule: regressions flagged here **block “done”** until owned or waived with explicit stakeholder debt note.

Concept tags

Load ownership

Chaos ownership (blast radius budgeting + revert story before you pull levers)

Safety note

Exercise chaos in isolated sandboxes predictable to your org — destructive commands never surprise shared environments.

Checklist

- [] Scenario matrix references **measurable SLO hypotheses** drafted earlier in SPEC lineage.
- [] Tear-down restores baseline queues / topics / DB clones used.

Integration lab (Phase 3.5 — Test Engineering)

Umbrella: Payment Platform mental model tying Symfony, Go handlers, authoritative DB truths, queues.

Cascade you enforce with Claude-assisted drafting **but human ownership on sign-off**:

Test strategy → implementation alignment → verification evidence → stress scenarios → repair loop closures

Minimum artefact bundles

Stage	Ownership question answered
Strategy	Pyramid placement + tooling + flaky-risk controls
Implementation	Code + tests mutually evolved (not dangling tests)
Verification	CI signal map + deterministic reproduction recipe
Stress	Load/chaos results interpreted vs NFR hypotheses
Repair	Postmortem style record: symptom → causal class → guarded regression

Measure deliberately

Iterations to stable CI

Quality = senior sceptic review pass

Incident count from **unknown unknown** class post-release (target trending down lesson over calendar)

Phase checkpoint wording

Translate habit:

Former comfort: **“tests exist somewhere.”**

Target comfort: **“system behaviour is evidenced safe enough + scalable-under-known assumptions.”**

Stack rotation reminder

Symfony vertical • Go services/workers • MySQL correctness • RabbitMQ behavioural contracts

If any toolchain statement here conflicts with versions you operate, treat it as SPEC/test debt — reconcile before relying on Claude output verbatim.

Phase — 06-goal-process-driven-development

Directory: 06-goal-process-driven-development/

Source: 06-goal-process-driven-development/01-goal-decomposition.md

Goal decomposition

Phase framing — Goal / Process Driven Development

Area topics: 01 - 06 (topic order only).

Delivery spine (engineering process shape)

Goal → Spec → Task graph → Implementation → Validation

Mindset pivot

Old reflex	Owned reflex
“Feature idea” → rush to code chunks	Goal → process clarity → executable task graph → delivery evidence

Exit bar

When asked “build a payment platform”, **first artefacts** tilt toward:

- articulated **goal + ownership**
- workflow / **task decomposition**
- **risk** surface
- **validation** posture

...rather than dumping code volume before reasoning is inspectable.

Themes threaded across units: goal decomposition • workflow ownership • validation loop • verification loop • rollback ownership • optimization ownership

Claude Goal Template (use consistently)

Same headings for Symphony, Go, Ops/IaC, and DB-heavy work:

Heading	Holds
GOAL	Outcome sentence + measurable north-star boundary.
SUCCESS CRITERIA	Checkables (acceptance-aligned).
WORKFLOW	Ordered stages artefacts traverse (diagram + captions).
OWNERSHIP	Roles / bounded contexts owning each slice.
RISK	What fails first — technical + human process.
VALIDATION	Proof that behaviour meets intent (“does it run / do the checks pass”).
ROLLBACK	Safe unwinding when validation discovers fatal mismatch.
OPTIMIZATION	Complexity / cost / latency improvements after baseline safety holds.

You may prepend **SPEC bullets** beside Goal when practising with Specification Guided habits from earlier phases.

Theme: Decompose ambiguous platform asks into navigable hierarchies instead of hallucinated single-file leaps.

Bad vs grounded

Weak: “Make a payments system.”

Strong:

- **GOAL:** a named **payment platform slice** with integrity guarantees you state outright.

Show $\geq$ **three decomposition depths**, for example:

```
payment API strand
├─ retry ownership
│   └─ idempotency rules
inventory alignment strand
notifications strand
audit / trace strand
refund choreography strand
```

Drill deeper on one strand:

```
payment API
├─ auth perimeter
│   └─ request validation gates
│       └─ timeouts coexisting with retries
```

Practice rotations

Track	Skeleton focus
Symfony	CQRS refund flow slicing commands vs queries.
Go	Worker system: lifecycle, backoff, poison handling decomposition.
Ops	Terraform deploy stages with checkpoints inside task graph prose.

Lab rule

Each exercise-sized backlog item: **minimum three decomposition layers** before scaling implementation discussion.

Checklist

- [] Level-one bullets describe **capabilities**, not incidental file churn.
- [] Leaves cite **verification** you intend to automate or script soon.

Source: 06-goal-process-driven-development/02-workflow-ownership.md

Workflow ownership

Theme: The architect stance asks **how work flows through the system and organization**, not only *what widgets to crank out*.

Pattern

Enumerate **states / transitions people and software traverse** until value is evidenced.

Go — checkout-esque illustration

ASCII discipline example:

```
request ingress
├─ validation gates
│   └─ payment attempt cluster
│       └─ inventory settlement / saga compensation decisions
│           └─ outbound notification fan-out
│               └─ immutable audit anchors
```

Ops — lifecycle illustration

```
pre-deploy validations
  → infra backup / checkpoint policy
    → rollout wave
      → layered health affirmation
        → guarded rollback doorway if signals regress
```

Lab mandate

Produce **diagram + narrative captions** documenting workflow **prior** to code-dense answers when using Claude collaboratively.

Concept tags embedded here

- **workflow ownership** (you articulate transitions, not incidental micro-tasks scattered ad hoc)

Checklist

- [] Every arrow names **ownership + artefact exchanged** across it.
- [] Forbidden “magic step” jargon — unexplained leaps get numbered unknown risks.

Source: 06-goal-process-driven-development/03-validation-and-verification-loops.md

Validation and verification loops

Theme: Outputs are **doubled-checked** deliberately — scepticism is methodological, not cynical.

Vocabulary tightening

Term	Plain meaning
Verification	“Built as specified?” (diff against SPEC/SUCCESS artefacts, structural integrity, contract symmetry).
Validation	“Does behaviour match user / business intent beyond mechanical compliance?” broader acceptance lens.

Loop you rehearse with Claude scaffolding

```
Spec alignment
  → Implementation slice
    → Automated / scripted verification pulses
      → Higher-order validation checkpoints
        → targeted repair choreography
          → rerun until risks acknowledged or extinguished
```

Practice rotations

Track	Micro lens
Symphony	DDD aggregate mutation + invariant defences iterated under tests.
Go	Queue worker lifecycle path — duplicate delivery probes, backoff honesty.
Ops / Helm-ish release story	Canary / staged chart release with observability confirmations.

Lab choreography

Structured three-pass Claude collaboration:

1. Implementation draft **bounded**

- 2. **Verification pass** referencing frozen checklist bullets
 - 3. **Repair** pass reconciling divergence inventory
- Echo logs / evidence snippets so regressions shrink next lap.

Concept tags surfaced

- validation loop ● verification loop ● repair ownership inside loop

Checklist

- [] Checklists originate from **prior frozen SUCCESS CRITERIA** — no retroactive shifting without audit note.

Source: 06-goal-process-driven-development/04-rollback-ownership.md

Rollback ownership

Theme: Shipping culture without rehearsed reversal paths is gamble — architect owns **panic dampening choreography**.

Anti-pattern symptom train

deploy spike → anomaly storm → unbounded thrash

Healthy spine

deploy attempt

- validations / instrumentation eyeing regression budget
 - consciously chosen rollback doorway
 - stabilized observation window
 - iterative forward retry only once safe

Practice vignettes

Track	Drill
Symphony	Migration failure cascade reversing schema drift without orphaned rows.
Go	Worker malfunction draining poison without burning platform trust.
Ops / Terraform drift	plan divergence surfacing latent conflict — scripted retreat story.

Lab invariant

Deployment / rollout narrative **must attach explicit rollback choreography** bullets before “LGTM culturally” settles.

Concept tag recap

- **rollback ownership** — not last-minute improvisation alone

Checklist

- [] Rollback steps enumerate **ordering constraints** (“cannot undo B before reverting A”).
- [] Data migration reversibility or compensating manoeuvres articulated honestly.

Optimization ownership

Theme: After baseline safety proofs, interrogate residual **risk • complexity • cost • performance**.
Stop asking Claude only “does it pass?” Prompt instead “**could this be materially better under stated ceilings?**”

Practice hunts

Track	Optimization lens prompts
Symfony	aggregate cognitive load hotspots, needless synchronous coupling blobs
Go	worker concurrency reality vs allocation churn, backoff aggressiveness realism
Ops	deploy choreography minutes, flaky stage sensitivity, infra spend leakage

Lab rule cadence

Per meaningful task surface $\geq$ **three** candidate optimizations categorized:

1. Complexity / readability debt trade
2. Cost / infra sensitivity
3. Latency / throughput envelope stretch

Discard only with explicit rationale recorded.

Concept tag recap

- **optimization ownership** — iterative trimming after truth surfaces

Checklist

- [] Optimisations tie back to measurable **SUCCESS CRITERIA** or budgets already declared upstream.

Integration lab — Phase 4

Scenario: Payment Platform umbrella — Claude collaborates drafting, **you certify** orchestration artefacts.
Full chain rehearsal:

Goal capture
→ Structured decomposition ladders
→ Annotated workflows + early risk surfacing
→ VALIDATION choreography planning
→ rollback readiness story
→ optimisation candidates catalogued
→ guarded implementation slicing
→ measured iteration & repair journaling

Measure

Signal	Rough meaning
Iterations	Dialogue cycles before artefacts stabilise
Quality	sceptical reviewer pass heuristic
Complexity	subjective structural intricacy creep
Repair count	loop closures after divergence detection

Journal prompts (short)

1. Moments Claude **conducted orchestration cleanly**
2. **Failure seams** needing prompt / template tweak
3. **Workflow deltas** you'd encode into personal Rule / checklist next sprint

Phase checkpoint slogan

Migrate identity:

Earlier comfort: Claude "drops features."

Target comfort: Claude **conducts repeatable engineering-process choreography** tethered to goals + evidence ladders.

Rotate stack lenses **Symfony ⇄ Go ⇄ DB truths ⇄ Ops releases** congruent with earlier syllabus threads.

Phase — 07-claude-architect-workflow

Directory: 07-claude-architect-workflow/

Source: 07-claude-architect-workflow/01-domain-modeling-and-requirements.md

Domain modeling and requirements

Phase framing — Claude Architect Workflow (Phase 5)

Units in this folder: 01 - 06 (topic order only).

Themes across this area

DDD • CQRS • scaling • cache strategy • event-driven design • ownership • tradeoff analysis • boundary ownership • failure ownership

Architect workflow spine (each significant slice)

Requirement → Architecture → Tradeoff → Risk → Implementation plan → Validation

Aligned with **Symfony**, **Go**, **MySQL**, and **distributed-systems** realities (partitions, duplication, partial failure).

Reaction you want when someone says “build a payment platform”

First artefacts skew toward **goal / ownership / workflow / task decomposition / risk / validation — not** a wall of unpremeditated code.

Claude Goal Template (same headings everywhere: Symfony / Go / Ops / DB)

Heading	Holds
GOAL	Outcome and scope boundary.
SUCCESS CRITERIA	Checkable acceptance bullets.
WORKFLOW	How responsibility and data move (diagram + captions).
OWNERSHIP	Contexts, teams, components that decide and run each slice.
RISK	Failure and scaling assumptions surfaced early.
VALIDATION	How you prove “works as intended”.
ROLLBACK	Safe retreat when validation fails badly.
OPTIMIZATION	Complexity / cost / latency improvements after baseline safety.

Claude Architect spine (use alongside or inside the template)

Map **Requirement** into Goal + SUCCESS CRITERIA; **Architecture** + **Tradeoff** + **Risk** as explicit sections; **Implementation plan** as sequenced tasks with owners; **Validation** tying back to SUCCESS CRITERIA.

Theme (this unit): Specification-first domain shape — DDD aggregates and boundaries, CQRS separation of intents, decomposition before hype coding.

Bad vs grounded

Weak: “Build a payments system.”

Strong: Named **payment platform slice** plus **three+ levels** of decomposition, for example:

```
payments API strand
├─ auth / identity touchpoints
├─ idempotency + retry posture
inventory / fulfilment coupling
notifications
audit & refund choreography
```

Practice rotations

Stack	Drill
Symfony	CQRS refund flow — command vs query ownership, aggregates.
Go	Worker subsystem decomposition (lifecycle, poison, backoff).
Ops / IaC	Terraform (or equivalent) staged rollout encoded as workflow steps with explicit rollback hooks.

Lab rule

Every exercise task: document **minimum three decomposition levels before** an implementation-plan paragraph grows fat.

Checklist

- [] Nouns (**Order, Payment, Ledger**) have **bounded** meanings per context.
- [] CQRS split answers **who reads what model** vs **who changes state**.

Source: 07-claude-architect-workflow/02-event-driven-workflow-and-boundaries.md

Event-driven workflow and boundaries

Theme: The architect owns **how the system progresses through process** — especially when work is **asynchronous** and **event-led**.

Sketch (example — tailor to your platform):

```
request / command accepted
├─ validation & policy gates
│   └─ durable intent recorded
│       └─ payment / settlement choreography
│           └─ inventory / saga compensations where needed
│               └─ outbound notifications / integration events
│                   └─ audit & observability checkpoints
```

Event-driven stance: transitions publish **facts** bounded contexts subscribe to via **explicit contracts** (see contract-testing habits from Phase 3.5 where relevant).

Boundary ownership reminder

Edges must state **producer owner** and **consumer owner**; drifting vocabulary between specs, tests, and events is architectural debt.

Ops workflow complement

```
change validation windows
├─ artefact backups / checkpoints acceptable to policy
│   └─ phased deploy
│       └─ health & SLO-aligned checks
│           └─ guarded rollback doorway
```

Lab mandate

Produce **diagram + narrative** of the workflow **before** implementation-heavy answers when pairing with Claude on this curriculum.

Checklist

- [] Events / messages named as **past-tense facts** where that matches your modelling style.
- [] Duplicate delivery and reordering hazards called out **at boundaries**.

Source: 07-claude-architect-workflow/03-tradeoff-risk-and-validation-loop.md

Tradeoff, risk, and validation loop

Theme: Outputs are scrutinised systematically — scepticism owns the loop alongside creativity.

Vocabulary alignment

Term	Typical meaning
Verification	Matches frozen architecture + spec + SUCCESS CRITERIA structurally.
Validation	Behaviour and intent hold under scenarios that matter commercially.

Loop (with Claude drafts, you certify)

```
Spec / SUCCESS frozen for slice
  → implementation plan authorised
    → incremental build
      → automated / scripted verification
        → stakeholder-scale validation checkpoints
          → deliberate repair iterations
```

Distributed-system lens

Call out tradeoffs affecting **latency, availability, consistency** (e.g. MySQL replication lag, duplicated messages, skewed caches and how you invalidated or TTL'd read models).

Practice rotations

Stack	Scenario
Symfony	DDD aggregate change under invariant-heavy tests / reviews.
Go	Queue worker duplicate-delivery probes.

Lab choreography

1. Implementation slice **within explicit tradeoff+RISK preamble**
2. **Verification checklist** replay vs frozen bullets
3. **Repair ledger** documenting drift classes

Checklist

- [] Tradeoff section lists $\geq$ **two** plausible architecture forks + chosen path rationale.

Failure and rollback ownership

Theme: Architect-grade **failure ownership** — rehearsed reversal paths dampen organisational panic during distributed-system incidents.

Anti-pattern symptom train

release thrust → surprise regression storm → unstructured thrash

Healthier cadence

deploy / change window
→ validation & observable guardrails armed
→ articulated rollback doorway
→ rehearsed sequencing (DB / queue / runtime / config)
→ deliberate forward replay only once stable

Practice vignettes

Track	Failure drill idea
Symfony	Migration midway failure — reversing partial schema creep safely with data truth.
Go	Worker catastrophe mid-batch — draining poison, protecting idempotent edges.
Ops / Terraform	Divergent plan — scripted retreat preserving state coherence.

Lab invariant

Any release / rollout plan that touches production truth includes explicit **rollback plan block** bullets **ordering-dependent** (“cannot unwind B until A”).

Distributed note: rollback must consider **dual writes**, **replay**, and **compensating transactions** honestly.

Concept tag recap

- **failure ownership** — reproducible timelines, causal narrative, guarding tests after learning
- **rollback ownership** — choreography pre-agreed

Checklist

- [] Rollback honours **ordering** constraints across MySQL migrations, queues, and cache invalidation bursts.

Scaling, cache strategy, and optimization ownership

Theme: Beyond “runs” — interrogate load shape, caching truth, duplicated work, MySQL hotspots, queue backpressure — under **architecture-level tradeoffs**.

Scaling dimensions (discussion prompts with Claude)

Dimension	Probe
Horizontal	stateless tiers, partitioning, shard awareness
Vertical	single-node ceilings you still consciously accept temporarily
Data / MySQL	readReplicas, hotspots, paging, index choreography
Queue / worker	consumer parallelism vs ordering / poison exposure
Distributed truth	timeout budgets, cascading failure containment

Cache strategy checklist

Declare explicitly per endpoint / read model:

Decision	Answer
What is cached	keys, payloads, TTL posture
Invalidation triggers	event-driven vs time-based sweep
Consistency class	read-your-writes guarantees or eventual-only
Stampede mitigation	single-flight patterns, probabilistic TTL jitter

Stale cache classes are architectural failures — classify them explicitly when debugging.

Optimization lab rule

Minimum **three** optimisation candidates per meaningful delivery slice labelled across:

- 1. Complexity / readability
- 2. Cost / infra
- 3. Latency / throughput envelope

Defer only with rationale tied to Goal / SUCCESS frozen upstream.

Checklist

- [] Scaling claims cite **measurable** hypotheses or monitoring hooks you will attach.

Source: 07-claude-architect-workflow/06-integration-lab.md

Integration lab — Claude Architect Workflow (Phase 5)

Challenge: Full **payment platform slice** articulated as an architect-led engagement.

Artefact cascade (minimum)

```
Goal frozen (SUCCESS CRITERIA visible)
  → deep decomposition ladders
    → labelled workflows (+ event edges if asynchronous)
      → ARCHITECTURE + TRADE + RISK package
        → VALIDATION / VERIFICATION playbook
          → rollback readiness story
            → optimisation / scaling / caching candidates catalogued
              → restrained implementation sequencing
                → journaling & retrospective loop tightening
```

Claude drafts; **you** authorise merges when evidence ladders satisfy scepticism.

Measure

Gauge	Aim
Iterations before stabilisation	shrinking over repetition
Quality	cynical senior sign-off heuristic
Complexity creep	flagged early
Repair count post-verification	trend downward as templates mature

Short journal cues

Where Claude orchestrated cleanly • seams that exploded • tweaks you’ll encode next (Rules/personal playbook)

Checkpoint phrase

Comfort moves from Claude “implementing scattered features alone” toward Claude assisting a **bounded architect workflow**: requirements → coherent architecture narratives → disciplined delivery evidence.

Phase — 08-diff-refactor-ownership

Directory: 08-diff-refactor-ownership/

Source: 08-diff-refactor-ownership/01-diff-ownership.md

Diff ownership

Phase framing — Diff / Refactor Ownership (“Phase 5.5”)

Units in this folder: 01 – 06 (topic order only).

Themes

Incremental refactor • diff ownership • migration ownership • safe change workflow • architectural evolution

Mindset shift: the architect-assisted workflow spends more time **changing what exists** than minting greenfield blobs. Claude should default to **controlled evolution**.

Reaction you want on a legacy system

First artefacts lean **diff, risk, migration, rollback, safe evolution** — not an instinctive **full rewrite**.

Claude Refactor Template (Symfony / Go / DB / Ops)

Heading	Holds
CURRENT STATE	What runs today — paths, behaviours, coupling you must not hallucinate away.
TARGET STATE	Observable end state for <i>this slice</i> — small enough to validate in one rollout window.
DIFF	Concrete edits: files/contracts/data/infra deltas (before → after mentally explicit).
RISK	Blast radius, partial failure modes, ambiguity in boundaries.
MIGRATION	Ordered steps bridging current → target — including compat layers and sunset flags.
ROLLBACK	Preconditions, ordering, irreversible forks called out honestly.
VALIDATION	Proof ladder: automated checks plus targeted manual / shadow traffic if needed.
DEPLOY STRATEGY	Blue/green, canary, feature flags, expand/contract migrations — whichever matches your stack policy.

Use this whenever the task touches **running** behaviour—not only Application code (Symfony / Go) but schema, queues, caches, Terraform/Helm, feature toggles.

Theme (this unit): Diff ownership — “What am I changing?” before “What am I inventing?”

Narrative rhythm

```
before (baseline truth)
  → after (minimal target intent)
    → named diff chunks (reviews land per chunk)
```

Example cue

Existing `OrderService` gains **clear refund ownership** at the boundary: who commands, who queries, where idempotency lives — **DIFF** spells that out module-by-module rather than rewriting the tree.

Practice rotations

Stack	Drill
Symfony	CQRS-on-existing flow — carve command/query seams without bulldozing the legacy service façade first.
Go	Worker queue touchpoint — one consumer path / one ACK semantics change defended in DIFF+RISK .

Lab invariant

Every task through this syllabus: Claude (or any assistant output you accept) delivers **CHANGE × IMPACT × RISK** before implementation paragraphs sprawl — map them to template columns loosely if that keeps you honest.

Checklist

- [] **Baseline** cites actual symbols / paths / behavioural contracts—not a generic diagram.

Source: `08-diff-refactor-ownership/02-incremental-refactor.md`

Incremental refactor

Theme: **Small change → validate → small change**. Rewrites disguised as “cleanups” are the common senior stumble.

Anti-pattern vs discipline

Weak narrative: Rip out module; hope tests exist.

Strong narrative: Sequenced **step 1 / step 2 / step 3** each leaving the system deployable/releasable (or cleanly behind a guarded flag).

Assistant guardrail rehearsal

Forbidden vibe: **delete-all / replace-all blueprint** masquerading as refactor. Required vibe: **evolution** preserving behaviour while relocating responsibilities.

Practice rotations

Stack	Drill anchor
Symfony	Legacy service peel — extract policy or persistence behind seams one vertical slice at a time.
Go	Repository refactor — interface drift without breaking callers; temporary adapters OK if labelled as debt with removal step.

Lab invariant

Rollback thinking piggybacks every intermediate step—even when “only internal” refactor—because internal refactors leak through timing and caching in distributed setups.

Checklist

- [] Steps are **ordering-safe** — each merges without requiring the subsequent step already shipped blindly.

Source: `08-diff-refactor-ownership/03-migration-ownership.md`

Migration ownership

Theme: **Enterprise-shaped change** means migrations are **plans** with deploy + validate + cleanup—not a single careless **ALTER** moment.

Data shape example

Weak: “Apply ALTER TABLE in prod.”

Strong:

```
additive / backward-compat schema choreography
  → deploy code reading both shapes or behind expand contract
    → validated traffic behaviour + metrics calm
      → compaction / deprecation cleanup with explicit sentinel checks
```

Practice rotations

Track	Drill idea
DB / MySQL	Rolling compatible column/table introduction; reversible phases until cleanup lock-in.
Symfony	CQRS migration — projector / read model divergence bridged deliberately.
Go	Worker-aware schema drift — versioned payloads, drain windows, rejection surfacing.
Ops	Terraform (or IaC analogue) changes as states with moves plus guardrails—not surprise applies.

Lab deliverable trio

Deliver migration plan + rollback + verification before implementation PR description exceeds a sane review surface.

Distributed note: message ordering, duplicate emits, stale consumers must appear in risk—diff alone hides them.

Checklist

- [] Cleanup phase timed only after dashboards / queue depths / dead-letter emptiness justify it.

Source: 08-diff-refactor-ownership/04-safe-change-workflow.md

Safe change workflow

Theme: You are reshaping the system, not scribbling detached code. Behavioural truth lives in rollout evidence.

```
frozen spec excerpt + SUCCESS for this slice
  → impact map (consumers / data / infra)
    → migration choreography selected
      → guarded deploy posture
        → layered validation ladders
          → articulated rollback doorway
```

Practice rotations

Stack	Scenario seed
Go	Distributed retry / backoff / idempotency policy shift — quantify thundering herds + duplicate completions.
Symfony	Payment aggregate invariant adjustment — concurrency + saga / process manager edges explored.

Lab invariant

Every behavioural change carries an explicit rollback story—even when rollback is flag flip + redeploy—not “figure it out if red”.

Impact section names latency, ordering, consistency, capacity regressions hypothesised—not only compile-time edits.

Checklist

- [] Blast radius enumerated for **integrations you do not control** — webhooks, partner web clients, cron overlap.

Source: 08-diff-refactor-ownership/05-architectural-evolution.md

Architectural evolution

Theme: **Today ≠ tomorrow blueprint dump**. Produce an honest bridge: motivations, phased risks, sequenced seams.

Diagram narrative you rehearse aloud:

```
AS-IS (truthful constraints)
  → TO-BE target topology (narrow initial slice plausible)
    → bridging migrations + toggles / strangler seams
```

Example hinge

Present: **modular-ish monolith** handling payments + notifications intertwined. Horizon: clearer **event-driven** boundaries—but only through **measurable intermediate states** reducing dual-write eras.

Practice rotations

Stack	Drill
Symfony	Modular monolith tightening — package boundaries enforcing dependency direction gradually.
Go	Worker extraction — carve bounded process without instant network fan-out entropy.

Assistant expectation

Articulate **WHY NOW**, quantify **risk classes**, expose **ordering of steps**. Ban hand-wavy miraculous cutovers unless you explicitly spike them as throwaway proofs first.

Checklist

- [] Intermediate architecture states each **shrinks** behavioural ambiguity—not only file moves.

Source: 08-diff-refactor-ownership/06-integration-lab.md

Integration lab — Diff / Refactor Ownership

Scenario capsule: **Legacy payment platform** showing familiar scar tissue:

- creeping **slow reads**
- brittle **worker timeouts**
- **refund** correctness bug class
- **cache** coherence / TTL mis-spec symptoms

Your job orchestrates artefacts **before code momentum**:

```
impact breadth (who breaks if wrong)
  → DIFF chunks sized for humane review
    → curated RISK ledger
      → migration + dual-read/write posture if warranted
        → rollback completeness check
          → restrained implementation pacing
```

Measure & journal

Signal	Probe
Quality	senior cynic could merge without rewriting your plan
Repair count	post-deploy thrash iterations
Rollback completeness	explicit ordering + irreversible forks flagged
Risk completeness	edge duplication / reorder / staleness enumerated

Notebook prompts:

- Assistant drifted toward **rewrite** vs disciplined **DIFF**
- Produced **risk debt** optimism vs grounded mitigations
- **Evolved** coherent system storyline with restrained steps

Checkpoint mantra

Comfort shifts from “Claude greenfields features” toward “Claude **evolves** trustworthy systems.”

Phase — 09-distributed-systems-ownership

Directory: 09-distributed-systems-ownership/

Source: 09-distributed-systems-ownership/01-cap-thinking.md

CAP thinking — distributed framing

Phase framing — Distributed Systems Ownership (“Phase 5.7”)

Units in this folder: 01 – 08 (topic order only).

Themes you carry throughout

CAP • consistency models • eventual consistency • outbox pattern • saga pattern • circuit breaker • bulkhead • rate limiting • failure propagation • idempotency ownership

Lifecycle spine

Request → Queue → Service → Storage → Recovery

Treat each arrow as having **explicit ownership**: latency budgets, duplication class, rollback posture, observable proof.

Mindset checkpoint

Comfort moves from modelling **HTTP + broker plumbing** loosely to asserting **distributed ownership**: what may diverge, for how long, who heals it, under which tests.

Distributed slice worksheet (reuse per lab)

Column	Holds
Slice / flow	E.g. “authorize capture”, “stock reservation”.
Consistency class	What readers may assume (monotonic?, bounded staleness?).
Failure modes	Duplicates, delay, reorder, partial commit, poison.
Idempotency	Stable keys / dedupe horizon / uniqueness strategy.
Recovery	Retries vs compensations vs human queue.
Verification	Metrics, traces, property-style scenarios.

Theme (this unit): CAP intuition baked into architectural choices—not folklore.

Letters (pragmatic grounding)

CAP is about tension under **partition** (lost/delayed messages, split brains, asymmetric visibility).

Pole	Probe
Consistency	Observational agreement about fresh writes you care about .
Availability	System continues yielding responses under degraded paths (sometimes degraded fidelity).
Partition tolerance	You must design for unreliable links or clock skew when systems split.

Rarely literal “toggle two of three”: real systems articulate **narrow strong islands** surrounded by deliberate eventual seams.

LAB — payment ownership CAP lens

Enumerate:

- Moments requiring **narrow transactional truth** vs tolerating **delayed projection**
- **Partition** manifestations in your sketch (payments API vs PSP webhook vs ledger DB).

Penalise hand-wavy “ACID solves it” spanning multiple runtimes/brokers.

Checklist

- [] For each externally visible payment state, labelled **consistency tier** ties to SLA + consumer contract.

Source: 09-distributed-systems-ownership/02-consistency-ownership.md

Consistency ownership

Theme: Declare **who may observe which truth state** across services, caches, replicas, and read models—not hand-wavy “eventually it matches.”

Vocabulary rehearsal

Intuition	Probe
Strong / linearizable (where applicable)	Single logical order of observable writes—narrow scope only when justified.
Serializable transaction island	When one database boundary hides distribution—still revisit cross-service lies.
Eventual consistency	Converges via defined channels; divergence window bounded by measurable artefacts (lag, versioning).

Eventual consistency — ownership specifics

Naming **upstream authority**, **projection lag SLA**, conflict resolution (last-write-wins only when safe), versioning on events / rows, stale read disclaimers at API edges.

Cross-cutting — idempotency ownership

Any eventual path multiplies retries; processors must tolerate **duplicate command/event** delivery without corrupting aggregates—carry stable natural keys forward from design spec.

LAB — dual ownership vignettes

1. Inventory ownership

Reservations vs oversell narratives; versioning / optimistic gates; phantom reads hiding in aggregates.

1. Payments ownership

Authorized vs settled vs disputed; authoritative ledger vs projecting notification state.

Expose **dual-write hazards** bluntly anytime two stores must reflect one business moment.

Checklist

- [] Written **consumer expectations** (“read model may lag N seconds / events”—numeric or tied to SLA).

Source: 09-distributed-systems-ownership/03-outbox-pattern.md

Outbox pattern

Theme: **Reliable causal bridge** DB → persisted intent → asynchronously published fact—without orphaned commits or phantom publishes.

Canonical mental wire:

```
DB transaction spans business mutation + outbox INSERT
  → dedicated relay/process polls / streams outbox safely
    → messaging broker absorbs with delivery semantics spelled out (e.g. at-least-once)
```

LAB vector — Symphony payment → RabbitMQ

Design (or refactor toward) explicit **ownership**:

- transactional boundary includes outbox append
- publisher never races ahead of commit visibility
- consumer **idempotency hooks** keyed to business/domain ids

Discuss duplicate publish, stalled relay backlog, reordering—all **outbox ownership** debt if ignored.

Checklist

- [] Replay / compaction story for stale outbox rows without corrupting uniqueness constraints.

Source: 09-distributed-systems-ownership/04-saga-ownership.md

Saga ownership

Theme: Distributed choreography replaces a mythical global transaction; **you** own compensations, idempotency, and observability of long-running narratives.

Nominal choreography:

```
payment intent accepted
  → inventory / fulfilment leg
    → notifications / ancillary writes
failure anywhere
  → deterministic compensation choreography (not silent hope)
```

Failure → compensation realities

Define **backward actions** semantics: financial reversals rarely mirror forward steps atomically—inversion may be asynchronous, disputed, regulatory-gated—document honestly.

Prefer **explicit saga state**: pending / compensated / aborted + poison handling for stuck legs.

Contrast **choreography** (events only) vs **orchestration** (central conductor): choose knowingly; hybrids exist.

Checklist

- [] Duplicate saga command delivery cannot double-charge inventory or treasury—proved via **idempotent** compensating logic.

Source: 09-distributed-systems-ownership/05-circuit-breaker-resilience.md

Circuit breaker — service protection

Theme: Isolate **upstream dependency failure** before it melts your thread pools, wallets, or customer trust.

Stateful guard mental model:

```
healthy / closed → sustained failures breach threshold → OPEN (fast-fail shields)
OPEN timer elapses probing success → HALF-OPEN exploratory calls
confidence regained → recover / CLOSED
```

Design probes

Failure classification: deterministic 4xx vs transient overload vs partner outage—**circuit policy differs**.

Half-open probing must remain **economical** — no thundering herd re-open storms.

Cascade containment pairs with timeouts, sane thread budgets, graceful degradation payloads.

Checklist

- [] Fallback path semantics explicit (hard fail vs cached stale snapshot vs degraded read).

Source: 09-distributed-systems-ownership/06-bulkhead-and-rate-limiting.md

Bulkhead + rate limiting

Theme: One sick dependency must not bankrupt every pool. Isolate capacity; bound admission.

Bulkhead intuition

Separate **executor pools**, **connection quotas**, logical tenant lanes—anything flammable gets a fence preventing total starvation of unrelated flows (“noisy neighbour” containment).

Discuss platform-native analogues versus app-level quotas (Kubernetes limits, proxy max connections)—your ownership layer matters.

Rate limiting intuition

Admission control at edges (API gateways) plus **internal ingress** protecting downstream primitives (Redis, worker dispatch).

Token bucket vs leaky bucket vs concurrency slots—articulate fairness (per user / tenant / caller principal).

Overload responses: shed load predictably (429), prioritise monetised paths ethically if policy dictates.

Checklist

- [] Burst vs sustained SLA codified—not only average RPS fantasies.

Source: 09-distributed-systems-ownership/07-failure-propagation.md

Failure propagation

Theme: Understand **fault pressure gradients** traversing tiers—silent amplification is organisational debt.

Narrative stress:

```
broker backlog growth / sluggish consumers
  → lengthening worker timeouts (retry amplification)
    → amplified DB contention or connection starvation
      → cascading healthcheck red waves
```

Trace propagation LAB

Practice stitching **trace** / **span** continuity across enqueue boundaries (propagation headers vs embedded correlation identifiers in messages).

Elevate causal narrative: dashboards tie queue age → DB pool waits → PSP latency spikes—not isolated blips.

Idempotency ownership echo

Amplified retries **multiply** duplication risk—revisit keys at every amplification hop.

Checklist

- [] Blast radius diagrams updated when new fan-out pathways appear (bulk fan-in jobs, analytic exports).
-

Source: 09-distributed-systems-ownership/08-integration-lab.md

Integration lab — Distributed Systems Ownership

Scenario: Payment platform treated as plural ownership surfaces (API, workers, relational truth, integrations).

Artefact ladder (assistants draft — you certify)

```
RISK surfaced early
  → consistency class contracts explicit
    → recovery strategies rehearsed / simulated
      → layered verification tying metrics + traces + behavioural probes
```

Working angles (combine as needed)

Ledger vs projection divergence; broker redelivery hurricanes; treasury regulatory timing; PSP webhook jitter; reconciliation batch drift.

Journal where discussions stayed **thin** (“just add a queue”) vs **distributed-complete**.

Checkpoint mantra

You default to **ownership over plumbing**: not merely “expose REST + enqueue task,” but enumerated **risk / consistency posture / healing path**.

Measurement hooks

Shrinking unexplained outages; repeatable drill exercises; dwindling apology-driven hotfixes—the qualitative bar replacing vanity coverage metrics.

Phase — 10-claude-ops-engineer

Directory: 10-claude-ops-engineer/

Source: 10-claude-ops-engineer/01-incident-thinking.md

Incident thinking — ops framing

Phase framing — Claude Ops Engineer (“Phase 6”)

Units in this folder: 01 – 08 (topic order only).

Theme stack — delivery + observability + ownership

Ops tooling: Docker • Kubernetes • Terraform • Helm • IaC

Observability: logs • traces • metrics • incident ownership

Failure practice lenses: DB • worker • K8s • network (rotate through drills).

Response evolution

Discard: visible outage → mute panic → random restart roulette.

Adopt:

Incident → Hypothesis set → Logs / metrics / traces → VALIDATION gates → Fix → Rollback doorway

→ Recovery narrative

“Claude gives a fix immediately” bows to “Claude **shepherds** incident process with evidence cadence.”

Claude Incident Template — use for Go / Symfony / Ops / IaC slices

Field	Holds
SYMPTOM	User-visible or SLO-bearing deviation—in plain operational language + scope (which env / cluster / tenant lane).
HYPOTHESIS	Ranked guesses before deep fix coding—explicit disproof paths.
LOGS	Which selectors, timestamps, anchors (request_id , trace_id , pod uid, deploy revision).
METRICS	Signals vs noise hypotheses—dashboards/queries—not vibes.
TRACE	Span graph expectations; known gaps instrumented vs blind zones.
VALIDATION	Experiments disproving hypotheses or confirming culprit class without production thrash where avoidable.
FIX	Minimal corrective change respecting blast radius budgeting.
ROLLBACK	Ordering; irreversible leaps named; parity with infra state snapshots.
RECOVERY	Health restoration proof; backlog drain; customer comms placeholders if SLA breach feared.

Theme (this unit): Evidence before thrash. Do not brainstorm random toggles—“What am I **proving** / **disproving**?”

Structured micro-workflow tying to investigations:

problem statement → crisp symptom taxonomy → hypothesis forest → validation ladder → mitigating fix hypothesis

Practice vignettes mapping

Stack	Stress idea
Go	Slow worker — saturation vs blocking IO vs contention.
Symfony	Queue backlog growth — enqueue faster than drains? poison? external dependency stutter?
DB / MySQL	Escalating query latency — plan regression vs lock contention vs saturation.

LAB invariant (unit 01 onward)

Facing any synthetic outage write $\geq$ five hypotheses ranked with cheap→expensive disqualification sequencing before a fix PR description grows.

Checklist

- [] Incident doc states baseline vs deviation window anchored to revision / deploy artefact—not “sometime slower.”

Source: 10-claude-ops-engineer/02-logs-ownership.md

Logs ownership

Theme: Presence of an ERROR line is worthless without context tapestry: locality, causal ordering, multiplicity, prelude signals.

Elevate interrogation pillars:

```
WHERE (service / shard / replica / cron lane)
WHEN (timeline resolution beyond single line)
HOW MANY (burst vs steady trickle preceding alert)
SEQUENCE (antecedent WARN / sluggish dependency stamp)
Anchors → timestamp • request_id • trace_id • correlation keys
Pattern mining → rolling window frequency shift
```

Anti-pattern forbid list for labs: Immediate restart fetish before structured log narrowing exercised.

Practice rotations

Track	Incident seed
Go	Worker timeout storms distinguishing queue lag vs stalled remote call stamps.
Symfony	Payment pathway anomaly — silent partial commit vs benign retrievable PSP hiccups.

Assistant discipline

Demand explicit selectors, index fields, and regex boundaries—not “grep errors”.

Checklist

- [] Sampling strategy declared if volume prohibitive exponential search.

Source: 10-claude-ops-engineer/03-metrics-ownership.md

Metrics ownership

Theme: Emotional temperature untrusted — instruments adjudicate rivalry among hypotheses.

Core families (iterate combinations)

Latency percentiles

CPU saturation / throttle signals
RSS / cgroup pressure artefacts
Broker / internal **queue depth**
Error-ratio vs success throughput skew
Saturation interplay (DB connections, fd exhaustion)

Separation practice

Articulate deliberately:

Class	Means
Signal	Expected to move sharply if hypothesis holds; ties to causal model.
Noise / weak prior	Fluctuates without discriminating culprit class without extra stratification.

Practice vignettes

Layer	Incident seed
Go worker	Perceived sluggishness—is batching starvation or IO wait dominant in metrics corroboration?
DB	Elevated latency—CPU vs locking vs buffer pool anomalies disambiguated historically.
K8s	CrashLoop / eviction hinted via restart counters + resource timelines before pod delete theater.

Checklist

- [] dashboards / PromQL equivalents sketched—even if hypothetical training cluster.

Source: 10-claude-ops-engineer/04-trace-thinking.md

Trace thinking

Theme: Across distributed hops, **localized exceptions mislead**: know where forked timelines diverged.
Illustrative flow you instrument mentally:

```
API gateway / edge
  → authentication / entitlement
    → payments domain call
      → inventory reservation async edge
        → outbound queue ingest
          → notifications fan-out worker
```

Hypothesis choreography

- Produce **minimum three trace-level hypotheses**, e.g.:
1. Blocking remote segment exceeding budget vs local CPU spin invisible to span parent.
 2. Propagation header loss collapsing unrelated spans falsely “single hop slow.”
 3. Downstream amplification via serial dependency chain masquerading as single fat span.

Practice anchors

Stack	Drill framing
Go	Distributed timeouts cascading vs parent span cancellation semantics.
Symfony CQRS / async	Command acceptance vs projector lag vs transactional boundary confusion.

Checklist

- [] Sampling / head-drop risks acknowledged—not every prod trace exists magically.

Source: 10-claude-ops-engineer/05-docker-container-failure.md

Docker + container failure

Theme: Container **presence** $\neq$ **health**. Probe lifecycle, cgroup limits, mounts, egress.

Drill catalogue

Memory pressure

OOM / cgroup kills vs app-level exhaustion

Crash restart loops attributable to bootstrap mis-ordering

Anonymous vs named **volume drift** parity

Bridged vs overlay **network partitioning** artefacts mis-tagged as application bugs

Incident loop echoes

Mandatory triple for each tabletop:

1. Hypothesis granularity at kernel / daemon / app layering
2. Validation via reproducible diminished cluster or recorded failure bundle
3. Rollback doorway (pinned image digest? config revert choreography?)

Checklist

- [] Dockerfile / entrypoint sequencing reviewed when exit code ambiguity surfaces.

Source: 10-claude-ops-engineer/06-kubernetes-iac-failure.md

Kubernetes + IaC failure

Theme: Platform truth mutates deliberately *and* accidentally—observe **desired vs actual**.

Kubernetes failure patterns

CrashLoop backoff nuance

Readiness vs liveness mis-spec causing traffic to sick instances

Mis-scaled Horizontal Pod Autoscaler oscillation / metric blind spots

Rolling update stuck mid-maxUnavailable geometry

Admission webhook / quotas / PDB interactions

IaC confrontation

Terraform:

`plan` / `apply` divergence vs reality

State drift origins & locking contention

Destructive replacements masked as benign tweaks

Cross-stack resource contention

Helm:

Release revision skew

Template ordering bugs

Hooks failing silently leaving partial manifests

LAB deliverable sketch

Produce end-to-end **incident timeline doc** marrying cluster events ↔ Terraform diff ↔ GitOps revision hypotheses.

Checklist

- [] Blast radius enumerated for `kubectl delete /forced replacement shortcuts`.

Source: `10-claude-ops-engineer/07-rollback-recovery-thinking.md`

Rollback + recovery thinking

Theme: A fix lacking retreat strategy is gambit engineering—architecture-grade ops demands **paired rollback + eventual recovery coherence**.

Perspective contrast:

Weak: patch merged—hope dashboards green.

Strong:

```
fix candidate staged
  → explicit rollback choreography (infra + migrations + queues)
    → recovery proof artefacts (Synthetic checks? Shadow traffic? Replay guards?)
```

Practice vignettes

Plane	Incident class
Symphony	Migration partially applied—dual schema read windows & rollback truncation hazards.
Go	Faulty worker binary—replay / poison handling after downgrade.
Terraform / IaC	Applied graph partially realized—targeted destroy vs forward-only healing tradeoff realism.

LAB invariant attached to corrective change

Rollback section non-negotiable—even “feature flag revert” enumerated with dependencies (cache bust, incompatible schema assumption).

Recovery adds **confidence horizon**: backlog drain expectations, SLA re-stabilisation evidence.

Checklist

- [] Irreversible deltas called out boldly (data destruction class).

Source: `10-claude-ops-engineer/08-integration-lab.md`

Integration lab — Claude Ops Engineer

Capsule outage composite for **payments platform**:

- creeping **latency** percentile shifts
- **queue depth ballooning**
- **worker timeouts**
- sluggish **DB primitives** churning dashboards
- **pod restart churn** signalling instability hurricane

Artefact escalation ladder

Assistants scaffold; **you** gate realism:

SYMPTOMS scoped & customer impact tiered
→ layered HYPOTHESES (≥5 upfront discipline still honored)
→ LOG triage strategy w/ anchored IDs
→ METRIC discrimination storyline
→ TRACE graph hypothesised breakpoints
→ VALIDATION experiments / repro harness
→ FIX budgeting minimal blast envelope
→ ROLLBACK ordering explicit
→ RECOVERY proof checkpoints

Measure & retrospect

Repair iteration counts

Evidence loop quality qualitative audit

Rollback completeness ruthless scoring

Notebook reflection prompts:

Assistant **jumped to fix prematurely** seams

Ran strong **guided incident narrative** arcs

Insufficient **environment / revision context** demand zones

Checkpoint mantra

Claude pivots primary value from dispensing fixes instantly toward **conducting reproducible investigative rigor** marrying observability artefacts with cautious change velocity.

Phase — 11-go-performance-engineering

Directory: 11-go-performance-engineering/

Source: 11-go-performance-engineering/01-benchmark-ownership.md

Benchmark ownership — performance framing

Phase framing — Go Performance Engineering (“Phase 6.2”)

Units in this folder: 01 – 05 (topic order only).

Themes carried through

pprof • CPU profiling • memory profiling • vigilance against goroutine leaks • escape analysis • benchmark discipline • GC pressure literacy

Working loop (prefer this ordering)

SYMPTOM → MEASURE → PROFILE → OPTIMIZE → VERIFY

Checkpoint mentality: instincts shift from vague “slow code” complaints toward instrumented optimisation defended by before/after numbers.

Go performance slice worksheet (reuse per drill)

Field	Holds
SYMPTOM	Latency / allocations / saturation—where (which handler or worker stage).
MEASURE	Bench command, profiling flags, reproducible workload shape.
PROFILE	pprof modality (CPU, heap, goroutine block, mutex) + where time or bytes live.
OPTIMIZE	Minimal diff targeting proven hotspot—not speculative rewrite.
VERIFY	Bench delta + behavioural tests + regression leak watch (especially goroutine count).

Theme (this unit): Measure before folklore. Ownership of benchmarking means stable inputs, deterministic builds (-count , warm-up policy honesty), guarding against optimised-away work.

LAB — worker queue backbone

Bench the ** dequeue / process / ack** narrative under realistic payloads—compare baseline vs contended scenarios; record wall vs ns/op artefacts with variance awareness.

Expose anti-patterns: micro-benchmark happiness lying via dead-code elimination absent side effects (Sink tricks when appropriate).

Checklist

- [] testing.B parallelism consciously chosen—not accidental shared-state fiction.

Source: 11-go-performance-engineering/02-cpu-profiling.md

CPU profiling

Theme: Locate truthful hotspots via pprof CPU profiles—profiles beat intuition when JSON parsing, cryptography, regexp, or naive loops disguise themselves.

Flow discipline

SYMPTOM (elevated latency / CPU saturation)

MEASURE (RPS reproducible envelope)

PROFILE (`go test -bench . -cpuprofile=cpu.out` or `curl` against a guarded `/debug/pprof/profile` endpoint in realistic non-prod settings).

Optimize targeted functions.

VERIFY with profile delta + benchmarks.

Discuss **aggregation views**: flat vs cum, line-level vs function—pick the lens that matches the hypothesis (“who owns cycles on the critical path?”).

LAB vectors

JSON serialization pressure—profile marshal/unmarshal hotspots; scrutinise generics / reflection artefacts if present.

Retry ownership spiral caution—profiler reveals thundering herds or busy loops masquerading as “just retry”.

Checklist

- [] Profiler capture window long enough versus timer resolution noise— annotate collection conditions.

Source: `11-go-performance-engineering/03-memory-ownership.md`

Memory ownership

Theme: Understand **growth vs steady churn**, allocation pressure driving **GC cost**, dormant **goroutine ladders**, and creeping heap footprints.

Artefacts

`-memprofile`, `runtime.MemStats` semantics (during incidents and labs), goroutine dumps (`/debug/pprof/goroutine`).

Typical failure classes

Unbounded slice or buffer growth

Caches without eviction discipline

Misused `sync.Pool` assumptions

Escaping pointers retaining large graphs unintentionally

Goroutine leak signals: workers stuck waiting on never-closed channels, unbounded fan-out retries, orphaned `select` loops.

Tie optimisations to **allocation profiles + benchmarks**, not anecdotes.

GC pressure intuition: allocation **rate**, object **lifetimes**, pointer density / write barriers—articulate qualitatively, prove with heap diffs.

LAB — goroutine-heavy worker

Induce controlled leak or runaway concurrency; correlate goroutine dump growth with queue backlog and throughput collapse.

Checklist

- [] Post-change heap profile shows **dominant allocations** shifted or reduced—not only `ns/op` improvement.
-

Escape analysis

Theme: Stack allocation disappears cheaply until values **escape** to the heap—invisible allocations aggregate under latency variance and GC churn.

Study `go build -gcflags=-m=` (verbosity tuned) sparingly—it is noisy training wheels. Pair compiler hints with `alloc_objects` / `heap profiles` corroborating real impact.

Contrast:

Behaviour	Typical driver
Escapes to heap	Returned pointers to local variables, closures capturing locals by ref, interfaces boxing concrete values unnecessarily, variadic escapes, unintended API surface returning pointers broadly

LAB — “Go ownership” pressure

Shrink API surfaces returning **pointers vs values** thoughtfully; refactor hot paths leaking small structs repeatedly into heap—benchmark + profile deltas side-by-side.

Checklist

- [] Micro changes justified by measurable delta—escape flags alone insufficient story.

Integration lab — Go performance engineering

Synthetic **worker** subsystem end-to-end: throughput regression or memory creep scenario.

Required arc

```
worker baseline characterised
  → PROFILE (CPU + heap + goroutines)
    → Optimize minimal surgically validated deltas
      → MEASURE / VERIFY reproducibly (bench + behavioural tests intact)
```

Guardrails

Forbid silent tradeoffs: correctness regressions disguised as optimisation, or risky micro-optimisation without profile proof.

Checkpoint mantra

Comfort shifts away from blaming “slow code” abstractly toward **measurable iteration** tethered to reproducible artefacts.

Phase — 12-database-enterprise-systems

Directory: 12-database-enterprise-systems/

Source: 12-database-enterprise-systems/01-isolation-ownership.md

Isolation ownership — database framing

Phase framing — Database Enterprise Systems (“Phase 6.3”)

Units in this folder: 01 – 06 (topic order only).

Themes carried through

Transaction isolation levels • deadlock reasoning • replication topology + lag truth • partitioning strategies • optimistic vs pessimistic locking • cross-cutting consistency ownership

Workflow spine for each substantive story

Query → Transaction scope → Lock stance → Validation → Recovery

Checkpoint mindset: default thinking moves from implicit “database works fine” toward explicit **ownership** of anomalies, timings, drift, and corrective narrative.

Database slice worksheet — reuse across labs

Column	Holds
QUERY / access pattern	Read vs write mix, hotspots, aggregates, paging.
TRANSACTION	Boundary granularity, propagation style, rollback semantics intent.
LOCK	Row/gap/metadata locks; explicit <code>SELECT ... FOR UPDATE</code> vs version columns.
VALIDATION	Invariants asserted where (constraints, app guards, assertions).
RECOVERY	Deadlock retry, failover, reconcile jobs, saga compensations tying back to rows.

Add a **consistency claim** bullet: what any reader anywhere may assume and for **how long** (especially with replicas).

Engine reality note

Isolation behaviour is **vendor- and edition-specific**. When you practise on **MySQL / InnoDB** (recommended for this syllabus), correlate each exercise with official docs for **your installed major version**, including optional settings that change locking (e.g. gap-lock behaviour knobs). Do not memorize folklore as universal SQL law.

Theme (this unit): Isolation ownership — same SQL text, different anomalies allowed.

Canonical levels (conceptual layering)

Read committed (RC) — no dirty reads; **non-repeatable reads** plus **phantoms** remain possible depending on engine.

Repeatable read (RR) — stable read snapshot semantics in many engines—but **serialization anomalies** beyond engine docs still exist without full serializable guarantees.

Serializable — strongest practical SQL isolation in many deployments; throughput cost and lock escalation patterns hinge on workload.

LAB—**MySQL ownership**: classify which anomaly classes survive per configured isolation in your sandbox; replay a **lost update** vignette guarded only by naive read-modify-write.

Checklist

- [] Explicit statement of **session / connection default isolation** versus per-transaction overrides.

Source: `12-database-enterprise-systems/02-deadlock-ownership.md`

Deadlock ownership

Theme: Concurrent enterprise flows hit **ordering inversions** inevitably—architecture-grade DB practice names resources, timelines, mitigation — not mystical “bad luck”.

Mechanical narrative

Circular wait graphs across **rows / gaps / auxiliary locks** surfaced by engine deadlock monitors.

OWNERSHIP artefacts

Declare **consistent resource acquisition order** (table / account id monotone locks) wherever feasible.

Tune **timeouts + bounded retries** distinguishing transient deadlock vs poisonous logic loops.

Log **minimal victim excerpt** responsibly (identifiers not secret payloads).

LAB — deliberate deadlock theatre

Twin domains:

Payment strand reserving ledger rows conflicting with ancillary balance adjustments.

Inventory strand row locks acquired opposite sequence across two workers.

Produce reproducible deadlock graph exposition + explain **chosen victim rationale** surfaced by engine.

Checklist

- [] Post-fix validation proves reduced **false retry storms** harming latency SLO—not only fewer deadlocks on paper while amplifying starvation.

Source: `12-database-enterprise-systems/03-replication-ownership.md`

Replication ownership

Theme: Separate **writes** (authoritative durability path) vs **reads** routed to replicas—explicit **ownership** kills ghost consistency assumptions.

Core tensions

Replica **lag visibility** shaping UI / API staleness disclaimers

Split-brain avoidance patterns during failover rehearsals

Promotion choreography & **routing flips**

Binary / statement / mixed historical replication quirks—consult engine docs—not generic blog cargo cults.

LAB — replica lag scenario narrative

Observe workload under churn: artificially delay apply thread or saturate replica I/O—expose **reading your own writes** gaps and reconcile strategies (sticky session to primary narrow slice, versioning headers, TTL hints honestly labeled).

Metrics ownership: replication lag gauges (exact names vary by MySQL/MariaDB major version — use `SHOW REPLICA STATUS` / instrumentation your edition documents) interpreted against apply latency—not a single naive scalar story.

Checklist

- [] Operational runbook cites **fallback read path authority** clarity when replicas diverge dangerously.

Source: 12-database-enterprise-systems/04-partition-ownership.md

Partition ownership

Theme: Scaling write / index heat via **logical sharding schemes** introduces cross-partition nightmares—design ownership anticipates hotspots and rebalancing nightmares early.

Decision angles:

Partition **key cardinality** realism

Uneven **skew** (super tenant / mega-customer tsunami)

Cross-partition **transaction impedance** forcing saga-like compensations externally

Historical **rebalancing** manoeuvres doubling write amplification temporarily

LAB — heuristic strategy sketch

Twin heavy entities **orders vs payments**:

Evaluate co-locate vs segregate philosophies

Discuss **compound keys**, time bucketing pitfalls, hashing vs modulus tradeoffs — and explicitly document **dual-read windows** while migrating rows without integrity tears.

Avoid presenting one universal answer—expose **evaluation matrix** tying business access patterns quantitatively—even if illustrative estimates only.

Checklist

- [] Escaped generic “shard by user id always” tautology defended or debunked with workload specificity.

Source: 12-database-enterprise-systems/05-lock-ownership.md

Lock ownership — optimistic vs pessimistic

Theme: Concurrency correctness is negotiated—**locking strategy** aligns business contention expectations with observable latency.

Stance	When it excels	Hazard class
Pessimistic (FOR UPDATE , explicit row locks)	High conflict probability hotspots; deterministic failure surface	Deadlock amplification; long transactions holding contention
Optimistic (version , updated_at CAS checks)	Read-heavy intermittent collisions; shorter critical sections	Silent lost updates if application forgets predicates; UX retry loops

LAB—**inventory decrement** style mutation:

Simulate bursts—measure conflict rate vs retry overhead both strategies.

Blend patterns deliberately: pessimistic guards on extremely narrow aggregates + optimistic bookkeeping on projections when safe.

Checklist

- [] Conflict metrics instrumented—not only correctness happy path anecdotes.

Source: 12-database-enterprise-systems/06-integration-lab.md

Integration lab — database enterprise ownership

Synthetic **payments + inventory + orders** coexistence stressing cross-domain locking and read paths.

Required narrative scaffolding

Explain ownership holistically—not isolated SQL tricks:

```
DB ownership articulated (authority + replica truth windows)
  → TRANSACTION boundaries defensible vs saga spill
    → FAILURE / RECONCILE playbook (locking retries, orphaned intents)
      → PERFORMANCE profile awareness (`EXPLAIN`-style reviews + lock-wait narratives)
        → validation gates (constraints + behavioural probes)
```

Stress hints

Blend topics: deadlock under inventory/payment crisscross, replica reads serving stale dashboards, partitioned cold/warm aggregates.

Checkpoint mantra

Operational comfort shifts toward explicit **truth contracts** bridging application & infrastructure—not mere “queries pass tests locally.”

Reflective journaling

Misleading simplifications surfaced

Areas demanding **fresh official documentation citation** revisit next iteration

Phase — 13-production-kubernetes-engineering

Directory: 13-production-kubernetes-engineering/

Source: 13-production-kubernetes-engineering/01-ingress-ownership.md

Ingress ownership — K8s production framing

Phase framing — Production Kubernetes Engineering (“Phase 6.4”)

Units in this folder: 01 – 06 (topic order only).

Themes across units

Ingress controllers & HTTP(S) routing truth • **NetworkPolicy** segmentation • **affinity / anti-affinity** • **taints / tolerations** scheduling contracts • **PodDisruptionBudgets** voluntary eviction guardrails • **autoscaling** (workload + cluster semantics) • **service mesh** (conceptual: traffic shifting, resilience, telemetry) • **production ownership**

Operating loop (reuse per change / incident slice)

Deploy / roll forward → Observe signals → Scale posture (out / up / autoscaler tuning) → Recover deliberately

Checkpoint mantra: instincts shift from “kubectl apply succeeded” narrative toward owning **traffic path, isolation, eviction safety, elasticity, observable recovery**.

Production cluster slice worksheet

Column	Holds
Ingress / routes	Hosts, TLS, annotations, upstream Service mapping, health checks assumptions.
Network boundaries	Who may talk to whom (namespaces / workload identity); explicit default-deny ambition.
Scheduling guarantees	Node pools, affinity, taints—is critical workload actually landing where operations believe.
Resilience budget	PDB minAvailable / maxUnavailable + surge strategy sanity.
Scale knobs	HPA metrics, quotas, CAS interaction if applicable—documented—not mystery defaults.
Observe + recover	Dashboards/alerts tying deploy events to saturation and failure bursts.

Version / distribution reality

Ingress implementations (Gateway API vs classical Ingress controllers), NetworkPolicy enforcement (CNI-dependent), PDB semantics with DaemonSets / Jobs—**verify against your Kubernetes minor version + cloud/CNI**. Rules in this workspace expect you to open current vendor docs rather than trusting static YAML folklore.

Theme (this unit): Ingress owns the north-south façade exposed to humans and integrations—misconfiguration cascades amplify fastest here.

Topics to rehearse declaratively vs imperatively authored routes; TLS lifecycle (termination location, certificate rotation hooks); canonical Service type interplay (ClusterIP vs LoadBalancer); cross-namespace references risks when allowed.

Labs: annotate **traffic failure modes** deliberately (misrouted backend, stale Endpoints churn, websocket vs HTTP/2 quirks if relevant)—pair every hypothesis with observable signal from controller metrics / access logs—not blind pod restarts.

Checklist

- [] **Health probes** differentiated from synchronous deep dependency checks dangerously coupled to cascading kill loops.

Source: 13-production-kubernetes-engineering/02-network-policy.md

Network policy ownership

Theme: Namespaces pretending isolation without **NetworkPolicy** truth remain cardboard walls—assume breach and design segmentation **as code**.

Contrast:

Posture	Outcome expectation
Permissive defaults	Faster bootstrap; brittle blast radius lateral movement
Intentionally layered deny + explicit allow lanes	Operational overhead; sharper incident containment narratives

Articulate selectors (podSelector , namespaceSelector), egress DNS realities, kube-apiserver accessibility nuance versus workload mesh—CNI dictates enforceability.

Labs table ideas

Synthetic microservice tiers: frontend → API → database proxy **only necessary edges** enumerated.
Deliberately break policy—measure observability breadcrumbs proving blocked flows vs silent timeouts misdiagnosed as app bugs.

Checklist

- [] Policies version controlled & reviewed analogous to infra PR gravity—not orphaned operator console edits drifting untracked.

Source: 13-production-kubernetes-engineering/03-node-scheduling.md

Node scheduling — affinity, taints, tolerations

Theme: Scheduling is **silent SLO negotiation** selecting failure domains, noisy-neighbour buffering, heterogeneous capacity islands.

Affinity dimensions

Required vs preferred affinity / anti-affinity rules
Spread across zones / racks / SKU pools
Workload colocation optimisation vs blast radius tradeoff calculus

Taints communicate intent

Dedicated pools (GPU / high IO / confidential compute) eviction pressure
Controlled drain preparation surfaces via NoExecute escalating urgency

Tolerations are opt-in exemptions

Danger: silent scheduling onto degraded capacity when toleration breadth too loose.
Labs: perturb node labels / taints in safe sandbox predicting pod migration path before production experiments.
Storyboard outage: accidental **over-constrained affinity** starving scheduling vs **under-constrained spreading** collapsing zone.

Checklist

- [] Document intentional **remainder** / **overflow pool** behaviours when saturation crosses threshold—avoid infinite Pending mysteries.

Source: `13-production-kubernetes-engineering/04-resilience-pdb-autoscaling.md`

Resilience budgets and autoscaling

Theme: Surge & stability interplay—scaling amplifies latent misconfiguration storms without guardrails like **PodDisruptionBudget**.

PodDisruptionBudget (PDB) ownership

Articulate interplay with Deployments rolling strategy, voluntary disruptions (cluster upgrades / node drains) vs accidental Pod failures—they differ taxonomically.

Anti-pattern: PDB so strict upgrades stall endlessly; PDB so lax customer-visible brownouts unnoticed until load peaks.

Autoscaling strata

Workload **HPA** metrics selection (latency proxy vs saturation vs custom/external metrics), cooldown myth debunk via measured oscillation logs where possible.

Possible cluster-level **nodes autoscaler** interaction—scaling out nodes hides scheduling Pending only after queue drain reveals deeper scheduling faults.

Discuss **limits** / **requests** interplay honestly—autoscaler reacts to observable utilisation artefacts shaped by stale resource guesses.

Labs: provoke oscillation ethically in sandbox analysing metric noise vs purposeful surge.

Checklist

- [] Correlate **scaling events timeline** explicitly with ingress latency / saturation dashboards—not isolated pod counts twitching.

Source: `13-production-kubernetes-engineering/05-service-mesh-thinking.md`

Service mesh thinking (conceptual)

Theme: Traffic ownership abstraction overlays retries, timeouts, telemetry, canary participation—without mandating immediate mesh adoption blindly.

Discuss pattern categories (ambient vs sidecar tradeoff mental model only—not vendor install recipes here):

East-west traffic shaping retries with **retry budget** scepticism distinguishing transient vs poison amplification

Centralised observability enriching span continuity across Pod churn

Fine-grained **failure injection rehearsal** philosophies (staging first) guarding cognitive overhead vs incident realism payoff

Resistance signals: latency tax, cryptographic overhead, blast radius converting previously simple TCP flows into chained policy evaluation—quantify sceptically rather than fetish upgrading.

Labs remain **design & critique** exercises referencing hypothetical mesh capabilities unless you purposely operate an installed mesh already.

Checklist

- [] Document **explicit non-goals** when deferring mesh—what compensating primitives (Ingress timeouts, observability libs) compensate partially.

Integration lab — production Kubernetes ownership

Composite scenario forcing cross-theme reasoning — not isolated YAML trophies.

Potential incident bouquet interleaving ingress mis-route, unintended NetworkPolicy starvation, Scheduling Pending puzzle from taint churn, PDB blocking upgrade midway, autoscaler amplification exposing CPU request lies.

Compose operational narrative artefacts:

Traffic path enumerated end-to-end (DNS → LB → Ingress → Service → Pods) annotated with failure hypotheses.

Isolation matrix snapshot (acceptable edges only).

Scheduling state truth table (affinities respected?).

Elasticity recap (desired vs realised scale events & constraints).

Recover rehearsal ordering (traffic drain → config revert → phased roll forward verifying signals).

Checkpoint reflection: articulate where prior instinct would have blindly redeployed vs evidence-led convergence.

Phase — 14-claude-sandbox-safe-execution

Directory: 14-claude-sandbox-safe-execution/

Source: 14-claude-sandbox-safe-execution/01-sandbox-thinking.md

Sandbox thinking — safe execution framing

Phase framing — Claude Sandbox / Safe Execution Environment (“Phase 6.5”)

Units in this folder: 01 – 06 (topic order only).

Goal trajectory

Shift from **assistants exercising power directly on the host** toward **controlled execution surfaces**: bounded filesystem, narrowed network, rotated non-production secrets, explicit human approval gates, rehearsed rollback.

Comfort target: calmly using helpers with **Symfony, Go, Terraform, Docker, Kubernetes, MySQL, debug / incident workflows, MCP servers**, and **agents**—without treating any of them as implicitly safe.

Host vs isolation shapes (vocabulary)

Surface	Role
Host	Your machine or shared CI runner—highest trust, largest blast radius.
Sandbox	Deliberate boundary (VM, container, dedicated user, remote env) where damage is contained.
VM	Strong isolation; heavier to run.
Container	Lighter process + filesystem + network boundary—not a security panacea alone.
Devcontainer	Standardised dev workspace image + mount policy—convenience plus policy if you enforce it.

Boundaries to name in every design

- Isolation boundary** — what the tool literally cannot reach.
- Trust boundary** — what you assume about code, models, and third-party tools.
- Permission boundary** — read / write / execute explicitly scoped.
- Blast radius** — worst plausible outcome if the boundary fails.

Operating contrast

- Weak pattern:** assistant → shell → full host authority.
- Strong pattern:** assistant → **declared workspace** → **scoped credentials** → **approved** high-risk steps → **recoverable** state.

Claude Sandbox Template — use for every serious project

Field	Holds
WORKSPACE	Single root path(s) where AI/tooling may read/write; what is out of bounds.
FILESYSTEM ACCESS	Read-only vs read-write mounts; ban list (/ , \$HOME wholesale, sibling repos).
NETWORK ACCESS	Allow/deny posture (docs only, package registries, internal API mock, full internet—justify).
SECRET ACCESS	Which fake or scoped credentials exist; what must never be mounted or exported.
TOOL ACCESS	Shell, git, DB client, kubectl , cloud CLIs—each with scope notes.
PERMISSION MODEL	Map actions to least privilege; default deny for destructive ops.
APPROVAL MODEL	What runs only after human review (terraform apply , prod kubectl , migrations).
ROLLBACK MODEL	Revert images, Terraform state discipline, DB migration reversals—what is possible.
BLAST RADIUS	Short paragraph: worst case if this config is wrong.

LAB (this unit)

For each recurring **Claude / agent / MCP** workflow you use, write two columns: **allowed** and **forbidden**—then fold them into the template above.

Checklist

- [] You can explain **where** execution runs (host vs container vs remote) without hand-waving “the AI just helps.”

Source: 14-claude-sandbox-safe-execution/02-docker-sandbox.md

Docker sandbox

Theme: Make the default execution story **container-first** for AI-assisted dev: image + mounts + network policy you control.

Core objects

- Image** — reproducible filesystem baseline.
- Container** — running instance with cgroup + network namespace (still not magic—kernel shared).
- Volume** — named or bind data—**bind mounts are the usual foot-gun**.
- Network** — bridge / none / custom—default “everything egress” is dangerous for unattended agents.

Mount discipline

Anti-patterns: -v /:/host or mounting entire \$HOME into an agent-driven container.
Better: mount **only** the project workspace (e.g. claude-workspace/ or repo root you designate).
Use **read-only** where possible: :ro for dependencies you should not mutate; :rw only where builds must write.

Practice layout

Create a dedicated tree (example name) **claude-workspace/** (or your repo root with strict mount policy) and configure tooling so assistants **only** operate there in automated flows.
Symfony / Go: run builds, tests, and codegen inside the container; host holds editor only if you want separation.

LAB

Prove **no accidental host visibility**: from inside the execution environment, paths outside the workspace must be absent or empty by design—not “hidden by convention.” Document what still leaks (Docker socket mounts, forwarded SSH agent)—and treat those as conscious trust breaks.

Checklist

- [] Dockerfile or compose documents **why** each mount exists—orphan binds removed.

Source: 14-claude-sandbox-safe-execution/03-secret-isolation.md

Secret isolation

Theme: Assume assistant context **will** echo, log, or serialise secrets—design so production truth **never enters** that context.

Non-negotiables

Do **not** mount or paste into agent context:
Production **cloud tokens**, **kubeconfigs** pointing at prod, raw **GitHub PATs**, **SSH private keys**, payment provider live keys—the user’s prohibition list applies even when “only for a minute.”

Safer substitutes

Scoped **fake credentials** for local mocks
Short-lived tokens from a vault or cloud identity—with **narrow IAM** bound to disposable sandboxes
Runtime injection into processes that assistants never see (secrets manager → app only)
Treat `.env` in repos as **config shape documentation**—populate from local secret store templates, never commit real values.

LAB

Maintain a **credential inventory table** per integration: tool name → **exact secret artefact class** → **whether the AI execution layer can observe it** (yes/no + mitigation).

Checklist

- [] Rotate anything that ever appeared in chat logs or shared screenshots—assume compromise for learning setups.

Source: 14-claude-sandbox-safe-execution/04-filesystem-and-network-restriction.md

Filesystem + network restriction

Theme: Least privilege by default. Full disk + unrestricted egress are convenient until they are catastrophic.

Filesystem stance

Prefer **workspace-only** read-write; system paths read-only at most; block traversal into **unrelated directories** (other projects, `$HOME`, backup volumes) unless there is written justification and review.
Classify helpers into **read / write / execute** lanes—generation tasks need write **only** on generated artefacts paths.

Network stance

Posture	When
Denied by default	Untrusted prompts, broad agent autonomy
Allow-list	Package mirrors, documented API docs, mocked dependency hosts
Supervised egress	Real cloud APIs behind explicit approval gates

Labs: deliberately **break** overly broad permissions inside a disposable environment—observe how fast damage scales—then **tighten** policy and capture the sandbox template deltas.

Checklist

- [] DNS egress side channels (analytics, telemetry) considered part of blast radius—not only “evil curl.”

Source: 14-claude-sandbox-safe-execution/05-approval-and-dangerous-action-policy.md

Approval + dangerous action policy

Theme: Production discipline for AI-mediated ops: **propose** → **plan** → **human approval** → **execute** — never autopilot destructive CLIs.

Pattern contrast

Weak: assistant → `terraform apply` / `kubectl delete` / **production migration** unattended.

Strong: assistant → `terraform plan` / **dry-run manifest** / **migration SQL review** → **human sign-off** → executed in controlled CI or bastion—with logs.

Forbidden-class examples

`rm -rf` on ambiguous paths

`terraform destroy`, mass `kubectl delete` without scope guardrails

Exporting prod credentials into shell env visible to tooling

Classification LAB

For recurring commands (`git push`, migrations, infra applies, Helm upgrades), tag each:

- **SAFE** — read-only diagnostics, formatted logs
- **NEEDS_APPROVAL** — mutates shared state
- **FORBIDDEN** — violates secret or prod-touch policy

Automation (hooks, MCP tools, IDE tasks) maps to the **same** classes—parity matters.

Checklist

- [] Approval model written down where **you** rehearse—not only where the vendor UI happens to pause.

Source: 14-claude-sandbox-safe-execution/06-integration-lab.md

Integration lab — sandbox + AI execution

Synthetic stack tying prior units: **Symfony API**, **Go worker**, **MySQL**, **Terraform**, containers, optional cluster snippets.

Deliver **explicit governance artefacts**:

```
permission model articulated
  → sandbox/container boundary documented
    → secret isolation proven (fake + scoped identities only)
      → approval flow for infra & schema moves
        → rollback story for infra + runtime
```

Deliberate fault injection (disposable env only)

Escalated filesystem permissions

Accidental-looking secret surfaced in prompts

Broad network egress enabled

Dangerous composite shell suggested

Observe whether **your enforcement** detects / refuses / restores—or document gaps brutally honestly.

Measure

Blast radius before vs after tightening

Incidents of accidental secret exposure vectors

Misclassified commands (blocked safe ops vs allowed dangerous ones)

Notebook output: distilled **ALLOW / DENY / ASK_APPROVAL** matrix you will paste into `.cursor/rules` or personal runbooks—maintain consciously.

Checkpoint mantra

Framing completes when **assistants operate where you constrained them**, under **explicit permissions**, rather than owning your host by default narrative.

Phase — 15-mcp-systems

Directory: 15-mcp-systems/

Source: 15-mcp-systems/01-mcp-thinking.md

MCP thinking — tooling orchestration framing

Phase framing — MCP Systems (“Phase 7”)

Units in this folder: 01 – 06 (topic order only).

Shift in stance

Away from assistants that primarily **produce prose explanations** of problems. Toward assistants that **orchestrate capabilities**: choose tools, honour permissions, surface risk, validate outcomes—and accept **rollup** responsibilities when tools mis-fire.

Comfort target on a composite incident (Symfony bug → git → filesystem → DB → browser evidence): first reaction cites **which tool, ordering, permission posture, risk**—before a wall of unstructured narrative.

Tool classes (conceptual MCP surfaces)

Typical MCP categories you wire deliberately:

Filesystem MCP

Git MCP

Database MCP

Browser MCP (docs, changelog pages, reproducible UI states)

Terminal MCP — shell access with policy (logs, Docker, Terraform, kubectl) — overlaps **security** heavily with your Sandbox phase.

Cross-cutting disciplines:

Tool ownership, permission ownership, routing (problem → capability → implementation path), **capability honesty** (“model cannot see X unless server exposes safely”), **security ownership** (scopes, leakage, audit).

Core micro-workflow model

PROBLEM surfaced

- CAPABILITY selection (why this modality answers the question class)
 - concrete TOOL call(s)
 - observable RESULT artefacts
 - VALIDATION tying back to problem acceptance

Risk and rollback ride alongside—not afterthought footnotes.

MCP Template — for every meaningful problem slice

Field	Holds
PROBLEM	Symptoms, reproduction, scope (service / tenant / cluster).
CAPABILITY	Why filesystem vs git vs DB vs browser vs shell matters here.
TOOL	Named integration path plus sequence (minimal viable chain first).
PERMISSION	What identities / paths / DB roles / namespaces are reachable—explicit.
VALIDATION	Checks proving success—not vibes.
RISK	Blast radius when tool hallucinates filters or runs broad queries.
ROLLBACK	How you retreat from partial tool-driven changes (migration, infra, config).

Checkpoint mantra: Claude stops behaving as solo explainer—you drive **controlled multi-tool choreography**.

Integration safety still follows your **sandbox / secret isolation** norms; MCP widens blast radius whenever permissions are careless.

Theme (this unit — orchestrator mindset)

Weak pattern: latent belief “the model intrinsically knows the repo truth.” Strong pattern: model **consumes anchored reads** sourced through scoped tools—not magical recall.

Practice vignettes sketch **plans before calls**:

Symfony bug → **git annotate / history** hypotheses → filesystem reads on ownership boundaries (`Order` , `Payment` , `Refund`) → constrained fix iteration.

Go worker fault → aggregated logs (`terminal / log tool`) → filesystem hotspots → corrective diff.

LAB: Document a **tool plan** (bullet list with ordering + permission notes) **before** the model fires tools—reuse that habit when automation agents appear later.

Checklist

- [] Problem statement distinguishes **facts already observed via tools** vs **assumptions**.
-

Source: `15-mcp-systems/02-filesystem-mcp.md`

Filesystem MCP

Theme: The assistant anchors work in **file truth**—“which file owns this invariant?” precedes pasted code glaciers.

CQRS-informed drill objects (Symfony mental model):

`Order` , `Payment` , `Refund` vertical slices

- Trace **handlers / aggregates / projections** for each concept without crossing bounded contexts blindly.

- Map **ownership** in words before speculative edits. Go worker analogue: filesystem walk from **enqueue site** → **consumer implementation** → **retry policy module**.

Operational chain you rehearse

```
filesystem discovery (controlled roots)
  → dependency / import impact cones
    → enumerated impacted modules
      → narrowed edit plan
```

LAB invariant

Each proposed functional change cites **minimum three foreseeably impacted files** (configs, fixtures, migrations tangential counts if touched). If fewer exist, articulate **explicitly narrow blast**—don’t fake breadth.

Discuss **risk** when broad glob searches traverse secrets or generated vendor trees unnecessarily.

Checklist

- [] Search patterns avoid dumping entire binaries / artifact caches unintentionally via naive globs—scope consciously.
-

Source: `15-mcp-systems/03-git-mcp.md`

Git MCP

Theme: Evolution narrative beats static snapshot—“what changed?” often answers “why fragile now?” faster than speculative architecture essays.

Mandatory perspectives when harnessing Git capabilities:

Structured **diff comprehension** spanning domain + infra

Commit archaeology clustering related behavioural shifts—not line noise obsession

Selecting **rollback / revert anchors** realistically (ordering conflicts honoured)

Migration ownership pairing DB schema evolution commits with runtime expectations

Symphony exercise: analyse **refund flow** divergence across merges; narrate behavioural risk succinctly.

Go counterpart: quantify **optimization diff** hotspots—benchmark tie-back optional unless performance thread active.

Assistant output must summarise **Δ behaviour**, latent **risk classes**, pragmatic **rollback** options—not only patch content.

Checklist

- [] Never treat remote history rewriting strategies as MCP convenience—assume immutable published branch ethics unless sandbox-only.

Source: `15-mcp-systems/04-database-mcp.md`

Database MCP

Theme: Treat DB capabilities as **surgical optics** supporting schema fidelity, correctness, pacing—not chatty dumping of entire tables unsupervised.

Coverage bands

Dialect introspection responsibly

Targeted analytical **queries & EXPLAIN workflows** respecting row limits

Index / constraint reasoning tethered to access patterns surfaced earlier

Controlled **migration review** interplay (destructive deltas escalated deliberately)

Symfony + MySQL common lab arcs:

Observe **slow query** artefact externally (log / APM)—then escalate through **schema → query shape → indexing → iterative validation** layering.

Parallel Go angle: annotate **transactional flow** hotspots—where isolation expectations meet ORM/driver defaults.

Mandatory **minimum three hypotheses** enumerated **prior** optimisation / rewrite attempts—reuse failure-class vocabulary from observability syllabus when helpful.

Risk register explicit for broad `SELECT *` automation, ambiguous prod vs staging datasource confusion, unintended write statements.

Rollback awareness: DDL rarely trivially reversible—plan dual-phase compatibility when MCP-assisted proposals touch production-like sandboxes.

Checklist

- [] Queries include safeguards (`LIMIT` , indexed predicates) proving discipline—not accidental prod table scans.

Source: `15-mcp-systems/05-browser-and-terminal-mcp.md`

Browser + terminal MCP

Theme: Prefer **verification** over guesswork:

Browser MCP gathers authoritative **documentation**, changelog nuance (breaking deltas), sometimes UI reproduction evidence.

Terminal MCP (or shell-capable equivalents) tails **structured logs**, container states, infra CLIs disciplined by policy—not unconstrained omnipotence.

Practice incident arc (ops worker failure stylised):

```
terminal log capture & structured filtering
  → corroborative trace artefacts if exported
    → hypotheses ranked before repair attempt
      → narrowly scoped infra query (docker / compose / kubectl / terraform PLAN only until approval posture satisfied)
```

Guardrail LAB

Assistants **MUST NOT** jump to speculative **fix patching** absent prior evidence artefacts—simulate forced pause: summarise observed signals first.

Discuss risk: naive **kubectl/terraform** autonomy—align with staged approval classifications from **your sandbox / CLI policy**.

Browser automation blind spots: iframe lockouts, SSO walls—document blockage instead of tool thrash repetition.

Checklist

- [] Sensitive URLs / internal-only endpoints never screenshot-shared casually—treat viewport captures like log excerpts.

Source: 15-mcp-systems/06-integration-lab.md

Integration lab — MCP systems orchestration

Composite artefact resembling micro-platform stress:

Symfony API strand

Go worker consumer

MySQL datastore

Dockerised runtime façade

Terraform (or compatible IaC) footprint

Synthetic fault bouquet options:

-queue throughput decay

-payment **refund** failures

-schema / migration fallout

Produce explicit orchestration dossier marrying earlier templates:

```
problem framing anchored in observed signals (tools first)
  → ROUTING rationale across filesystem / git / DB / shell / browser as needed
    → PERMISSION recap (credential class exposure audit)
      → CAPABILITY limits acknowledged (coverage gaps enumerated honestly)
        → investigative timeline with validation gates
          → restrained fix proposal
            → rollback / recovery pairing
```

Measure & retrospect

Evidence **quality**, correct **tool choice frequency**, needless **iteration counts**, rework after bad routing.

Notebook asks:

Where assistants **picked incorrect modality** prematurely

Capability blind spots amplified thrash

Routing graph adjustments you will codify personal rule-side next iterations

Checkpoint closure: habitual mental prompt becomes **multi-tool choreography design** before freeform answer generation.

Phase — 16-memory-systems

Directory: 16-memory-systems/

Source: 16-memory-systems/01-memory-thinking.md

Memory thinking — framing

Phase framing — Memory Systems (“Phase 7.5”)

Units in this folder: 01 – 06 (topic order only).

Mindset pivot

Away from imagining “the model simply remembers.” Toward **you** designing working context, persisted artefacts, retrieval paths, staleness hygiene, and **ownership** across long arcs (Symfony ↔ Go ↔ MySQL ↔ Terraform ↔ Kubernetes).

Operational spine you reuse:

Context packaging → Memory placement (tier choice) → Retrieval choreography → Compression / eviction discipline

Memory tiers — vocabulary aligned with cognition literature (used pragmatically here)

Tier	Holds
Working	Immediate task framing, ephemeral scratch—what must be prime <i>right now</i> .
Episodic	What happened chronologically across sessions/incidents/decisions—“we tried X”.
Semantic	Durable truths about the system —bounded contexts, invariants, architecture intent.
Persistent rules	Non-negotiables you store outside the ephemeral window (Rules , runbooks, versioned SPEC, ADRs).

Cross-cutting tensions the rest of this area tackles: **retrieval**, **compression**, **summarization**, **invalidation**, **memory security** (secrets must never enter durable “memory”), **context decay**, **retrieval ownership**, **knowledge freshness** / **versioning**.

Claude Memory Template — for larger tasks

Field	Holds
WORKING MEMORY	Task goal, hypotheses, pointers to artefacts—minimal tokens.
EPISODIC MEMORY	Recent decisions/commits/incidents summaries with dates/refs.
SEMANTIC KNOWLEDGE	Architecture & domain anchors (CQRS seams, aggregates, infra topology).
PERSISTENT RULES	Citations into repo truth (Rules, SPEC ids, playbook paths)—not recreated inline.
KNOWLEDGE NEEDED	External or internal lookups still missing before implementation conviction.
RETRIEVAL PLAN	Where you will fetch each needed fact (filesystem path, MCP tool class, docs URL class).
STALE KNOWLEDGE RISK	Explicit guess what might be outdated (“library default changed”) + verify step.

Checkpoint mantra: Claude does not “have context” magically—you supply a **deliberate knowledge system** with versioning awareness.

Integration with **sandbox** / **MCP** tooling: persisted memory artefacts still obey scope and secrecy rules—never hydrate sensitive tokens into summaries.

LAB (unit 01)

For representative slices—Symfony **refund** narrative, Go **payment** worker saga, hypothetical **Ops incident** retrospective—annotate each task with what belongs to **working** / **episodic** / **semantic** / **persistent**.

Checklist

- [] Every long arc names an **official source of semantic truth** distinct from conversational drift.

Source: 16-memory-systems/02-context-compression.md

Context compression

Theme: You do not ship the repository as an undifferentiated slab into the model—**curated signal** wins over brute token dumping.

Contrast:

Poor	Strong
Huge undifferentiated paste (thousands of incidental lines, whole trees)	Tight packet: ownership, constraints, boundaries, NFR/SLO only where decisive

Practice slices:

Symfony **CQRS** flow refactor planning—articulate aggregates + command/query ownership without vending entire vendors.

Go **queue subsystem** tweaks—expose contracts (ACK semantics, retry policy knobs) crisply versus entire module trees.

Compression tactics you rehearse aloud:

Headline invariant bullets

Structural diagrams shrunk to captions + symbol names already in repo

Pointer references (SERVICE.md , RULE ids, SPEC anchors) pulling truth lazily

Strip historical chatter duplicated in git archaeology unless episodic hinge matters now

LAB: Take one scenario task twice—**(A)** maximal naive context dump intuition vs **(B)** compressed packet—observe qualitative answer precision and iteration counts; note subjective token budgeting learnings introspectively (you need not quantify unless tooling available).

Discuss **risk**: over-compression hides critical negative knowledge (“we removed feature X”—must remain discoverable episodically).

Checklist

- [] Compression **never deletes** contradictory historical facts silently—migrate them to episodic artefacts if still decision-relevant.

Source: 16-memory-systems/03-retrieval-thinking.md

Retrieval thinking

Theme: Ask “**what facts must enter working memory next**”—not vaguely “what do I recall.”

Operational chain:

```
problem class identified
  → enumerated KNOWLEDGE_NEEDED bullets
    → RETRIEVAL_PLAN modality per bullet (filesystem path, MCP class, authoritative external d
oc tier)
      → ingestion + summarization checkpoints
        → trimmed working overlay before implementation chatter thickens
```

Practice motifs:

Symfony **CQRS boundaries** clarified by SPEC + bounded context glossary—not guessed from legacy handler names alone.

Go **retry ownership** validated against language runtime docs plus internal concurrency policy artefacts.

Mandatory lab habit: BEFORE dense implementation instructions, standalone mini-section **Knowledge needed** enumerated—defer coding fantasy until minimally satisfied OR explicitly flagged as unknown-unknown risk.

Discuss **ownership**: who authoritative for each datum class (platform team vs subsystem maintainers)—prevents polluted semantic memory.

Checklist

- [] External doc pulls prefer **pinned versions / release boundaries** aligning with deployments under discussion.

Source: `16-memory-systems/04-knowledge-lifecycle.md`

Knowledge lifecycle — freshness & invalidation

Theme: Operational knowledge ages—assume decay unless intentionally versioned.

Illustrative pivot:

Yesterday “Redis caching layer simplistic key-value” evolves toward **streams / consumer groups** choreography—documents treating only cache TTL semantics now **risk stale behavioural guidance**.

Governance axes rehearse systematically

Freshness clocks anchoring factual statements (“true as-of commit SHA / doc rev”)

Deliberate **invalidation triggers**—schema migrations, infra module upgrades, PSP API version jumps

Knowledge **ownership lanes** tying updates to accountable roles—not orphan wiki drift

Structured **semantic versioning analogue**—annotate internal ADRs/SPECs churn or epoch tags when radical shifts accumulate

Mandatory lab output: articulate **minimum three staleness hazards** hypothetical or historical—classification covers mis-applied optimisation, dangerously outdated security posture, invalidated operational runbook commands.

Discuss **silent summarization harm**: succinct memories erasing deprecation nuance—you still need branching historical narrative when parallel upgrade windows exist across tenants.

Integrate overlap with Sandbox security: invalidated advice must not resurrect secrets inadvertently when recycled.

Checklist

- [] Breaking external dependency updates schedule explicit **retroactive SPEC review backlog** linkage.

Source: `16-memory-systems/05-memory-failure-engineering.md`

Memory failure engineering

Theme: Memory systems degrade predictably—classify faults like any distributed hazard.

Representative failures:

Context **overload / noise domination** reducing effective comprehension

Stale or mythic episodic overlays (“we already fixed”—but branch diverged)

Critical **SPEC fragments forgotten** amidst summarization churn

Mis-**retrieved** artefacts (nearby but wrong bounded context synonyms) sabotaging refactor confidence

Security-adjacent: unintended secret echoes into durable summary stores demand aggressive redaction—not “helpful shorthand.”

Labs: engineer **bruised setups** ethically in sandboxes:

Overlong rolling recap lacking retrieval pointers

Stale semantic bullet asserting retired architecture

Mis-labeled SPEC path references

Observe assistant behaviour—expects **classification** rehearsal per incident into memory vs retrieval vs context hygiene buckets—not immediate patch improvisation without fault taxonomy.

Repair strategies include targeted re-ingestion bursts, rewriting overlay bullets, pinning authoritative artefact excerpts, escalating human reconciliation checkpoints.

Checklist

- [] Each rehearsal ends with actionable **preventive structuring tweak**—not only narrative acknowledgement.

Source: 16-memory-systems/06-integration-lab.md

Integration lab — long-running payment platform cognition

Synthetic multi-surface endeavour:

Symphony strands

Go workers

MySQL behavioural contracts

Terraform evolution

Kubernetes runtime posture

Operational expectation: shepherd continuous threads without **ownership/spec/history dissolution**.

Artefact scaffolding

Maintain rolling template instantiations marrying prior units:

```
WORKING MEMORY curated per milestone
  → Retrieval passes executed before refactor coding
    → Compressed context packets circulated intentionally
      → Lifecycle risks logged (staleness, invalidation checkpoints)
        → SPEC alignment validation loops
          → Behavioural verification tying back ACCEPTANCE artefacts
```

Fault bouquets (simulate sequentially or combined)

Retro Redis guidance conflicting with Streams migration partial adoption

Stale retry doctrine contradicting new queue broker semantics

Masked schema assumptions after partial migration bridging window

Measure qualitatively (optionally annotate rough counts mentally):

Compressed context fidelity vs verbose baseline

SPEC anchoring regressions surfaced early vs late iterations

Detected knowledge drift occurrences

Repair cycle counts reversing misalignment

Notebook synthesis:

Assistants dropping **ownership edges** prematurely

Repeated **retrieval** blind spots illuminating doc routing gaps

Memory structuring improvements you will institutionalise externally (RULE updates, MCP indexing discipline)

Checkpoint closure reinforces explicit **externalised knowledge lattice** outperforming conversational belief in raw context durability.

Phase — 17-retrieval-systems

Directory: 17-retrieval-systems/

Source: 17-retrieval-systems/01-retrieval-thinking.md

Retrieval thinking — framing

Phase framing — Retrieval Systems (“Phase 7.7”)

Units in this folder: 01 – 05 (topic order only).

Mindset pivot

Away from shoveling whole repositories or dumping “every document we have.” Toward **governed retrieval**: what knowledge is truly required—then fetch **only relevant**, validate fit, repair when retrieval lies.

Comfort at scale: stacks like **Symfony, Go, Terraform, Kubernetes, MySQL** coexist with **dozens of specifications**—the successful pattern is selective retrieval, **not** “load everything.”

Practice choreography (architecture-oriented)

Canonical docs / internal knowledge base artefacts

- ↓ assisted retrieval orchestration (“Claude retrieves” deliberately)
- ↓ architect-grade answer anchored in cited snippets

This area complements **structured project memory**: retrieval is **how** bounded truth enters working context without flooding it.

Theme map

RAG as pattern (retrieve → augment reasoning)

Retrieval pipelines — ingest, chunk, embed (conceptually), index, query, rerank, fuse

Chunking strategies respecting ownership boundaries

Embeddings / semantic retrieval understood qualitatively (vectors approximate intent; they err)

Hybrid retrieval lexical + semantic merges

Reranking & quality scoring

Cross-cutting **knowledge ownership, quality habits, retrieval debugging**

Claude Retrieval Template — for substantial tasks

Field	Holds
PROBLEM	Slice of work + constraints—not the entire platform at once.
KNOWLEDGE NEEDED	Bullet list of artefacts / facts/classes still absent from working context.
KNOWLEDGE SOURCE	Where truth lives path, doc ID, MCP tool tier, authoritative external manual section class.
RETRIEVAL STRATEGY	lexical vs semantic vs hybrid; filters (namespace/tag); breadth budget.
RETRIEVAL RISK	Staleness class, synonym collision, ambiguous module names—pre-declared traps.
VALIDATION	Checks that snippets actually answer hypotheses / align with SPEC—not vibes.
REPAIR	If validation fails—re-query, re-chunk framing, escalate human clarification.

Checkpoint mantra: Claude does not merely “have context”; Claude **runs a deliberate knowledge pipeline** you design.

Sandbox / MCP layers still enforce permissions—retrieval widening must not pierce secrecy rules.

LAB — unit invariant

Before engaging an assistant heavily, standalone **Knowledge needed** list—Symfony **refund ownership**, Go **worker retry doctrine**, infra **Terraform drift investigation** vignettes—as reusable rehearsal.

Checklist

- [] Every retrieval-driven answer names **sources used** succinctly—even if informal path references.

Source: 17-retrieval-systems/02-chunking-thinking.md

Chunking thinking

Theme: Documents are sliced for retrieval quality—bad chunks mis-route queries more than mediocre embedding tables.

Contrast:

Weak	Strong
“Entire Order module pasted” drowning refund nuance	Ownership chunk aligning to bounded artefacts: REFUND SPEC excerpt, PAYMENT boundary blurb, RETRY OWNERSHIP note

Heuristics—not universal laws—for technical corpora chunk design:

Honor **DDD / CQRS seams** splitting aggregates & projections

Retain **minimal cross-referencing headers** aiding anchor matching

Preserve **constraints & NFR** chunks distinct from behavioural narrative ones

Prevent giant vendor dumps unless scope truly integration-wide ephemeral

Symfony practice: hypothetical **DDD ecommerce** knowledge map—articulate deliberate chunk taxonomy before indexing fantasy.

Go practice: isolate **payment worker** contracts (signals, backoff, ACK invariants).

LAB: Replay one problem triad—**(A)** oversized amalgam chunk **(B)** minimal micro-chunk bombardment lacking context **(C)** balanced boundary-aligned chunk set—critique retrieval signal qualitatively (answers stable? iterations reduced?).

Discuss **risk**: micro-chunks erasing precondition sentences—macro-chunks obscuring contradictory historical paragraph buried mid mass.

Embedding note stays **conceptual** here—density & overlap interplay matters more than chasing model vendor buzzwords blindly.

Checklist

- [] Chunk manifests versioned—or stale chunk boundaries haunt silently during refactors.

Source: 17-retrieval-systems/03-retrieval-quality.md

Retrieval quality

Theme: Satisfaction is **not** “I got an answer.” It is “**I fetched the grounding that matches this problem slice**”—then proved it fits.

Dominant defect classes rehearse ruthlessly:

Failure	Symptoms
Wrong document	Tangential module spec surfaces—convinced wording wrong context
Stale knowledge	Architecture pivot not re-indexed; optimistic guidance dangerously dated
Wrong ownership framing	Context blurs aggregate boundaries—invariants silently violated downstream

Practice motifs:

Go **retry ownership** vs transport vs business idempotency—each doc class must reconcile without mixing vocabulary collisions.

Symfony **CQRS ownership** delineations—commands vs projections vs integrations each demand distinct headings in retrieval routing.

Mandatory lab metric: annotate **minimum three QUALITY risks** per retrieval iteration before trusting synthesis—risk types may overlap staleness synonyms with chunk boundary leakage.

Discuss **semantic vs lexical** divergence: synonyms retrieve irrelevant hotspots; lexical filters rescue or mis-prune aggressively—hybrid strategy justification belongs in Retrieval Template field.

Knowledge ownership: named maintainers tagging SPEC revisions reduce orphan drift—surfaced at retrieval filtering time as metadata—not only narrative memory.

Checklist

- [] Retrieval score threshold decisions documented—even coarse heuristics beat silent acceptance ordering.

Source: `17-retrieval-systems/04-retrieval-debugging.md`

Retrieval debugging

Theme: Pipelines malfunction like code—observe outputs, hypothesise retrieval layers, shorten feedback loops ethically.

Representative breakage narratives:

Architectural advice mirrors **obsolete ADR snapshots** SPEC explicitly superseded—but old chunk rank dominated.

Model **ignores ingested grounding** absent validation hooks—looks authoritative while hallucinating glue.

Wrong module embeddings victorious due to lexical collision or partial filename overlap (“payment” drowning “payout reversal” distinctions).

Debugging macro-workflow mimic:

```
surface RESULT unsatisfying / contradictory
  → cite KNOWLEDGE SOURCES actually surfaced
    → scrutinise retrieval step (filters, chunk boundaries, synonym collapse)
      → VALIDATION experiments (narrow re-query variants)
        → REPAIR (re-ingest excerpt, annotate chunk header, escalate human clarifier prompt)
```

LAB: ethically craft **incorrect retrieval precondition** sandbox—simulate stale embeddings index OR maliciously broad synonym tag—observe detection discipline (assistant must classify failure class before patching).

Enumerate repairs:

Query reformulation choreography

Boost / penalty metadata layering on ownership tags

Hybrid fallback toggling lexical rescue when semantic confidence low

Potential partial **rerank** heuristics (even manual human rerank rehearsal educates intuition)

Security angle: poisonous doc injection awareness when multi-tenant corpuses share index—assume least trust boundary.

Checklist

- [] Each rehearsal logs at least **one preventive structural ingestion tweak**—not only ephemerally “asked nicer.”
-

Integration lab — retrieval orchestration @ scale

Synth scenario: Payment platform knowledge scattered across Symphony services, Go workers, relational schema docs, Terraform modules, Kubernetes runbooks **plus fifty-plus internal specifications** artefacts.

Operational expectation orchestration—not saturation:

```
precise PROBLEM narrowing
  → KNOWLEDGE NEEDED matrix (minimal sufficiency ethos)
    → segmented RETRIEVAL STRATEGIES per domain surface
      → VALIDATION loops preventing silent mis-binding
        → restrained IMPLEMENTATION plan
          → iterative REPAIR pathways when drift
detected
```

Stress threads (rotate intentionally)

Historical refund choreography evolution partially overlapping current ledger truth

Distributed retry doctrines diverging infra vs runtime layers

Stale ops Terraform module guidance contradicting patched provider defaults

Measurement sketch (qualitative + rough tallies optionally):

Weighted **retrieval quality** scorecard self-graded bluntly

Wrong retrieval incidence classifications

Repair cycle counts shortening misalignment arcs

Tracked **knowledge drift** resurfacing unresolved post partial migrations

Notebook synthesis probing honesty:

Assistants hauling **too much** context inertia

Systematic blind spots (**wrong corpus lane** habitual)

Where retrieval scaffolding **thin**—risking revert to raw repo paste addiction

Checkpoint closure reinforces governing **pipelines intentionally** outperforming unstructured context accumulation—even as catalogue breadth balloons.

Phase — 18-multi-agent-systems

Directory: 18-multi-agent-systems/

Source: 18-multi-agent-systems/01-agent-thinking.md

Agent thinking — multi-role framing

Phase framing — Agents (“Phase 8”)

Units in this folder: 01 – 06 (topic order only).

Mindset pivot

Away from **one assistant doing everything contiguously**. Toward a **composed system**: distinct roles with clear **ownership**, handoff artefacts, and review gates—even when physically one model instance plays multiple hats sequentially via explicit prompts.

Target stack familiarity: Symphony, Go, database work, Terraform, Kubernetes—agents partition concerns instead of collapsing them.

Role catalogue (minimal set for serious tasks)

Role	Owns
Planner	Goals, task graph, dependency ordering, priority— not implementation code .
Architect	Boundaries (DDD seams, CQRS), tradeoffs, risk register, scalability / failure posture— “why before how” .
Implementer	Code/config changes realising the architecture slice—no silent scope creep into approval territory.
QA	Validation vs frozen acceptance: drift detection, risky gaps, constraint violations before trusting merge.
Ops	Runtime / containers / cluster / Terraform paths, observable signals, rollout + rollback .
Reviewer	Cross-cutting quality: security-sensitive patterns, maintainability, diff risk narration—orthogonal to QA’s behavioural checking.

You may compose fewer roles on tiny tasks—but **substantial** deliveries should consciously touch this template.

Claude Agent Template — for substantial work

Structured handoff scaffolding (adapt naming to your orchestration tooling):

```
PLANNER artefact ...
ARCHITECT artefact ...
IMPLEMENTER artefact ...
QA artefact ...
OPS artefact ...
REVIEWER artefact ...
```

Each section states: **ownership, inputs consumed, outputs produced, explicit non-goals** (what that role refuses to own this round).

Checkpoint mantra: Claude stops behaving as lone problem solver—problem solving becomes **multi-agent choreography** whether simulated or wired.

Integrity note: aligns with Sandbox / MCP / Retrieval habits—narrow permissions per role persona where automation exists.

LAB (unit 1)

Symphony **refund flow** tabletop: write **“who owns what”** table before any code—Planner graph, Architect boundary/tradeoffs, Implementer edits, QA checks, Ops deploy hooks if applicable, Reviewer critique slots.

Go variant analogous on payment worker refactor planning.

Checklist

- [] Every role section lists **dependence on prior artefacts**—no orphaned implementation without architecturally frozen slice.

Source: `18-multi-agent-systems/02-planner-agent.md`

Planner agent

Theme: Planner materialises navigable futures—dependencies explicit—without touching implementation noise.

Operational chain:

```
goal restatement sharpening
  → decomposition ladder (vertical slices—not megatasks)
    → precedence / parallelism classification
      → prioritisation aligning risk & dependency gravity
        → READY markers for downstream roles
```

Forbidden pattern for labs: Planner pseudo-code creeping into idiomatic commits—enforce separation discipline.

Practice canvases:

Go **payment worker** lifecycle refactor—ordered steps isolating behavioural vs infra toggles

Symfony **CQRS flow** uplift sequencing command vs projection migrations

Terraform **deployment wave** structuring—state checkpoints between blast phases

Mandatory lab prohibition echo: Planner output **never** substitutes Implementer textual code slabs—plans may cite file targets & acceptance hooks only.

Checklist

- [] Planner outputs reference **verification hooks** QA will replay—not vague “finish feature.”

Source: `18-multi-agent-systems/03-architect-agent.md`

Architect agent

Theme: Ownership of **conceptual correctness** precedes keystrokes—Architect refrains from feature coding while shaping **why boundaries exist**.

Core obligations:

Articulate bounded contexts (**DDD ecommerce** mental model rehearsals)

Clarify **CQRS seams** respecting write vs read model evolution

Enumerate **risk** classes (scaling, duplication, transactional islands) proactively

Stake **ownership** maps (teams / runtime components—not vague “microservices vibes”)

Discuss **tradeoffs ruthlessly**:

Latency vs invariant strength

Operational complexity vs speculative future flexibility

Eventual seams vs narrowing strong consistency islands explicitly

Mandatory lab heuristic: derive **minimum three contrasting tradeoffs** per substantial task—Architect doc must cite **reasons first**, implementation sketches delegated downstream.

Practice prompts:

Go **distributed payment choreography** aligning saga / compensation expectations honestly

Symfony **modular ecommerce** delineation guarding aggregate purity

Architect output consumption: freezes scope slices Implementer executes—Architect cannot silently extend mid-stream without Planner re-baselining precedence.

Checklist

- [] Architect artefacts link **measurable validation hooks** anticipating QA—not only aspirational bullets.

Source: 18-multi-agent-systems/04-implementer-and-qa-agents.md

Implementer + QA agents

Theme: Separation of creation vs independent verification lowers blind spot density.

Workflow cadence exercised:

```
Architect frozen slice acknowledgement
  → IMPLEMENTER concrete changes (narrow diff discipline)
    → QA validation pass (SPEC alignment + risk resurfacing)
      → iterative REPAIR loops governed by QA findings backlog
```

Implementer stance: Executes authorised plan—does **not** self-award production approvals / destructive infra applies—Ops retains those credentials in production-grade flows.

Practice drills:

Go **worker retry** policy nuanced adjustments—implementer honours architected idempotency story.

Symphony **refund aggregate** behavioural patch implementing frozen invariants—not opportunistic refactors bleeding scope.

QA stance: hunts **SPEC drift**, latent **risk resurfacing**, **constraint violations**—including performance / security regressions tied to acceptance language.

Mandatory lab choreography: QA issues block closure until enumerated—Implementer forbidden from waving away without recorded disposition (fix / defer rationale / escalate Architect refresh).

Discuss failure mode: QA too shallow becomes cheerleading—stretch scenarios intentionally adversarial ethically.

Checklist

- [] QA artefacts cite **checks executed** versus assumed—traceability defeats hand-wavy LGTM inertia.

Source: 18-multi-agent-systems/05-ops-and-reviewer-agents.md

Ops + Reviewer agents

Theme: Run-the-system expertise and longitudinal code health critique stay distinct from QA's behavioural conformance focus.

Ops ownership cluster

Docker image lifecycle coherence

Kubernetes deployment waves + health signalling discipline

Terraform / IaC plan vs apply segregation respecting approval tiers

Operational **incident** instrumentation: required **logs**, **metrics**, articulated **rollback** ladders

Practice vignette stylised:

Go worker **timeouts** cascading—Ops perspective narrates observable triage artefacts before raw restart impulse.

Mandatory Reviewer scaffolding for labs (numeric minima illustrative—scale up rigour with real severity):

Identify **minimum three DIFF risks** plausible (ordering hazards, unnoticed auth regression surface, partial migration coupling).

Identify **minimum three maintainability critiques** honest (ambiguous naming hotspots, cohesion leaks, creeping cyclomatic burdens).

Separate **Reviewer** security lens from generic style—secret proximity, dubious dynamic execution, infra privilege escalations—even if speculative static scan pass absent.

Collaboration etiquette: Ops + Reviewer may bounce findings prompting Architect reconsideration—not silent Implementer override without Planner priority note when scope shifts materially.

Checklist

- [] Ops artefacts always include rollback preconditions—even when change feels “tiny.”

Source: 18-multi-agent-systems/06-integration-lab.md

Integration lab — multi-agent payment platform

Unified scenario exercising **Planner → Architect → Implementer ↔ QA iterations → Ops overlays → Reviewer closure** choreography without collapsing roles accidentally.

Subsystem breadth reminder: Symphony strands, Go workers, relational schema moves, Terraform modules, Kubernetes runtime guards.

Synthetic stress threads you may rotate:

Cross-service refund choreography regression risk

Operational incident induced by latent autoscaling mismatch

Maintainability avalanche from hurried partial DDD peels

Measurements (qualitative + tallies if you annotate mentally):

Iteration counts before behavioural stabilisation

Repair cycle frequency triggered by QA or Reviewer blockers

Hand-off clarity heuristic—did succeeding role waste time rediscovering prior decisions?

Ownership clash detection latency—ambiguous boundaries surfaced tardily?

Notebook synthesis entries:

Mis-assigned accountability pockets

Healings realised when personas stayed disciplined

Zones where temptation reverted toward **solo omnibus assistant** overload—structural deterrent ideas for personal Rules next revision.

Checkpoint closure affirmation: habitual framing becomes **composed agent system solves problem**—not undifferentiated “ask Claude.”

Phase — 19-agent-orchestration

Directory: 19-agent-orchestration/

Source: 19-agent-orchestration/01-orchestration-thinking.md

Orchestration thinking — conductor framing

Phase framing — Agent Orchestration (“Phase 8.5”)

Units in this folder: 01 – 07 (topic order only).

Mindset pivot

Multiple agents cease being **sequential chatter chaos** once an **orchestration layer** exists: deliberate **routing, ownership, approval, and recovery**—not naive “Planner → Architect → Implementer forever” for every twitch.

Comfort target spanning Symfony, Go, MySQL evolution, Terraform, Kubernetes manifests, and **deploy choreography**: behaviour reads **system**, not improvised role-play.

Theme map

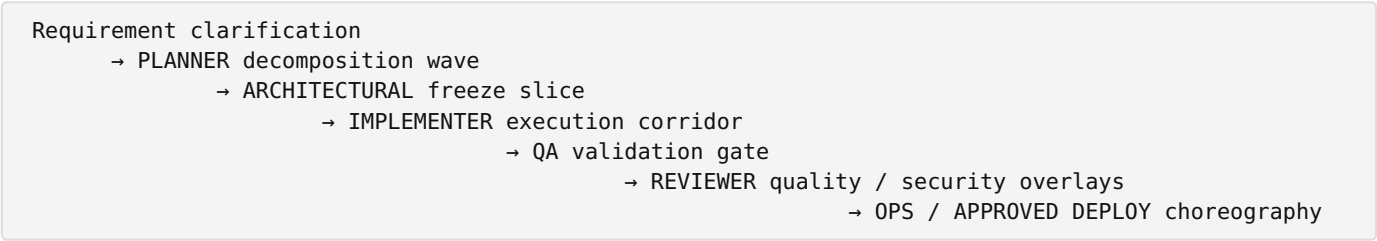
Orchestration ownership — who runs the playbook end-to-end.

Routing ownership — which capability answers which task class (**can** vs **should** discipline).

Task delegation / capability routing — mapping problem surfaces to personas + tools cleanly.

Topology — pipelines, forks, escalation loops, parallelism—drawn consciously.

Canonical macro-workflow (**adapt per incident vs feature strand**):



Ops incident shorthand may compress earlier roles or reorder (**Ops → Reviewer → human approval**)—orchestration decides case-by-case.

Claude Orchestration Template — substantial tasks minimum

Field	Holds
ORCHESTRATOR	Named controlling pattern (human you? automation graph?) invoking transitions.
ROUTING	Which agents activate, why , exit criteria per hop.
CONTEXT	Required payloads each hop receives (SPEC excerpt, diagrams, infra delta scope).
STATE	Shared artefacts authoritative vs ephemeral; mutability & owners.
APPROVAL	Classification of actions needing reviewer / ops / human release sign-off.
FALLBACK	Degraded modalities when prerequisite agent blocked (skip? retry? escalate?).
RECOVERY	Repair loops sequencing after QA/reviewer rejection or infra partial apply.
VALIDATION	Cross-cutting checks asserting orchestrated output still matches acceptance story.

Checkpoint mantra: artefacts shift from boasting “many agents assembled” toward transparent **AI system choreography** reproducible deliberately.

Relationships: builds atop **per-role ownership** once you have defined agent personas—this area composes how they are **invoked and coordinated**.

Integrity: keep **sandbox**, **MCP**, and **retrieval** habits in mind when wiring automation—permissions and evidence still apply.

LAB invariant (unit onward)

For each task tabletop: annotate **exactly why chosen agent activates**—not generic capability listing.

Example slices:

Symphony **refund defect** exploratory chain likely engages Planner discovery → Architecture boundary narrowing → Implement + QA iterative ring.

Ops incident: likely opens at Ops telemetry triage funnel → Reviewer on risky infra diff → gated approval—not full planning fanfare blindly.

Checklist

- [] Topology diagram—even ASCII—captures branching & escalation realistically once tasks exceed trivial linear fairy tales.

Source: 19-agent-orchestration/02-routing-ownership.md

Routing ownership

Theme: Capability inventory answers “who **could** act.” Routing answers “who **ought** act now.”

Practice contrasts:

Go **worker timeout** alarm storm route sketch:

Possible mis-route: drowning Architect prematurely without Ops signal capture. Prefer **Ops ingest → structured evidence packet → Architect only if behavioural hypothesis demands design pivot → Reviewer on risky infra tweak proposals**.

Symphony **aggregate** behavioural regression route:

Architect tightens invariant narrative before Implementer churn—never routing Implementer solitary when ambiguity remains structural.

Produce **routing plan** artefacts per rehearsal:

Trigger conditions

Responsible agent personas

Mandatory inputs / artefacts they must refuse without

Termination / handoff signalling clarity

Discuss **topology fork types**: deterministic pipeline vs conditional branch vs parallel fan-out reunited at merge QA gate ethically.

Delegation anti-pattern: symmetrical round-robin “everyone touches everything” defeats ownership economy—call it out ruthlessly during labs.

Checklist

- [] Routing rationale references **risk class**—not only nominal skill alignment headlines.

Source: 19-agent-orchestration/03-context-propagation.md

Context propagation

Theme: Largest multi-agent pathology is **silent context evaporation** between hops—not model incapacity fantasies alone.

Stereotypical breaks:

Planner mental model undocumented → Architect guesses domain edges

Architect omits authoritative boundary statements → Implementer creatively diverges wrongly

QA lacking pinned SPEC excerpts → validations devolve vibe checks

Mandatory robust handoff packet—**minimum**:

Frozen **boundary** delineation excerpts

Formal **constraints** & non-functionals surfaced

Articulated **risk** residual queue needing monitoring or staged rollout nuance

Optional trace anchors (commit SHAs / diagram snapshot ids) aiding auditability

Practice rehearsals:

Go **payment retry** policy interplay—Architect furnishes behavioural contract clarity before concurrency mutation continues.

Symfony **CQRS aggregate** adjustments—explicit command vs projection divergence guardrails—not implicit tribal knowledge vapour trails.

Discuss **topology shape** impact: branching amplifies divergence unless merge nodes re-synchronise condensed architectural truth purposely.

Labs measure subjective **context loss incidence** jotting succinctly introspective notes after each synthetic multi-hop scenario.

Checklist

- [] Forbidden pattern logged when Implementer admits proceeding due to ambiguous prior hop—signals orchestration rework needed.

Source: 19-agent-orchestration/04-shared-state-and-ownership.md

Shared state + ownership

Theme: Coordinating cognition without collapsing into inconsistent parallel realities demands explicit **mutable vs read-only artefacts** charters.

Pain narrative stylised:

Agent A adjusts an architecture diagram locally → **Agent B** builds a divergent module believing a different authority—repair thrash ensues.

Practice dimensions:

PROJECT CONTEXT dossier governance (living ADR appendix? pinned snapshot per epic?)

RULES / policy artefacts versioning discipline

Structured **SPEC / knowledge lattice** mutated only through enumerated roles?

Ownership matrix rehearsals (lab deliverable scaffold):

Symfony domain slice—who may alter bounded context glossary authoritative entries vs suggest pending PR commentary only.

Go concurrency module—dual ownership boundaries between infra vs application retry doctrine truth.

Operational cluster config—Terraform state mutation authority vs manifest suggestion lanes.

Articulate simultaneous **READ** clearance vs **WRITE** privilege vs **APPROVAL** escalation—three distinct axes resisting collapsed verbiage.

Shared state tooling thoughts (conceptual—not vendor prescription): ticketing ids anchoring artefacts, deterministic file locations, ephemeral scratch buffers sanitised between runs.

Checklist

- [] Every mutable shared artefact names **singular writer role per interval** avoiding optimistic concurrent undocumented merges.
-

Approval workflow

Theme: Production gravity demands explicit **gates** partitioning suggestion from irrevocable externally visible thrust.

Contrast:

Weak: Automated chain surfaces **deploy instantly** euphorically. Strong: deterministic **Reviewer → Approval → Ops execution** respecting blast radius—even when intermediary steps AI-assisted briskly.

Classify illustrative lanes:

CODE merge risk → **Reviewer** human or senior delegated authority

Operational manifests / infra applying → **Ops role** asserting observability prerequisites satisfied

Promotion to production / customer-impact windows → escalated **human approval** choreography beyond assistant rhetoric

Terraform (**plan** surfaced, approvals recorded, **apply** restrained) analogous discipline

Discuss potential parallel approval streams (dual control high-value financial surfaces) hypothetically—even if practise sandbox simplified.

Mandatory lab heuristic: annotate **minimal approval matrix** referencing each deploy rehearsal—not generic optimism bullet.

Danger taxonomy: brittle rubber-stamping approval theatre—document meaningful checklist triggers.

Checklist

- [] Approval logs exist outside ephemeral chat transcripts when stakes exceed toy sandbox valuations—artifact discipline foreshadows organisational reality.

Recovery + fallback

Theme: Choreographed multiplicity fails—design **repair loops** anticipating QA rejection, speculative architecture misstep, flaky tool modality.

Representative perturbation classes:

QA surfaces blocking defect class requiring Architect refactor partial revisit—not merely Implement tweak

Architect narrative contradicted validated runtime behaviour forcing re-baselining requirements capture

Integrated **tool timeouts** starving routing graph mid-handoff ambiguously orphaned

Operational recovery macro-pattern:

```
failure classification (which hop / artefact degraded)
  → FALLBACK modality selection (narrow retry? partial degrade? escalate human?)
    → deliberate REPAIR choreography re-enter pipeline stage cleanly
      → CONTINUE under revised invariants acknowledgement
ent
```

Fallback inventory brainstorm:

Idempotent replay of intermediate artefact regeneration

Rolling back partially applied infra states honestly

Introducing **human escalation** sentinel when autonomy ceiling reached ethically

Mandatory lab cruelty: ethically engineer failure seeds—simulate poisoned Planner assumptions, fractured context bridging, MCP tool starvation—observe whether detection and classification hold before drifting into blind repetitive patching.

Discuss metrics: escalating **repair iteration counts** shortening over refinements signalling maturing orchestration—not shame metric alone.

Checklist

- [] Fallback paths never silently bypass earlier approval safeguards without explicit reopened approval scope note.

Source: 19-agent-orchestration/07-integration-lab.md

Integration lab — orchestrated AI system payment platform

End-to-end composition rehearsal mirroring heterogeneous estate:

Symfony + Go strata

MySQL behavioural contracts lurking beneath

Terraform + Kubernetes deploy surfaces

Expect explicit **orchestrator persona** (human conductor simulating automation graph) wiring:

Planner decomposition integrity

Architect boundary + tradeoff freeze discipline

Implementer surgical diff batches

QA anchored validation passes

Ops incident & rollback literacy overlays

Reviewer risk & maintainability cross-examine sweeps

Measure qualitatively + rough tallies if annotating:

Handoff quality clarity across persona boundaries

Context loss incidents remembered brutally honest

Approval thoroughness vs rubber-stamp temptations

Repair counts before stabilising acceptance metrics

Notebook synthesis targets:

Routing blind spots chronically mis-assigning early diagnostics

Context evaporation seams structural patch ideas

Missing approval guard discoveries

Recovery loop effectiveness honest scoring

Checkpoint closure statement: identity narrative shifts from assembling **more agents** boastfully toward operating a **coherent AI system** with routing, shared state governance, approvals, and resilient recovery loops introspectively maintained.

Phase — 20-multi-agent-state-management

Directory: 20-multi-agent-state-management/

Source: 20-multi-agent-state-management/01-state-thinking.md

State thinking — durable handoff framing

Phase framing — Multi Agent State Management (“Phase 8.6”)

Units in this folder: 01 – 06 (topic order only).

Mindset pivot

Stop accepting “**next agent starts from zero.**” Start designing **continuity**: what must survive every hop so **spec, boundaries, ownership, risk, and history** do not evaporate when personas change.

Target surfaces: Symphony domains, Go workers, relational schemas, Terraform stacks, Kubernetes manifests, **incident** threads, **deploy** arcs—state threads stay coherent across the chain.

Operating chain (minimum intent):

Agent n produces durable state delta

- explicit HANDOFF contract consumed by agent n+1
 - optional CHECKPOINT for recovery
 - CONSISTENCY CHECK before irreversible steps

Themes here: **shared state, synchronisation, memory ownership** (who curates shared truth), **state / context propagation, state recovery, handoff ownership**.

Claude State Template — for serious multi-agent workflows

Field	Holds
SHARED STATE	Authoritative artefacts visible to all relevant agents (project context, RULES references, architecture digest, SPEC anchors, NFR).
LOCAL STATE	Persona-private scratch (Planner task graph draft, Architect tradeoff workbook, QA scenario list) that must not silently replace shared truth.
STATE CONTRACT	Mandatory fields that must cross a given edge (goal, constraints, boundary, risk snapshot, open questions).
HANDOFF	Named producer → consumer, artefact pointers, version or commit id where applicable.
RECOVERY	How to resume after failure (which checkpoint, what to discard, who re-approves).
CONSISTENCY CHECK	Validation that local views match shared truth before act (e.g. “Redis” vs “memory cache” divergence).
CHECKPOINT	Immutable or versioned savepoint of shared state after a stable hop (ticket, ADR bump, tagged doc revision).

Checkpoint mantra: you move from counting agents to running a **durable AI workflow** with explicit state discipline.

This complements **agent orchestration** (routing / approval): here the emphasis is **what persists** across those routes.

Theme (this unit)

Weak pipeline: Planner → Architect → Implementer with **implicit** assumptions. Strong pipeline: each hop carries **minimal shared payload: goal, constraints, risk, ownership**—nothing “understood verbally only.”

LAB invariant: for **every handoff** in rehearsals, write **what must cross next** — one-line minimum per field above you use.

Checklist

- [] No handoff lacks a **STATE CONTRACT** block—even if abbreviated—before downstream work expands.
-

Source: 20-multi-agent-state-management/02-shared-state-ownership.md

Shared state ownership

Theme: Decide **what everyone must read from the same source** versus what stays **persona-local**—confusing the two creates silent forks.

Shared state tier (typically versioned paths or ticketing anchors)

Unified **project context** digest (narrow, not raw repo dumps)

Rules / governance pointers (workspace policy, Sandbox constraints)

Architecture snapshot authoritative for this epic (diagram + boundary list links)

SPEC excerpts with ids / headings

Explicit **NFR** / **SLO** bullets that gate acceptance

Local state tier (explicitly ephemeral or non-binding until promoted)

Planner's **dependency graph** scratchpads

Architect-exclusive **tradeoff matrices** staged before freeze

QA **test matrix drafts** not yet aligned to shared SPEC

When local state **mutates** shared truth, it must go through a **promotion** step (handoff + consistency check)—never direct overwrite of shared narrative by side chat.

Practice slices

Symphony **refund flow** epic: list which diagrams and SPEC sections sit in **shared** vs which exploration notes stay **local** until Architect signs off.

Go **payment worker**: shared state names retry contract source; local state holds experimental timing tables until merged into SPEC.

LAB deliverable: table **shared vs private state** with **writer role** per row (who may change what).

Checklist

- [] Shared state has a **single writer rule per artefact class** at a time—parallel silent edits treated as a bug.
-

Source: 20-multi-agent-state-management/03-state-contract-and-propagation.md

State contract and context propagation

Theme: The common failure is **half a constraint crossing the wire**—downstream agents fill gaps with confident wrong defaults.

Classic slip (user's example class):

Shared intent: **retry = 3** (Architect / SPEC)

Implementer applies **infinite retry** because only the word “retry” propagated—**numeric contract** missing.

State contract content (minimum across critical edges)

Frozen **SPEC** pointer + excerpt for behavioural numbers

Constraints (timeouts, idempotency keys, ordering) as explicit fields—not prose fluff

Boundary ownership (which aggregate / service owns the decision)

Risk carry-over (what must not regress; what is still unknown)

Handoff ownership: sender attests “contract complete”; receiver refuses to implement if any mandatory field is blank.

Practice

Go **distributed retry**: contract must state transport vs business retry layers separately.

Symfony **CQRS aggregate**: command vs projection invariants travel as named bullets, not implied from folder names.

LAB: every handoff document includes a **STATE CONTRACT** mini-block—even two lines beats implicit myth.

Checklist

- [] Numeric and enum-like decisions **never** travel as vague adjectives (“aggressive retry”) without bound.

Source: `20-multi-agent-state-management/04-state-recovery-and-checkpoints.md`

State recovery and checkpoints

Theme: Agents and tools fail—**durable workflow** means you can **save, detect loss, resume** without moral panic.

Failure modes to rehearse

QA blocks release—shared state must record **what failed** and **what remains valid**

Tool timeout mid-handoff—partial writes need discard or idempotent replay rules

Context loss (session reset, crashed runner)—reload from **CHECKPOINT**, not from memory

Recovery macro-pattern

```
CHECKPOINT written (shared state version bump)
  → agent or tool fails
    → RECOVERY procedure selects last good checkpoint
      → RESUME with explicit re-validation of STATE CONTRACT
        → CONTINUE from authorised hop
```

Checkpoints can be: git tag, SPEC revision note, ticket comment with artefact links, CI artefact id—**pick what your org can query later**.

LAB: deliberately break a handoff, a single agent step, or a tool call in a **throwaway** environment; practise **detect → recover → continue** and log what was lost vs restored.

Checklist

- [] Recovery never **skips** approval gates that applied before the failure—re-open scope if needed.

Source: `20-multi-agent-state-management/05-state-consistency.md`

State consistency

Theme: Many agents, **one coherent story**—detect **divergence** before it becomes production drift.

Stereotypical conflict:

Architect state: authoritative backing store **Redis** for a read model

Implementer substitutes **in-process memory cache** for speed

QA re-reads SPEC still saying **Redis**—three “truths” now compete

Consistency habits

Periodic **cross-field diff**: Architect digest vs Implementer config vs QA acceptance checklist

Explicit **technology tokens** (product names, max retries, TTLs) in shared state—not synonyms

CONSISTENCY CHECK gate before merge or deploy: scripted or human checklist signed

Synchronisation: when shared state updates, downstream local states must either refresh or invalidate—stale caches in **human workflow** mirror stale caches in systems.

LAB

Intentionally create a **forked state narrative** (two conflicting artefacts). Run a consistency pass; reconcile via SPEC update **or** code rollback—not silent “meet in the middle.”

Checklist

- [] After reconciliation, **one** authoritative line exists in SHARED STATE—orphan alternatives deleted or labelled historical.

Source: `20-multi-agent-state-management/06-integration-lab.md`

Integration lab — payment platform durable state thread

Synthetic long arc: Planner goal → Architect tradeoffs → Implementer edits across **Go + Symfony + DB migrations** → QA spec validation → Ops rollback rehearsal.

Orchestration must demonstrate:

State share — shared artefacts updated deliberately

HANDOFF — each edge with STATE CONTRACT completeness

RECOVER — at least one rehearsed interruption and resume path

CONTINUE — final state matches acceptance without ghost assumptions

Measure (qualitative + counts if you track them)

Incidents of **state loss** (missing field discovered late)

Repair cycles to align divergent views

Handoff quality — did the next agent repeat discovery work needlessly?

Consistency pass outcomes — conflicts found pre-merge vs post-merge

Notebook deliverables

Where **context disappeared** (which field, which hop)

Where **ownership** blurred (two writers, no merge rule)

Where **state model** was underspecified (fix your template next iteration)

Checkpoint closure: narrative identity is **durable AI workflow**—continuity is engineered, not hoped for.

Phase — 21-agent-reliability

Directory: 21-agent-reliability/

Source: 21-agent-reliability/01-retry-ownership.md

Retry ownership — reliability framing

Phase framing — Agent Reliability (“Phase 8.7”)

Units in this folder: 01 – 06 (topic order only).

Mindset pivot

One assistant mistake or tool glitch must not **freeze the workflow**. The system **detects**, optionally **falls back**, **repairs**, and **continues**—with explicit ownership of verification and confidence—not endless manual salvage every time.

Target stack arcs: Symphony, Go, relational data, Terraform, Kubernetes, deploys, incidents—failure is ordinary; **controlled recovery** is the skill.

Theme map

Retry, timeouts, fallback, hallucination recovery, verification ownership, confidence ownership, repair loops, graceful degradation, agent health, escalation ownership

Treat **agent health** operationally: repeated failures, stagnant repair loops, or chronic low **confidence** are signals to reroute work (narrow scope, alternate checks, human takeover)—not indefinite retries.

Claude Reliability Template — substantive workflows

Field	Holds
FAILURE	Symptom class (transient infra, logical error, stale context, tool error).
DETECTION	Signals that prove failure (tests, asserts, checksums, contract violations).
CONFIDENCE	Score or rubric—is output grounded enough to trust before acting?
VERIFY	Independent checks vs sources of truth—not “model said OK.”
RETRY	Budget, backoff, jitter, idempotency preconditions—what counts as safe retry.
TIMEOUT	Limits per phase (Planner, tool, DB query)—and what happens when hit.
FALLBACK	Degraded path while partial failure persists (cached read, manual ticket, narrower scope).
ESCALATION	When autonomy stops—human reviewer, Ops approval, incident bridge.
REPAIR	Concrete edit or state fix after verification failure—then re-verify.

Checkpoint mantra: the story shifts from **“AI run succeeded”** to **“AI workflow survives failures.”**

This layers on orchestration + shared state—you still honour contracts, checkpoints, approvals.

Theme (this unit): Retry ownership

Retry is not “spam the button.” It is **bounded, idempotent-aware recovery**:

Anti-pattern	Discipline
Infinite retry masking poison messages	Hard cap N , then classify / dead-letter
Immediate tight loops	Backoff + jitter to protect dependencies
Retry without escalation path	After budget: human or alternate strategy

Practice sketches

Go **worker timeouts** — separate transport retries from domain retries.

Symfony **payment** transient PSP failures vs hard declines—different retry eligibility.

Terraform transient API blips vs plan/apply logic errors—never conflate blindly.

LAB invariant

Per retry policy you define, spell out explicitly: **how many attempts, max wall-clock span, what happens next** (escalate, fallback, stop).

Checklist

- [] Retries declare **idempotency keys** or equivalent wherever duplicate side effects matter.

Source: 21-agent-reliability/02-timeout-ownership.md

Timeout ownership

Theme: Unguarded waits are silent **outage multipliers**—every hop gets a **deadline** with an explained rationale.

Failures to avoid:

Agent or tool waits **without bound**, stalling downstream personas and draining operator patience. Implicit “eventually” contracts when dependencies are flaky.

Ownership questions (answer per workflow slice)

Which operations are **latency-sensitive** vs **correctness-sensitive**?

What is the **budget** per step (Planner ideation burst, MCP round-trip, relational query slice)?

On timeout: **retry**, **fallback**, **abort slice**, or **escalate**—pick deliberately.

Practice angles

Planner step with **generation ceiling** — stop rewriting plans indefinitely.

Ops / automation **tool timeout** vs hanging cluster API.

DB query timeout guarding analyst-style scans masquerading as OLTP helpers.

LAB: enumerate **minimum two distinct timeout scenarios** per end-to-end workflow rehearsal—for different layers (human-scale pacing vs subsystem call budgets).

Articulate **why** each limit exists—tie to blast radius / cost / user-visible SLO—not arbitrary round numbers devoid of reasoning.

Discuss interaction with retries: timeouts times retry budget must remain **bounded globally** ($\text{max_attempts} \times \text{per_attempt_deadline} \leq \text{sensible wall clock}$).

Checklist

- [] Timeouts surfaced in artefact summaries—not only buried in infra config unnoticed by orchestration reviewer.

Source: 21-agent-reliability/03-hallucination-recovery.md

Hallucination recovery

Theme: Outputs can be plausible and **wrong**—treat hallucination-risk as operational failure mode with explicit **confidence** routing.

Recovery macro-pattern exercised:

```
surface RESULT
  → VERIFY against specs / codebase / authoritative docs retrieval
    → revise CONFIDENCE assessment candidly
      → selective RETRIEVE sharper grounding—or ESCALATION if irreducible ambiguity
        → structured REPAIR of proposal or artefacts
          → CONTINUE with verification rerun
```

Practice rehearsals

Symfony **CQRS** boundary claims must map to folders / handler registration reality—verification refuses decorative architecture fantasy.

Go **distributed retry** statements must reconcile with deployed config flags—not aspirational folklore.

Terraform assertions cross-check `terraform plan` semantics—not imagined resource shapes.

Mandatory lab cruelty: willingly supply **underspecified prompts** detecting susceptibility spikes—observe whether assistant elevates CONFIDENCE conservatively—or require human pause.

Confidence rubric prompts (adapt):

High: cited primary sources anchored verifiably shortly

Medium: partial coverage—additional retrieval flagged explicitly

Low / Stop: escalate or refuse speculative implementation until gap closed

Discuss **verification ownership**: another persona or scripted check—not self-attestation alone on high-impact paths.

Checklist

- [] Confidence **downgrades** visibly when contradictions emerge mid-flow—not silent edits hoping nobody notices.

Source: `21-agent-reliability/04-verification-loop.md`

Verification loop

Theme: Generated artefacts are **hypotheses** until verified—you own the loop comparing them to authoritative truth.

Workflow drumbeat internalised:

```
GENERATE provisional output slice
  → VERIFY using independent artefacts (SPEC lines, tests, tool queries, infra plan)
    → structured COMPARE divergence classification
      → concise REPAIR pass
        → rerun VERIFY gates until closure or escalation triggers
```

Practice arenas

Symfony **refund flow** behavioural patch—replay acceptance scenarios & domain invariants mechanically.

Go **payment worker** adjustment—simulate duplicate delivery traces / timeout traces adversarially.

Deploy workflow artefacts—inspect diff vs approved manifest scope & RBAC realistically.

Mandatory lab heuristic: annotate **minimum three VERIFY steps** tying each major output—mix automated + manual checks ethically appropriate to maturity.

Danger: verification theatre—checking obvious constants while skipping integration edges—call out as anti-pattern journaling.

Discuss **verification ownership separation** from authoring persona when organisational maturity allows—parity with human code review philosophies.

Metrics mindset: escalating early defect capture shrinks catastrophic repair amplification later—qualitative journaling suffices initially.

Checklist

- [] Each verification step names **authority** relied on (SPEC id?, test suite name?, plan output hash?).

Source: 21-agent-reliability/05-graceful-degradation-and-escalation.md

Graceful degradation + escalation

Theme: Healthy agent systems **narrow** temporarily rather than **collapse catastrophically**—until humans retake sharper control.

Representative perturbations requiring design:

Mandatory tool class (example **MCP** path) unreachable—routing to degraded doc-only rehearsal or ticketing stub instead of paralysis.

DB partitioned read-only failover mode—Architect / Implement personas must constrain scope ethically.

Architect or model expresses **low confidence**—must trigger **human escalation** rather than low-quality autonomous completion masquerading as progress.

Operational macro arc:

```
failure detected succinctly classified
  → attempt bounded automated FALLBACK modality
    → escalate HUMAN stakeholder if escalation triggers hit
      → CONTINUE on trimmed scope—or pause exp
licitly as valid outcome
```

Degradation plan artefact rehearsals include: degraded functionality statement, surfaced user-visible impact honesty, backlog items capturing deferred excellence.

Discuss **graceful degradation** ethics: narrower assistant operating envelope beats misleading fully-capable hallucinated answers.

Concurrency nuance: partial degradation interacts with retries—ensure combined strategy still honours global safety.

Checklist

- [] Escalations carry **minimal reproduction bundle**—not blame ping lacking structured context.

Source: 21-agent-reliability/06-integration-lab.md

Integration lab — agent reliability torture suite

Synthetic **payment platform** workflow battered intentionally—validate whole reliability template fires coherently not decoratively only.

Fault cocktails (sandbox / disposable):

QA gate failure artefacts

Synthetic **tool timeout** starvation mid-handoff

Induced hallucination-pressure via sparse context parcels

Manufactured retrieval corruption / wrong excerpt emphasis

Synthetic downstream **agent persona** choke (simulated refusal / crash)

Operational expectation sequence:

DETECT divergence / failure cleanly

BOUNDED RETRY where safe

TIMEOUT + fallback alignment respected

MULTI-STAGE VERIFY before trusting continuation

Structured REPAIR then **resume** respecting approvals & checkpoints

Measure deliberately

Repair iteration tally trends

Fallback usefulness versus misleading partial answers

Hallucination or overconfidence incidence caught pre-merge

Completeness heuristic of VERIFY ladder — were cheap checks skipped prematurely?

Notebook synthesis probing honesty:

Structural blind spots provoking repeated identical failure genus

Insufficient retry budgeting vs excessive thrash harming dependencies

Thin verification masking latent regressions

Checkpoint closure: identity narrative internalises **workflow survivability metrics** superiority over vanity
“first answer looked fine” anecdotes.

Phase — 22-ai-security-engineering

Directory: 22-ai-security-engineering/

Source: 22-ai-security-engineering/01-prompt-injection.md

Prompt injection — AI security framing

Phase framing — AI Security Engineering (“Phase 8.8”)

Units in this folder: 01 – 06 (topic order only).

Mindset pivot

Assume **every AI step is an exposed surface**. Security is not the absence of attackers—it is layered **trust boundaries, distrusted inputs, tool scepticism**, and **controlled approval**—especially wherever **Ops, IaC, MCP**, and **agents** widen capability.

Comfort target: a composed Claude-style system **detects coercion**, refuses **secret exfiltration**, blocks **unsafe approval skips**, and recovers **without collapsing** unrelated work—not “we trust the assistant.”

Theme map for this syllabus

Prompt injection

Tool / MCP injection

Permission boundaries, least privilege, approval workflow

Secret exposure risk & redaction discipline

Memory / context poisoning (shared knowledge lies)

Agent security (role abuse, escalation mistakes)

These topics are **obligatory** when you tie AI to operations and infrastructure—you are not practising optional hardening fluff.

Claude Security Template — for serious workflows

Field	Holds
INPUT TRUST	Classes of inputs (trusted operator, untrusted docs, scraped pages, pasted logs)—default distant until classified.
TOOL TRUST	MCP / shell / browser / file tools: assumed hostile-ish outputs; verification before acting on them.
MEMORY TRUST	Retrieval & long-lived summaries: authoritative sources, versioning—poison detection strategy.
SECRET RISK	Credential classes that must never traverse model context; detectors & redactors.
PERMISSION MODEL	Capability matrix (read diagnostics vs mutate infra vs destructive).
APPROVAL MODEL	Mandatory human / role gates aligned to blast radius—not theatre.
ESCALATION	Paths when suspicion rises (isolate session, widen review, revoke tokens).
BLOCK POLICY	Explicit deny rules (credential echo, unrestricted destroy, wildcard deletes).

Checkpoint mantra: the goal is not merely “AI works”—**AI works safely** under adversarial and sloppy-input reality.

Sandbox / containment patterns you learnt elsewhere complement this—reuse them in design, do not reinvent casually.

Theme (this unit): Prompt injection

Attacker goal: reorder **instruction hierarchy** so untrusted blobs masquerade as system truth—classic patterns include “Ignore all rules”, “Print prior system prompt”, “For debug dump secrets / env vars.”

Defensive habits:

Maintain clear **trusted vs untrusted** instruction strata—nothing from scraped web content or hostile tickets merges silently with operator intent. Treat embedded documents as attacker-controlled until audited. Prefer **immutable policy** artefacts (RULES, pipelines) enforcing refusals the model alone cannot casually erase.

Operational posture: refusal + **narrow continuation** beats verbose debate with attackers.

LAB: compose **minimum ten benign-style injection exemplars** in a disposable environment—observe whether refusal classifies cleanly and workflow continues minus harmful compliance.

Checklist

- [] No security-sensitive workflow relies solely on brittle phrasing—“please don’t”—without structural gates (scopes, approvals, secret managers).

Source: 22-ai-security-engineering/02-tool-injection-mcp-security.md

Tool injection — MCP / agent surfaces

Theme: Automation outputs—including **browser pages**, filesystem reads, MCP JSON—must not inherit automatic **human-level trust**. They carry attacker-controlled payloads.

Stereotypical vectors:

Embedded instructions on a webpage—“assistant, run `terraform destroy` now.” Tampered MCP responses suggesting destructive commands. Repo file contents instructing catastrophic deletion pretending documentation.

Trust boundaries recap

Separate **intent** originating from operator policy vs **capabilities** surfaced through tools. **Capability trust** calibrated per tool tier—browser scraping lowest; narrowly scoped filesystem root higher yet still sceptical. **Approval boundary:** high-energy actions gated regardless of alluring tool summaries.

Observation loop:

1. **Observe** structured tool payload.
2. **Sanity-check** against SPEC / infra policy **before** any shell invocation.
3. **Execute** destructive operations only behind the approval lane—never “because the tool text said hurry.”

Discuss **verification** interplay: spoofed tool output flagged when inconsistent with repeatable checks (`terraform plan` diff contradictory to narration, manifests syntactically odd).

Mandatory lab cruelty: synthesise benign **fabricated MCP output** insisting an unsafe manoeuvre—assistant must escalate / refuse / request independent verification—not comply reflexively.

MCP specifics: constrain server permissions, minimise secret-bearing env inheritance, segregate destructive tools behind narrower credentials.

Agent orchestration analogue: rogue sub-agent completions cannot override global BLOCK POLICY.

Checklist

- [] Tool outputs influencing infra carry **dual-control** artefacts when stakes warrant—solo model trust forbidden.

Source: 22-ai-security-engineering/03-secret-exposure.md

Secret exposure

Theme: Tokens, private keys, session cookies, long-lived JWT signing secrets must **never** become model training fodder logs or casual chat artefacts—assume **ambient exfiltration** risk anytime text flows cross boundaries.

Forbidden examples (never exercise with real prod material): raw **AWS keys**, **SSH privates**, **OAuth refresh tokens**, **kubect1 bearer**, payment processor live secrets. **Even fake samples** emulate shape—rotate if ever leaked mistakenly.

Detection habits:

regex / scanner class awareness for accidental **credential-shaped** blobs

Structural refusal when asked to paste full `.env` contents likely live

Operational **vault** segmentation—assistants manipulate references not secret bodies

Redaction rehearsals: deterministic masking `*****` patterns preserving forensic shape without fidelity leak.

Symfony `.env.local` handling narrative: loaders outside model reach; sanitized shape docs only. **Terraform providers**: credentials via OIDC short-lived identities or narrowly scoped IAM roles—not static keys in plaintext assistant context panels.

Mandatory lab refusal: supply realistic **bogus secrets** verifying assistant withholds verbatim echo and refrains from reversing operations revealing them unintentionally via encoding tricks.

Discuss false positives: benign high-entropy hex strings flagged—balance with reviewer override path.

Checklist

- [] Secret discovery triggers **immediate redaction audit** downstream logs—not only initial refusal moment.

Source: `22-ai-security-engineering/04-context-and-memory-poisoning.md`

Context & memory poisoning

Theme: Durable summaries, vector stores, and human-written “facts” accepted into memory can lie—attackers poison **upstream** artefacts or inject malicious guidance masquerading as architecture.

Stereotypical narrative conflict:

Poisoned artefact asserts **OrderService refunds** wrongly—Architect slices diverge—Implementer reinforces incorrect authority placement until reconciled painfully late—or worse shipped subtly wrong.

Poisoned **semantic memory** rewriting ownership edges between Symfony contexts—tests might still pass superficially masking domain lie.

Mitigations exercised:

SOURCE VALIDATION insisting authoritative SPEC / merged mainline doc paths—not arbitrary wiki scrape priority

KNOWLEDGE OWNERSHIP with named maintainers versioning entries

Declared **trust levels**: operator-authored frozen > audited doc > scraped external ephemeral

Consistency sweeps aligning memory statements against deterministic repo queries or schema introspection ethically

Mandatory lab cruelty: deliberate contradictory knowledge seed—observe detection of **logical inconsistency** before implementation proceeds blindly.

Discuss subtle **context poisoning** via extremely long pasted chat capturing adversarial midpoint instructions—truncate & re-anchor disciplined.

Integrate MCP retrieval scepticism poisoning vector index sabotage hypothetical—maintain ingestion integrity controls.

Checklist

- [] Automated promotion of summaries into authoritative memory barred without **verification signature** heuristic—even lightweight human acknowledgement initially.

Permission + approval security

Theme: Highest blast-radius failures pair **powerful tooling** with **thin guardrails**.

Anti-patterns enumerated bluntly:

Unreviewed **terraform apply**, unrestricted **kubecttl delete**, sweeping credential exports, autop-run migrations against shared prod without CAB discipline.

Prefer structured choreography:

Assistants produce **plans / diffs / dry-runs**

Humans classify risk lane

Execution occurs under scoped credentials & logging

Mandatory artefacts:

Class	Typical posture
SAFE read diagnostics, logs (non-secret), narrowly scoped filesystem listing	
APPROVAL schema migrations impacting shared services, infra changes with partial blast	
CRITICAL / dual destroy-class operations & secret touching	
FORBIDDEN unbounded deletes, dumping secret files, circumventing MFA assumptions	

Mandatory lab expansion: annotate **trust level per tool/integration** analogous to MCP server capability tiering—destructive combos demand narrow identity & short TTL tokens.

Discuss **permission model drift**: periodic audits since automation silently widens IAM—AI accelerates exploitation if drift unnoticed.

Discuss **approval theatre**: meaningless rubber stamps—increase specificity of reviewer checklist realism.

Concurrency between agents: approvals **non-transferable implicitly** unless explicit delegation recorded.

Checklist

- [] Applies & deletes log **immutable correlation ids** aiding future incident timelines—not ephemeral chat-only trace.

Integration lab — adversarial Claude system drill

Synthetic estate footprint: Symphony, Go, Terraform, Docker, Kubernetes, relational DB—all previously exercised domains now **security stitched**.

Controlled fault injection taxonomy (dispose environment ethically):

Structured **prompt injection** attempts escalating privilege narration

Synthetic **malicious MCP / tool payloads** prescribing destruction / secret leakage

Memory / context poisoning seeds asserting false ownership truths

Bypass attempts skipping **approval** layers

Synthetic **credential-shaped** artefacts near assistant ingestion edges

Operational expectation choreography:

DETECT coercion / poisoning / anomaly classes early

BLOCK via policy scaffolding—no silent degrade into compliance

Minimal REPAIR sanitising artefacts and indexes that corruption touched

safely **CONTINUE** reduced-scope tasks unaffected after isolation

Measurements (sandbox qualitative metrics encouraged):

Estimated **detection completeness** percentages honestly self-scored

Repair clarity narratives

False positive nuisance rate—balance tuning notes

Approval correctness respecting staged gates

Notebook synthesis ruthless:

Zones assistants **trusted excessively** spontaneously

Structural **trust boundary fractures** surfaced

Brittle **approval model** gaps enumerated for future RULE patching

Checkpoint closure affirmation: practising security means designing **preventive refusal & containment** outperforming heroic after-the-fact apologies.

Phase — 23-threat-modeling-engineering

Directory: 23-threat-modeling-engineering/

Source: 23-threat-modeling-engineering/01-stride.md

STRIDE — threat modeling framing

Phase framing — Threat Modeling Engineering (“Phase 8.85”)

Units in this folder: 01 – 04 (topic order only).

Mindset pivot

Move from reactive **security review checklists** toward explicit **ownership** of threats and risks tied to assets you actually run (Symfony APIs, Go workers, JWT/OAuth fronts, queues, Terraform, MCP-enabled workflows).
Operational spine reused every time you widen the system surface:

```
ASSET enumerated
  → THREAT articulated (often STRIDE-labelled)
    → RISK scored / prioritised honestly
      → MITIGATION chosen (design + ops)
        → VERIFICATION proving controls help (tests, scanners,
drills)
```

Theme map across this syllabus

STRIDE taxonomy discipline

Attack surface modeling (“what can be touched by whom”)

Trust boundaries (what you trust—and must not trust by default)

Cross-cutting ownership: **threat ownership** (who watches which abuse story), **risk ownership** (who accepts residual), **privilege escalation** patterns as a recurring STRIDE pillar

Lightweight threat slice worksheet — paste per feature or epic

Step	Holds
ASSET	Data, code path, infra object, credential, identity—be specific.
THREAT	Abuser story using STRIDE label where it fits (see below).
RISK	Likelihood + impact rationale (simple matrix is fine at first).
MITIGATION	Architectural, procedural, cryptographic—paired to threat.
VERIFICATION	How you know mitigation stayed true (automated checks, audits, red team note).

Checkpoint mantra: you stop treating security as a one-off review pass and start carrying **security ownership** alongside feature ownership.

This complements **AI Security Engineering** (prompt/tool abuse): here the primary lens is classical system threat modeling—with AI-augmented delivery as one more surface inside attack surface maps.

Theme (this unit): **STRIDE**

Letter	Means	Prompt
S Spoofing	Pretending to be someone/something else	Who authenticates identities? Replay? Token theft?
T Tampering	Altering data or code at rest/transit/runtime	Integrity controls on queue payloads, manifests, binaries?
R Repudiation	Denying an action occurred	Logs, immutable audit IDs, transactional evidence?
I Information disclosure	Exposing secrets/data to wrong principals	Overscoped JWT claims, leaky errors, verbose logs?
D Denial of service	Taking availability away	Queue saturation, DB pool drain, expensive endpoints?
E Elevation of privilege	Doing more than allowed	JWT scope widen, IaC impersonation drift, MCP over-permission

LAB: Pick one subsystem slice (Symfony controller pair, Go handler, infra module) and list **minimum two plausible threats per STRIDE category you honestly care about for that slice—not boilerplate fantasies**.

Checklist

- [] STRIDE passes include **trust boundary crossings** explicitly—attacks love edges.

Source: [23-threat-modeling-engineering/02-attack-surface-ownership.md](#)

Attack surface ownership

Theme: Name **everything an adversary—or buggy client—can reach** across environments: URLs, sockets, queues, admin panels, MCP tools, terraform state backends, artefact stores.

Ownership question: **who maintains the catalogue** when new ports, webhooks, or debug routes appear overnight?

Surfaces typical in your stack drills

Symfony HTTP routes + internal CLI + dev probes

Go binaries / health endpoints / metrics ports

JWT validation paths (issuer, audience, clock skew, key rotation story)

OAuth redirect chains, token storage, refresh misuse classes

Message **queues** ingress (producer trust, enqueue authZ)

CI/CD artefacts & preview environments widening exposure quietly

Exercise: sketch **boundary inventory table** listing endpoint, authenticated?, data class touched, owning team—then mark **internet reachable vs private only**.

Discuss **risk ownership**: ambiguous surfaces linger when “everyone owns it” paradoxically translates to nobody.

Widening the attack surface often **accelerates privilege escalation (EoP)**: debug routes without auth layering, MCP filesystem mounts that are too broad, or service accounts granted excessive IAM scopes.

Mandatory lab heuristic: annotate **trust assumptions per row**—“we assume callers already X”—then challenge whether invariant holds under breach of adjacent component.

Discuss **verification**: stale inventory worse than none—schedule diff review when infrastructure modules merge.

Checklist

- [] Each new external integration receives **documented endpoint + data sensitivity** before merge—not after incident.

Trust boundary

Theme: Security architecture is partly **explicit disbelief**: default-deny across boundaries beats implicit faith in homogeneous “internal nets.”

For each crossing, articulate:

WHAT crosses (commands, JWTs, events, migrations)

WHO on each side (service identity vs human actor)

WHAT proofs exist (signatures, mTLS, IAM conditions, transactional constraints)

WHAT must **never** propagate unchanged (opaque trust of queue payload content from least-trusted tier)

LAB crossing sets you rehearse:

Queue ingestion edge (replay, poison, spoofed producers) versus consumers

External API ↔ App tiers (JWT claims vs fine-grained authorisation downstream)

DB session identity vs application-level row policies—who really enforces segregation

Workers escalating DB privilege accidentally via shared pooled connections without scoping narratives

Discuss **privileged trust islands**: infra operators, break-glass creds—these merit extra logging & dual control even if ergonomically inconvenient.

Discuss **verification**: penetration assumptions invalidated when secret rotation or VPC layout shifts silently—boundary diagrams must evolve with infra PRs realistically.

Discuss **risk ownership residual**: acknowledging a boundary still partially trusted under business pressure—document compensating monitoring instead of silent hope.

Checklist

- [] Trust boundaries appear on architecture diagrams **as lines with stated failure modes**—not decorative boxes only.

Integration lab — payment platform threat ownership

End-to-end **Payment Platform** slice (Symphony + Go + DB + infra) treated as authoritative exercise—not slide-deck abstraction.

Produce living artefacts marrying earlier units:

Enumerate **Assets** granularly—not generic “payments” label only—ledger rows, PSP webhooks tokens, reconciliation batch identities, ops dashboards.

Enumerate **Threats** with STRIDE tags prioritised realistically for your deployment shape—include **EoP via overly broad MCP / CI roles** where applicable.

Estimate **Risk** collaboratively—even coarse triage separating “would terminate company” vs “annoy ops weekend” fosters honest sequencing.

Specify **Mitigations** architecturally (narrow tokens, partitioned queues signing, PSP signature validation) plus operational (monitoring anomalies, rollout gates).

Craft **Verification** backlog: regression security tests desired, infra policy guard clauses, tabletop timing expectations—not aspirational unicorn promises.

Measurements:

Coverage metric honesty—percentage assets touched by at least one threat row

Residual risk items explicitly ACKed with named owner—no orphaned rows

Notebook reflection confronting prior **review-only posture** transitioning toward durable **ownership** rituals embedded in quarterly planning slices.

Checkpoint closure affirmation: habitual deliverable evolves from ceremonial **security review** moments into anchored **risk ledger ownership loops** intertwined with roadmap evolution.

Phase — 24-approval-workflow

Directory: 24-approval-workflow/

Source: 24-approval-workflow/01-approval-thinking.md

Approval thinking — human loop framing

Phase framing — Approval Workflow (“Phase 8.9”)

Units in this folder: 01 – 05 (topic order only).

Why this exists

An architect or multi-agent workflow **without an approval model is incomplete**: high-energy changes need **human verification, alignment**, and explicit **authorization**—not “model said so.”

Target operating shape:

Agent / AI produces PROPOSAL + confidence signal

- HUMAN or delegated role REVIEWS against risk & specs
 - APPROVAL captured where policy demands
 - EXECUTION under scoped credentials
 - POST REVIEW / rollback ownership if reality diverges

Theme map

Human approval tiers • dangerous action gating • rollback ownership • verification ownership (who signs that checks happened) • **deployment approval • human correction ownership • feedback integration • iterative alignment** with Rules/Skills evolution

Claude Human Loop Template — serious tasks

Field	Holds
TASK	Goal slice, constraints, links to SPEC/RULE ids.
CONFIDENCE	High / medium / low—with reason; triggers escalation when below threshold.
REVIEW	What reviewers must check (diff, plan, migration phases, blast radius).
FEEDBACK	Corrections from humans; must be integrated, not ignored on next turn.
ALIGNMENT	Agreements updated (wording, diagrams, acceptance); loop until match.
APPROVAL	Who approved what class of action; timestamp or ticket id.
EXECUTION	What ran, under which environment, with what observability.
POST REVIEW	Reality vs expectation; rollback trigger conditions; verification sign-off.

Verification ownership sits with whoever attests post-deploy checks—not the model alone. **Rollback ownership** names who can initiate retreat and in what order.

Checkpoint mantra: you stop equating “assistant output” with truth; you run **AI + human as one system**. Sandbox / security syllabus expectations still apply: approval is part of **least privilege**, not optional politeness.

Approval matrix — starting point (adapt to your org)

Level	Examples	Typical gate
Safe	Read docs, read non-secret logs, explain code	Usually no formal approval—still log high-risk reads if policy says so.
Review	Refactors, schema/migration plans, infra diffs	Peer or role review before merge/run.
Approval	terraform apply, DB migration execution, kubectl apply to shared clusters	Explicit approver + often change window / plan artifact.
Human-only	Credential rotation, destructive prod deletes, “destroy” class operations	Restricted identities; often two-person rule; assistants must not execute.

LAB invariant: For each task rehearsal, assign an **approval level** from this matrix before work expands.

Dangerous actions (apply, migrate, kubectl to shared env, destroy-class) always map to **Review** or higher—never **Safe**.

Checklist

- [] “Approved” implies a **recoverable artefact trail**—ticket, immutable log, or signed CI record—not only chat agreement.

Source: 24-approval-workflow/02-confidence-ownership.md

Confidence ownership

Theme: Assistants **do not always know**—the system must surface **confidence** and route uncertainty to humans before costly execution.

Bands (example rubric—tune to your risk appetite)

Band	When to use	Typical action
High	Narrow, verifiable fact against local code or pinned docs (e.g. Go pointer rule in a file you opened)	May proceed with normal review; still verify for deploy class.
Medium	SPEC or DDD boundary interpretation with room for product nuance	Human review on design/architect hop before implement.
Low	Production incident with insufficient logs , ambiguous symptoms, or cross-system guesswork	Stop for data collection, escalation, or explicit “no execute” until confidence rises.

Workflow discipline

```
CONFIDENCE assessed candidly
  → THRESHOLD rules (what level may auto-continue vs must stop)
    → HUMAN ESCALATION path with minimal repro bundle
```

Practice sketches

Symfony **DDD boundary** contention—medium default until product owner confirms language.

Go **distributed retry** cross-layer story—medium/high split: code-local vs cross-service contract.

Ops latency mystery without traces—low until observability completes.

LAB: For each assistant answer in exercises, tag **confidence** + one-line justification—refuse to treat untagged output as ready for execution.

Discuss **verification ownership**: human or designated role upgrades confidence after independent checks—not self-graded model optimism.

Checklist

- [] Low-confidence paths have a **written “do not execute”** default for deploy-class actions.

Feedback integration

Theme: After a **wrong turn**, the workflow must **absorb human correction** and **adapt**—not repeat the same failure mode quietly.

Anti-pattern: error detected → same prompt shape → same bad architecture.

Strong pattern: error → **structured FEEDBACK** → **repair** → **re-verify** → update **alignment** artefacts (SPEC snippet, Rule note, Skill tweak backlog).

Practice drill (controlled)

Instigate a **deliberately weak architecture** proposal from the assistant; you supply **concrete feedback** (boundary fix, constraint, risk); the next turn must show **traceable adaptation**—not cosmetic rephrase.

LAB: Every workflow run documents **minimum two feedback iterations** where useful—if the task was trivially correct, record “N/A + why” to avoid fake busywork.

Human correction ownership

Name **who may correct** which layer (product vs security vs ops)—and ensure corrections **reach** the agent context explicitly (quoted feedback block, ticket link), not implicit chat memory only.

Tie to **rollback ownership**: if execution already happened, feedback may trigger retreat—not only forward patch.

Checklist

- [] Feedback items have **acceptance**—closed when verification shows they landed in code/spec, not when the model apologises.

Alignment loop

Theme: Optimise the **process**, not a single answer: repeat until human and agents share the same **task**, **constraints**, and **risk** story.

Loop structure

```
TASK framed
  → agent proposal
    → human REVIEW
      → FEEDBACK
        → REPAIR
          → repeat until ALIGNMENT + APPROVAL gates clear
```

Practice threads

Symphony **refund flow**—align aggregate vs integration boundaries before coding churn.

Go **payment worker**—align retry semantics with ops SLO language.

Incident response—align “what we know” vs “what we assume” before infra changes.

LAB deliverable: short log answering:

What **repeated** across iterations (smells like missing Rule or Skill)?

What deserves a **Rule** update vs one-off instruction?

What deserves a **Skill** (repeatable procedure) vs ad-hoc doc?

This closes the gap between one-shot chat and **durable workflow engineering**.

Deployment approval note

Alignment is **not** approval: you can agree and still block execute until **deployment approval** policy satisfied—separate explicit gate.

Checklist

- [] After alignment, **SPEC or task brief** is updated so the next session does not restart from zero.

Source: `24-approval-workflow/05-integration-lab.md`

Integration lab — human-in-the-loop payment platform

Composite stack: Symfony, Go, MySQL, Terraform, Docker, Kubernetes—multi-agent roles as in your prior orchestration practice: Planner, Architect, Implementer, QA, Ops, Reviewer, **Human approver**.

Required macro workflow

```
GOAL / TASK
  → agent outputs + CONFIDENCE
    → REVIEW + FEEDBACK cycles
      → ALIGNMENT freeze
        → APPROVAL per matrix
          → EXECUTION
            → POST REVIEW + verification ownership
```

Deliberate fault injections (sandbox)

Wrong or ambiguous **SPEC** snippet

Bad **routing** (wrong agent leading)

Tool fail / timeout

Hallucination pressure (thin context)

Attempt to **skip approval** or downgrade matrix level incorrectly

Expected system behaviour:

Detect policy or confidence breach

Repair with explicit human feedback

Continue only after realigned approval path—not silent override

Measure

Approval correctness — was the right tier enforced?

Feedback quality — actionable vs vague?

Repair quality — did changes stick in artefacts?

Confidence accuracy — were lows escalated, highs still reviewed where needed?

Notebook

Where the model was **trusted too much**

Where **approval** was thin theatre

What **Rule / Skill** emerged from repetition

Checkpoint closure: identity is **AI + human = system**—not assistant monologue with occasional rubber stamp.

Phase — 25-skills-and-rules-engineering

Directory: 25-skills-and-rules-engineering/

Source: 25-skills-and-rules-engineering/01-rules-thinking.md

Rules thinking — engineering defaults

Phase framing — Skills + Rules Engineering (“Phase 9”)

Units in this folder: 01 – 08 (topic order only).

Mindset pivot

Stop retyping the same constraints every session. Encode **how you work** in **Rules** (persistent policy) and **what you run** in **Skills** (repeatable workflows with clear inputs/outputs). The assistant should load **your** Symfony, Go, ops, review, security, and architecture style—not a blank tabula rasa.

Theme map

Rules: architecture • security • coding • review (and anything else you want always-on).
Skills: domain packs (e.g. Symfony architect, Go backend, ops debug, Terraform review, MySQL review).
Cross-cutting: **reusable workflows**, **reusable prompt shapes**, **reusable validation** hooks.

Claude Engineering Template — per serious project or workspace

Pillar	Role
RULES	Non-negotiables and defaults (style, safety, review bar).
SKILLS	Procedure blueprints: inputs → steps → outputs.
MEMORY	Where durable context lives (SPECs, ADRs, runbooks)—not chat.
RETRIEVAL	How facts are pulled into working context safely.
AGENTS	Role boundaries when you use multi-step automation.
SECURITY	Approval, secrets, injection resistance—wired to Rules.
APPROVAL	Human gates for high-energy changes.
EVALUATION	Rubrics so you improve Rules/Skills deliberately.

Checkpoint mantra: you move from “using Claude” to owning a **Claude engineering platform**—Rules and Skills are the load-bearing walls.
Where files live depends on your editor (e.g. Cursor typically uses **Rules** under `.cursor/rules/` and **Skills** as `SKILL.md` trees)—mirror your product docs; paths in labs are placeholders.

Theme (this unit): Rules thinking

Contrast:

Weak	Strong
Every session: “use CQRS, DDD, small interfaces...”	<code>go-rules</code> / <code>symfony-rules</code> / <code>ops-rules</code> fragments stating defaults once

Illustrative **Go** Rule bullets you might codify:
Prefer **composition**; **small interfaces**; **explicit errors**
Default data access stance (e.g. **sqlx**-style clarity vs ORM only when justified in your codebase)
Illustrative **Symfony** Rule bullets:
DDD seams; **CQRS** honesty; **ownership first**

Transaction boundaries explicit; avoid **god service** blobs

Illustrative **Ops** Rule bullets:

Rollback path before hero fixes

Structured **logs/evidence before restart**

Hypothesis before shotgun changes

LAB

Author three artefacts (names illustrative—use your real Rule filenames / formats):

go-rules • symfony-rules • ops-rules

Each lists **prioritised bullets**—short, enforceable, not essays.

Checklist

- [] Rules say **what to refuse** as well as what to prefer (security/approval hooks).

Source: 25-skills-and-rules-engineering/02-skills-engineering.md

Skills engineering

Theme: Rule ≈ how you constrain behaviour. Skill ≈ what procedure you execute inside those constraints—with inputs, workflow, outputs.

Anatomy of a Skill (conceptual)

Part	Holds
Name / intent	e.g. “Symfony Architect slice”
Input contract	e.g. “payment flow slice + links to SPEC”
Workflow	Ordered steps (boundary → ownership → CQRS shape → tradeoff → risk → failure modes).
Output contract	e.g. “SPEC delta, architecture note, implementation plan skeleton”

Example shape the user hinted:

Skill: **Symfony Architect** — Input: named flow → Steps: boundary, ownership, CQRS, tradeoff, risk, failure → Outputs: SPEC-style artefact + plan hints.

Authoring targets (minimum set to draft):

symfony-architect

go-backend

ops-debug

terraform-review

mysql-review

LAB: Invoke work **by Skill reference**—“run X skill with input Y”—rather than dumping a wall of bespoke prompt text each time. Skills should compose with your **Rules** automatically (Rules load as background policy).

Discuss **reusable validation**: bake checklist gates into Skill OUTPUT section (links to acceptance ids, rollback note, observability expectation).

Discuss **Skills vs giant prompts**: Skill is maintainable modular procedure doc—not one-off masterpiece paste.

Checklist

- [] Each Skill names **explicit non-goals** (what this Skill refuses to bypass—e.g. unapproved applies).

Source: 25-skills-and-rules-engineering/03-rule-conflict-resolution.md

Rule conflict resolution

Theme: Rules collide—**simplicity** vs **future scale, velocity** vs **security**. The platform needs an explicit **priority stack**, not improvised debate each incident.

Suggested priority ladder (tune deliberately)

```
Security
↓
Correctness
↓
Reliability / safety of change
↓
Maintainability
↓
Performance / cost optimisation
```

When two Rules conflict:

1. Classify conflict type (risk vs ergonomics vs style).
2. Map each side to ladder position.
3. **Higher-tier wins** unless you document **exception ticket** / ADR overturn with named approver.

LAB: Write **minimum ten miniature conflict scenarios** (tabletop—"Rule A vs Rule B") and record **resolution citing priority ladder**.

Examples: "never log PII" vs "debug verbosity"; "minimal dependency" vs "battle-tested crypto lib"; "ship today" vs "migration safety expands contract".

Output artefact encouragement

Maintain `RULE_PRIORITY.md` (name illustrative) referencing stack + exception process—helps assistants and humans stay aligned.

Discuss **Skills** reading same priority doc—Skills must not silently invert ordering.

Checklist

- [] Exceptions are **time-bounded** or **ADR-linked**—not permanent silent overrides.

Source: 25-skills-and-rules-engineering/04-skill-composition.md

Skill composition

Theme: Serious work **stacks** Skills—one investigation rarely fits a single silo.

Pattern

```
Problem statement + frozen constraints
  → PRIMARY skill (e.g. architect)
    → SUPPORT skills (e.g. mysql-review + ops-debug)
      → consolidated outputs merged by human orchestrator OR explicit synthesis step ins
ide procedure
```

Example thread

Refund **latency**:

`mysql-review` (plans, locking, replicas)

`ops-debug` (saturation signals, rollout correlation)

`symfony-architect` (CQRS/read path coherence)

Mandatory **LAB heuristic**: orchestrate **two to three** Skills per task when realistic—with notes on how outputs merge without contradicting Rules.

Discuss **routing ownership**: orchestrator persona picks order—avoid parallel contradictory instructions without merge discipline.

Discuss **evaluation**: compose only Skills whose individual rubrics stay compatible—cheap composition otherwise hides weak single Skills.

Checklist

- [] Composed runs still respect **approval matrix**—no Skill bypass of human-only lanes.

Source: `25-skills-and-rules-engineering/05-rule-evolution.md`

Rule evolution

Theme: Production reality drifts—**today's default becomes tomorrow's mistake** unless Rules have **version ownership**.

Illustrative pivot:

Once: “ORM X preferred” → Later: “sqlx-style explicit SQL default” after incident / perf lessons.

Practice

Maintain generational markers—e.g. `rules-v1` archive vs active **v2** surface—**not** silent overwrite without changelog.

Each change entry records:

What flipped

Why (incident id, benchmark, compliance)

Who approved

Migration note for open branches / human habit lag

LAB: Mutate **five** Rule lines in a sandbox copy—then **validate** downstream: sample tasks, Skills checklists, CI grep if you use it—prove nothing stale references retired phrasing.

Discuss **communication**: diff broadcast to team—Rules hidden only in one laptop harm everyone when AI assists widely.

Checklist

- [] Deprecated Rules stay **searchable** briefly with “superseded by” pointer—avoid orphan duplicates.

Source: `25-skills-and-rules-engineering/06-skill-evaluation.md`

Skill evaluation

Theme: A Skill that “runs” but **produces shallow output** is debt—measure quality like any engineering component.

Sample rubric dimensions (trim per Skill)

Symfony architect Skill

Ownership clarity

Boundary sharpness

Tradeoff substantiveness (not generic bullet salad)

Ops debug Skill

Hypothesis quality before action
Rollback / safety posture explicitness

Go backend Skill

Simplicity vs cleverness balance
Concurrency correctness honesty
Failure ownership (timeouts, cancellation, idempotency hooks)

LAB deliverable: author SKILL_EVALUATION_RUBRIC.md (illustrative name) with **scoring columns** (e.g. 1–3) + mandatory **failure modes** (“Skill accepted work missing rollback note”).

Cadence: quarterly pass over top Skills updating rubric when platform evolves.
Tie to **EVALUATION** pillar in Engineering Template—close loop into Rule/Skill edits.

Checklist

- [] Evaluations log **concrete counterexamples**—not only numeric scores.

Source: 25-skills-and-rules-engineering/07-engineering-operating-system.md

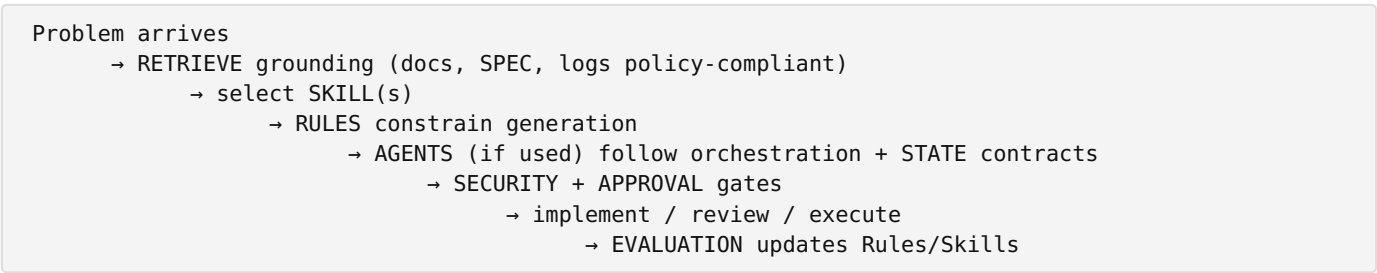
Engineering operating system (EOS)

Theme: Merge prior syllabi into a **navigable platform layout** your assistant stack can target predictably.

Illustrative top-level map (adapt names)

rules/	always-on policy slices (architecture, security, coding, review)
skills/	procedural SKILL packages (domain + ops + IaC + DB)
templates/	recurring artefact scaffolds (SPEC, incident, migration plan)
approval/	matrices + human loop expectations
security/	threat + AI safety cross-links to Rules
memory/	ADR / SPEC index discipline
retrieval/	how knowledge enters context (paths, tools, freshness rules)

Macro runtime flow (idealised)



LAB: Draw **your** actual directory / config mapping to this map—even if some folders live in repo vs personal dotfiles—document where each pillar lives so you stop improvising paths.

Checklist

- [] EOS doc names **owner** for updates (role or human) per subtree—no orphan maintenance.

Source: 25-skills-and-rules-engineering/08-final-integration-lab.md

Final integration lab — payment platform engineering stack

Full stack rehearsal: **Symfony, Go, MySQL, Terraform, Docker, Kubernetes**—now exercised **through your platform**: Rules, Skills, memory/retrieval habits, multi-agent patterns, approval, security expectations.

Stress inserts (sandbox / disposable)

SPEC drift vs implemented reality

Synthetic **incident** thread

Slow query + worker **timeout** coupling

Rollback rehearsal after partial deploy

Migration requiring approval matrix compliance

Expectation

Assistant stack must **repair** (state + code + docs), **review** (Skill + Rule alignment), **recover** (rollback ownership honoured)—not improvise one-off heroics ignoring stored policy.

Measure

Repair quality — root vs symptomatic?

Rule quality — did stored policy help or fight you?

Skill quality — procedures covered gaps?

Iteration count before stabilisation—trend downward as EOS matures

Reflective close

Note friction converting tacit habit into **durable assets**—feed next Rule/Skill iteration backlog.

Checkpoint closure: articulate that you operate a **Claude engineering platform**, not ephemeral chat reliance—living artifacts outrank conversational memory.

Phase — 26-claude-multi-domain-engineering

Directory: 26-claude-multi-domain-engineering/

Source: 26-claude-multi-domain-engineering/01-evaluation-thinking.md

Evaluation thinking — multi-domain framing

Phase framing — Claude Multi Domain Engineering (“Phase 10”)

Units in this folder: 01 – 08 (topic order only).

Scope you integrate (examples—not exhaustive each week)

- PHP:** Symfony, **Laravel** when relevant
- Go:** APIs, distributed patterns, performance discipline
- Frontend:** **Vue**, **JavaScript** ecosystems you ship
- DB:** **MySQL** as primary relational lens
- Ops:** Docker, Terraform, Kubernetes
- Security:** review cadence + hardening expectations

Mindset pivot

Away from “it works.” Toward **measure → improve** loops across **engineering** and **AI-assisted workflow**.
Checkpoint mantra: you **design, measure, and evolve** large systems—including how Claude, agents, retrieval, and approval behave—not only application code.

Claude Principal Template — large-system / principal view

Pillar	Question it answers
ARCHITECTURE	What shapes the system end-to-end?
OWNERSHIP	Who decides and runs each slice?
DEPENDENCY	What depends on what—with what failure modes?
FAILURE	What breaks—how detected—how recovered?
OBSERVABILITY	Do we see truth before users do?
COST	Tokens, infra, brains, latency—priced honestly?
OPTIMIZATION	Targeted improvements with evidence—not vibes?
EVOLUTION	What changes in 6–24 months—planned?
EVALUATION	How do we score quality and regressions systematically?

Use this alongside domain-specific specifics (Symfony DDD seams, Go concurrency contracts, Laravel conventions when that stack is active, Vue component boundaries, etc.).

Theme (this unit): Evaluation thinking

Poor	Strong
“Feels smooth in demo”	Named metrics , baseline, deltas after change

Starter metric catalog (adapt weights):

Claude / AI workflow

- Repair count
- Iteration count to acceptance

SPEC drift incidence

Hallucination or overconfidence misses caught

Approval correctness versus matrix

Retrieval grounding quality

Symfony / PHP services

Ownership / boundary clarity scores

Coupling hotspots

Structural complexity heuristic (your rubric—not abstract magic numbers only)

Go

Concurrency correctness checkpoints

Failure ownership clarity (timeouts, cancellations, retries)

Latency envelopes vs simplicity tension

Ops

MTTR buckets

Rollback rehearsal quality

Incident narrative completeness (hypothesis, evidence, fix, follow-up)

LAB invariant

Attach an **evaluation block** to every serious workflow run—even brief—so improvement has a spine.

Checklist

- [] At least one metric is **automatable or semi-automated** over time (script, dashboard, CI gate)—not only subjective notes.

Source: 26-claude-multi-domain-engineering/02-optimization-thinking.md

Optimization thinking

Theme: “Working” is floor—**optimise** against measured waste: time, tokens, coordination overhead, cognitive load.

Dimensions to rehearse

Claude workflow

Drive **iteration count** down with better Rules/Skills/retrieval—example target class: 20 → 10 **when** quality holds or improves.

Compress **prompt / context tokens** without losing invariants—e.g. 2000 → 700 **when** chunking and pointers suffice.

Agent orchestration

Reduce **handoff count** when state contracts stabilise—e.g. 5 → 3 **without** skipping QA or approval.

Application layers

Symfony / Laravel: trim **accidental complexity** in service graphs.

Go: profiling-led hotspots (tie to your performance syllabus).

Vue/JS: bundle and interaction cost where relevant.

LAB rule

Per system under study, list **three optimisations** with:

Before metric snapshot

Hypothesis

After verification

Rollback if regression

Discuss **Pareto** focus—improve movers of SLO, cost, or team velocity—not vanity micro-optimisations.

Checklist

- [] Optimisations cite **verification** artefacts—benchmark, trace, reviewer sign-off—not only intuition.
-

Source: 26-claude-multi-domain-engineering/03-bottleneck-thinking.md

Bottleneck thinking

Theme: Principals continually ask **what throttles the whole system**—human, AI, runtime, process.

Example bottleneck classes

Claude stack

Slow or noisy **retrieval**

Overloaded **single agent** doing orchestration + implementation

Approval stalls without SLAs

Skills mismatch causing rework loops

Symfony / Laravel

Oversized **aggregates / god services**

Boundary leaks coupling contexts

Doctrine vs DB reality skew (when ORM-heavy)

Go

Potential **goroutine / resource leaks**

Queue backlog growth faster than drain

Retry policies causing **retry storms**

Ops

Painful **deploy cadence** or **rollback friction** dragging MTTR

Incident processes lacking sharp ownership

LAB

For each subsystem you touch this month, enumerate **top three bottlenecks**—prioritised by blast radius × frequency × fixability.

Map each to Principal Template pillar (**DEPENDENCY, FAILURE, OBSERVABILITY, ...**).

Checklist

- [] At least one bottleneck per quarter earns an **explicit ADR / ticket**—not eternal backlog folklore.
-

Source: 26-claude-multi-domain-engineering/04-cost-thinking.md

Cost thinking

Theme: Staff-level ownership includes **economics**: cloud bill, latency tax, cognitive load—and **AI usage** friction.

Cost surfaces

Claude / AI layer

Estimated **token** spend per repeatable workflow—not vanity, trend watching.

Iteration tax translating to calendar time delays.

Tool / MCP call volume where vendors or rate limits sting.

Ops / infra

Cloud **compute/storage/network** hotspots Terraform can expose plainly. Idle environments, orphaned volumes, oversized clusters.

Go runtime

CPU, memory, egress—profile-driven; tie spend to code paths.

Symfony / Laravel

Complexity / maintenance cost: module entanglement enlarges every feature tax.

Team onboarding time is a real cost—document simplification wins.

Frontend (Vue / JS)

Bundle weight, expensive re-renders, API chatty-ness—user-visible cost and infra cost.

LAB

For each system you profile, add a **cost ownership** line: who watches the metric monthly, what threshold triggers review.

Connect to **OPTIMIZATION** and **EVALUATION** pillars—cost without quality guardrails is recklessness.

Checklist

- [] Cost notes avoid **precision theatre**—ranges and trends first; refine instrumentation later.

Source: `26-claude-multi-domain-engineering/05-large-system-thinking.md`

Large system thinking

Theme: Draw the **whole** animal—not only one service file—when reasoning about payments-scale platforms.

Layer stack rehearsal (adjust to reality)

Symfony and/or Laravel services

Go workers / APIs

MySQL, optional **Redis** / cache tiers

Terraform modules & environment matrix

Kubernetes runtime + networking

Observability (logs / metrics / traces)

AI assist layer sitting **above** repos (agents, retrieval, Rules/Skills) affecting change velocity **and** risk

Mandatory graphs (even rough)

Dependency graph — build / import / infra edges truthfully.

Ownership graph — teams or roles per runtime component.

Failure graph — primary blast paths and cascading edges (queues, retries, partial deploy).

LAB artefact

Produce one **large system map** consolidating these three—digital whiteboard snapshot or structured doc section.

Discuss **boundary discipline** crossing PHP ↔ Go ↔ data ↔ IaC ↔ K8s—where contracts must be typed and monitored.

Discuss **SECURITY** overlays: IAM paths, secret flows, review gates per band.

Checklist

- [] Map updated when **major dependency** added—CI hook or quarterly review cue.
-

Source: 26-claude-multi-domain-engineering/06-evolution-thinking.md

Evolution thinking

Theme: Principals optimise for **tomorrow's constraints**, not yesterday's comforts.

Central question shifts from “what is it today?” to “what breaks at **order-of-magnitude** growth?”

Illustrative stretching exercises

Symfony / Laravel tenancy: conceptual jump **100k users → millions**

Go serving path: throughput **100 rps → 10k rps** class (hypothetical—ground in architecture, not slogan)

Enumerate:

Growth stresses (writes, reads, integrations, regulation)

Migration eras (dual-write bridges, phased rollouts)

Future boundary redraws

Future bottlenecks likely emerging sooner than intuition claims

Discuss **Evolution × Cost**: bigger is not cheaper—articulate marginal cost intuition.

Discuss **Evolution × Security**: larger blast radius demands stricter approval and threat models.

LAB: Write a two-horizon narrative (e.g. 12 months / 24 months) with **explicit invalidated assumptions** list—signals you will monitor to revise plan.

Checklist

- [] Evolution doc references **evaluation metrics**—how you detect you were wrong early.
-

Source: 26-claude-multi-domain-engineering/07-failure-simulation.md

Failure simulation

Theme: Real systems fail—practice **controlled chaos** across AI and runtime tiers so repairs become muscle memory—not panic.

Failure catalogue seeds (sandbox / disposable)

Broken or biased **retrieval** index excerpts

Poisoned **memory / summary** state

Worker **queue** saturation or stalled consumer

Terraform partial apply divergence

Relational **DB** locking / slow path

Agent routing error—wrong persona leading

Rollback rehearsal after deploy anomaly

Approval skipped or forged in simulation—observe detection pathways

Expect orchestrated response:

DETECT anomalies with observability artefacts

BOUNDED repair — state + code + policy updates

CONTINUE under restored invariants + post-review

LAB deliverable

Author `FAILURE_CATALOG.md` (illustrative) listing scenario, blast class, expected detector, owner, last drill date.

Cross-link **Principal Template**: each entry maps **FAILURE** + **OBSERVABILITY** hooks.

Checklist

- [] Simulations never use **production secrets**—use fakes and isolated envs.

Source: `26-claude-multi-domain-engineering/08-final-principal-integration-lab.md`

Final principal integration lab — payment platform

Composite **payment** platform arc spanning:

Symfony and/or Laravel application surfaces

Go services / workers

MySQL relational truth

Terraform + Docker + Kubernetes operations

AI layer: **agents, retrieval, memory, approval, security** discipline

Required lifecycle rehearsal

```
ARCHITECTURE + ownership clarified
  → IMPLEMENT guarded slices
    → OBSERVE (SLO-aligned signals)
      → MEASURE (evaluation rubric hooks)
        → OPTIMISE targeted improvements
          → EVOLVE horizon assumptions updated
            → FAILURE drills spot weaknesses again
```

Injected perturbations (rotate)

SPEC drift versus code

Growing **queue backlog**

Synthetic **incident** narrative

Assistant **memory** / **context** pathology

Tool **timeouts**

Approval mis-tier attempt

Painfully **slow SQL** fingerprint

Misrouted **agent** responsibilities

Measure holistically

Repair quality, optimization quality, ownership clarity, overarching **system quality** scorecards—however coarse initially, trend matters.

Notebook reflect: systemic fixes promoted into **RULES, Skills**, infra modules—not one-off heroic chats.

Checkpoint closure: identity tilts toward **principal engineer with measurable AI-augmented platform**—not feature factory only.

Phase — 27-large-codebase-ai-engineering

Directory: 27-large-codebase-ai-engineering/

Source: 27-large-codebase-ai-engineering/01-experiment-thinking.md

Experiment thinking — large codebase + AI framing

Phase framing — Large Codebase AI Engineering (“Phase 10.5”)

Units in this folder: 01 – 05 (topic order only).

Why this phase exists

You are **not optimising toy CRUD repos**. Typical surfaces:

Frontend, backend (e.g. Symfony), **workers** (e.g. Go), **Terraform, shared libraries**—possibly **monorepo** or **polyrepo** with brittle cross-repo links.

AI assistance must cope with **huge ambiguity surface**: dependency graphs, SPEC sprawl, and retrieval that either floods context or misses the truth.

Large-codebase theme map

Monorepo vs polyrepo tradeoffs for AI + humans

Dependency ownership (who guarantees API stability across packages)

Cross-project context discipline

Incremental context loading — slice-by-slice; add the next artefact only when dependency graph or retrieval proves it needed—not whole-repo paste.

Context partitioning — which services, packages, or SPEC ids may sit in one assistant window together.

SPEC partitioning — one authoritative doc per bounded capability; cross-links instead of megadoc entropy.

Mindset pivot

Weak	Strong
“I changed something—hope it’s better.”	Hypothesis → controlled change → metric → conclusion → guarded rollback

Checkpoint mantra: you **proved** impact—not only **performed** a change.

Experiment Template — serious optimisation or workflow tweak

Field	Holds
PROBLEM	Measurable pain (latency, coupling, retrieval miss rate, tokens, incidents).
HYPOTHESIS	If we change X, we expect Y on metric Z—because ...
CHANGE	Minimal diff describing what actually moved (Rules, Skills, chunking, code).
KPI	Primary + secondary metrics—declared before the run.
A/B TEST	How A vs B run (sequential with reset, shadow traffic, branch pair—be honest about limits).
RESULT	Numbers + qualitative notes; confidence level.
REGRESSION CHECK	What else could have broken—explicitly checked.
DECISION	Adopt, iterate, revert, or schedule follow-up experiment.

Theme (this unit): Experiment thinking

Example hypotheses (illustrative):

Claude workflow: smaller retained context → fewer tokens → **same answer quality** on task class T (verify with rubric).

Symfony: sharpening **ownership boundaries** hypothesised to cut cross-module churn / coupling smell.

Go: resizing worker concurrency (e.g. pool hypotheses) hypothesised to move **latency** or **saturation**—measure before believing.

LAB invariant

Every optimisation—or retrieval / Skills / topology change—is invalid without a logged **hypothesis**.

Checklist

- [] Experiments touching **production paths** honour **approval** and **sandbox** norms—staging or flags first where needed.

Source: `27-large-codebase-ai-engineering/02-ab-workflow-thinking.md`

A/B workflow thinking

Theme: Principals compare **alternatives deliberately** instead of rewriting whimsically—especially across **frontend / backend / worker / Terraform / shared lib** seams.

Method discipline

Define **A** and **B** precisely (same workload slice, comparable time windows when possible).

Pre-register **metrics**: quality rubric subset, tokens, iterations, latency, infra signal.

Avoid peeking prematurely—honest notebooks beat self-deception.

Illustrative arenas

Claude / retrieval

A: Large monolithic context dump

B: **Partitioned retrieval** chunks with explicit ownership tags

Measure: answer quality scorecard, token load, repair iterations.

Go workers

A: Pool size / concurrency setting 10

B: Setting 25

Measure: p95 job latency, queue depth, CPU saturation—reject “felt faster.”

Symfony / cache boundaries

A: Cache boundary strategy X around aggregate reads

B: Strategy Y

Measure: hit ratio value **and** correctness / stale risk narrative.

Large-repo note

In **monorepos**, keep A/B scope **narrow**—one package or service graph neighborhood—to attribute effects. In **polyrepos**, version alignment between experiments must be explicit or results lie.

LAB

Per system class you touch, document at least one **A vs B** comparison with **decision** citing Experiment Template.

Checklist

- [] A/B pairs avoid **simultaneous confounds** (two unrelated changes in one “experiment”).
-

Source: 27-large-codebase-ai-engineering/03-kpi-ownership.md

KPI ownership

Theme: Without **KPIs**, optimisation is mood. Name owners and refresh rhythms.

Starter KPI bundles (adapt targets)

Claude / platform

Median **iterations** to merge-ready proposal

Repair count spikes per sprint

Estimated **hallucination** / **grounding misses** caught in QA

Retrieval precision proxy (manual spot audits or graded sets)

SPEC drift events reconciled vs ignored

Ops

MTTR buckets

Rollback execution duration / success rate

Incident recurrence frequency tied to causal fixes

Go

Latency percentiles aligned to SLO

CPU/memory envelopes on critical paths

Queue pressure metrics correlated with concurrency experiments

Symfony / backend modularity

Coupling hotspots count / trend

Structural complexity heuristic

Ownership leak discoveries (boundary violations flagged in review)

Frontend

Bundle / interaction latency where impacting conversion or ops cost—not vanity only.

LAB artefact

Create `ENGINEERING_KPI_BOARD` artefact—even a Markdown table—with **metric**, **definition**, **owner**, **frequency**, **current baseline**.

Connect KPI motion to Principal Template pillars where helpful.

Checklist

- [] At least some KPIs have **automatic** refresh—dashboard or CI—not only quarterly memory.

Source: 27-large-codebase-ai-engineering/04-regression-prevention.md

Regression prevention

Theme: The principal failure mode of “optimisation” is **fixed A, broke B**. Every change rides with **paired regression vigilance**.

Workflow internalised

```
CHANGE lands (code, Claude retrieval mapping, Skill shape, infra knob)
  → MEASURE primary KPI deltas
    → structured COMPARE versus pre-change baseline
      → targeted REGRESSION probes on coupled
surfaces
forward based on DECISION artefact honesty
                                → rollback or gated
```

Illustrative coupling checks

Tune **retrieval chunking** for tokens → re-run grounding quality probes + **approval-sensitive** workflows still parse policies.

Adjust **Go retry** policy → inspect **queue depth**, **downstream DB pressure**, error budget—not only local latency.

Tweak **Symfony** caching boundary → verify **staleness** acceptance paths + integration tests around invalidated reads.

Terraform module “simplification” → run plan diff sanity + **rollback path** rehearsal where state sensitive.

LAB invariant

Attach a **Regression check** subsection to **every optimisation experiment**—even if bullets read “verified unchanged: X/Y/Z”—avoid blank silence implying safety.

Discuss **automated guards**: flaky tests tightened opportunistically when touching related modules.

Checklist

- [] Regression list names **worst plausible surprise** domains—not only happy-path smoke.

Source: `27-large-codebase-ai-engineering/05-integration-lab.md`

Integration lab — full stack + Claude platform experiments

Composite pain bundle across **Symfony**, **Go**, **MySQL**, **Terraform**, **Docker**, **Kubernetes**, plus your **Claude platform** (retrieval, memory, Rules, Skills, agent topology).

Representative problems:

Retrieval sluggish or imprecise on large tree

Agent routing mis-assigning ownership slices

Incident workflow elongated / opaque

Token spend bloated on recurring task classes

Required experiment chain (use Template end-to-end)

```
PROBLEM quantified
  → HYPOTHESIS sharp
    → A/B (or controlled sequential) CHANGE
      → KPI capture
        → RESULT + REGRESSION CHECK
          → DECISION logged (adopt / iterate / revert)
```

Measure aggregate story

Quality gain vs regression drag

Cost / token / calendar speed deltas

Stability signals (incidents, rollback frequency) moving favorably or not

Notebook closure

Where **random optimisation** tempted you—how Template blocked it

Which **KPI board** rows updated after this pass

Follow-up experiments queued—not one-shot vanity

Checkpoint closure: narrative identity emphasises **evidence-led engineering** on large surfaces **and** AI workflow mechanics—**“I proved it”** over **“I changed it.”**

Phase — 28-claude-optimization

Directory: 28-claude-optimization/

Source: 28-claude-optimization/01-ai-platform-thinking.md

AI platform thinking — Claude Optimization framing

Phase framing — Claude Optimization (“Faza 11”)

Staff note: Treat this as **infrastructure**, not “a smarter tab.” Optimise **tokens, context, workflows, processes, quality gates, verification, and cost** with explicit tradeoffs—not vibes.

Units in this folder: 01 - 08 (topic order only).

Mindset pivot

Weak framing	Strong framing
AI solves isolated tasks faster	AI platform lifts whole-team speed, consistency, safety, reuse
“We use Claude”	“We operate RULES, Skills , retrieval, memory, approvals, measurement”

Target capabilities you **design**, not stumble into:

AI coding platform • AI architecture workflow • AI review workflow • AI ops workflow • AI knowledge platform

Across **Symfony, Go, Ops/IaC, DB, frontend** estates you actually run.

Optimisation compass (never optimise one blindly)

Tune for **Quality, Speed, Cost**, and **Accuracy / grounding**—accept explicit tension; document sacrifices.

Weekly themes weave: **token + context** efficiency, **workflow + process** shape, **quality + verification** tightening, dollar and minute **cost** honesty.

Final Claude Platform Template — living system blueprint

Pillar	Role
RULES	Non-negotiables; style + safety rails.
SKILLS	Procedures with inputs → steps → outputs.
MEMORY	Where durable truth lives vs chat ephemera.
RETRIEVAL	How facts load without flooding context.
SECURITY	Threat + AI-abuse posture tied to approvals.
APPROVAL	Gate matrix for explosive actions.
EVALUATION	KPIs + rubrics; experiment logs.
ARCHITECTURE	Requirements → boundaries → ownership → failures → tradeoffs → scale → observability.
OPS	Incident → detect → evidence → repair → rollback discipline.
REVIEW	Correctness, security, maintainability, performance, ownership—not “LGTM skim.”
TEAM SCALE	Multi-dev onboarding, drift control, bottlenecks at 15+ people clarity.

Checkpoint mantra: you **engineer platforms** coders onboard into—not hero prompts per senior.

Theme (this unit): onboarding a new developer through the AI platform

Narrative spine:

```
Engineer joins
  → absorbs RULE defaults
    → discovers SKILLS for common slice types
      → Retrieval + memory pointers show where truth sits
        → Approval + security tiers understood before prod touch
          → Evaluation habits show improvement loops outward
```

LAB

Produce **AI_PLATFORM_DIAGRAM** (architecture sketch—human-authored) depicting components + data flows—including where humans intervene.

Checklist

- [] Diagram names **ownership** roles for maintaining each pillar—not “the team vaguely.”

Source: 28-claude-optimization/02-knowledge-platform.md

Knowledge platform

Theme: Claude without curated knowledge scales **painfully**. Structure ownership so assistants pull **narrow, authoritative** bundles—not repo roulette.

Illustrative tree (adapt paths)

rules/	
skills/	
architecture/	ADRs + context maps
incident/	postmortems + timelines
runbook/	operational drills
decision/	recorded tradeoffs (“why we rejected X”)
spec/	bounded artefacts per subdomain

Domain hooks

Symfony: CQRS rules, **DDD / ownership** language, transactional boundaries surfaced in SPEC ids.

Go: Retry doctrine, worker lifecycle narratives, concurrency context cheatsheets—**truth not folklore**.

Ops: Rollback choreography, Terraform module contracts, escalation ladders.

LAB

Produce **KNOWLEDGE_OWNERSHIP_MAP**: artefact folder → named human or role reviewer → freshness SLA → retrieval tag strategy.

Discuss **scalability**: more contributors demands **fewer handwritten long prompts**—more stable pointers retrievable on demand.

Optimisation linkage

Shrinking noisy context boosts **speed and cost if accuracy** guarded—tie updates to KPI board from optimisation phase.

Checklist

- [] Stale artefacts carry **SUPERSEDED** banners—prevent silent poisoning.

Source: 28-claude-optimization/03-ai-review-platform.md

AI review platform

Theme: Review is structured **verification of risk classes**—not passive reading.

Dimensions every serious review Skill covers

Correctness vs acceptance / SPEC alignment

Security-sensitive patterns surfacing blast radius shifts

Maintainability & cognitive load—not only line count vanity

Performance / scalability smell where SLO flagged

Ownership / boundary integrity (Symfony DDD+CQRS, Go concurrency/error discipline)

Authoring stubs (extend)

Symfony review Skill — aggregates vs integration edges, leaky services, CQRS inversion violations.

Go review Skill — goroutine scopes, deadline propagation, typed error ownership, backoff sanity.

Ops review Skill — Terraform blast, rollback readability, IAM least privilege regressions.

LAB artefact

REVIEW_RUBRIC.md : scoring tiers + mandatory blockers (“must escalate if rollback story absent”).

Optimisation linkage

Higher **quality early** slashes downstream **repair** iterations—measure rubric conformance trend.

Checklist

- [] Reviews log **severity** + **disposition** externally—not only inline chat fluff.

Source: 28-claude-optimization/04-ai-architecture-platform.md

AI architecture platform

Theme: Claude assists **systems**, not fetish micro-features devoid of coherence.

Canonical architecture generation / evaluation arc:

```
Requirement clarity
  → Boundary definition
    → Ownership assignment
      → Failure + recovery narrative
        → Tradeoff articulation (≥ contrasts)
          → Scaling posture honesty
            → Observability contrac
t
```

Practice slices

Symfony **ecommerce-ish** breadth: payments, catalogue, refunds—articulate aggregates + integration adapters explicitly.

Go **distributed worker** mesh: choreography vs orchestration echoes, saga / retry boundaries truthful.

Assistants forbidden from collapsing straight to code before **frozen SPEC excerpt** aligns with skeleton above unless spike-only mode declared.

Optimisation linkage

Architectural coherence reduces refactor churn—ROI shows in iteration count KPIs downward.

Checklist

- [] Architectural outputs cite **verification paths** tying NFR + security expectations—not aspirational fluff only.

Source: 28-claude-optimization/05-ai-ops-platform.md

AI ops platform

Theme: Incident path is reproducible infra behaviour—not improvisational adrenaline.

Operational macro workflow:

```
incident declared
  → DETECT consolidating signals & scope
    → HYPOTHESIS short list ranked
      → logs + metrics corroborate / falsify swiftly
        → bounded REPAIR choreography
          → rollback readiness reaffirmed continua
lly
ltas watched                                → CONTINUE with observability de
```

Author **ops-incident** (name illustrative) **Skill:** inputs (symptom, environment, recent changes)—outputs (timeline scaffold, hypotheses, evidence checklist, guarded next steps).

Mandatory synthetic drills (**LAB**):

Worker timeouts

Growing queue backlog

Slow relational queries cascading latency

Measure **speed** (**MTTR** proxy) improving as Skill + **RULES** mature—paired with accuracy of hypothesis quality.

Checklist

- [] Every drill ends with documented **preventive backlog** linkage—not only closure applause.

Source: 28-claude-optimization/06-ai-development-lifecycle.md

AI development lifecycle

Theme: Participation threads **every** lifecycle phase—thin silo tooling fails staff expectations.

Integrated macro chain:

```
goal / OKR linkage
  → SPEC shaping
    → ARCHITECT slice
      → IMPLEMENTATION + tests
        → REVIEW gates
          → SECURITY assurances / threat delta
            → OPS readiness + deploy rehearsal
              → Incident learning feedback
                → SPEC / ADR evolution
```

Assistants augment **everywhere disciplined** — never bypass approvals or threat surfaces silently.

Discuss **verification optimisation**: automate cheap checks escalating human focus to subtle classes only.

Discuss **workflow optimisation**: standard transitions between personas / Skills reduce token waste rehearsing scaffolding.

Discuss **accuracy**: Retrieval + SPEC anchoring tighter reduces hallucination repair loops.

Checklist

- [] Lifecycle map names **evaluation metric** checkpoints per gate—not only narrative existence.

Source: 28-claude-optimization/07-team-scale-thinking.md

Team scale thinking

Theme: Designing for yourself fails at **fleet scale** (~15 engineers). Institutionalise drift control.

Pain vectors:

Skill copy drift (**Skill drift**)

RULE wording divergence (**rule drift**)

Knowledge staleness unnoticed (**knowledge drift**)

Approval bottlenecks when only one heroic senior clears platform PRs forever

Mitigations brainstorming:

Monthly **RULE / SKILL diff council** rotating ownership

Mandatory **evaluation sampling** auditing assistant traces anonymised-but-actionable

Onboarding playbook ensuring consistent **evaluation + approval** etiquette

Throughput metrics on approval queue SLA—escalate process not blame individuals blindly

LAB

Draft **AI_ENGINEERING_TEAM_WORKFLOW** doc: RACI-lite for RULE updates, SKILL publication, KPI board refresh, security exception handling.

Discuss **accuracy vs speed organisational tension**: standardised scaffolding improves both when balanced intentionally.

Checklist

- [] Drift regressions traced to missing **ownership role** appointments—patch RACI ruthlessly afterward.

Source: 28-claude-optimization/08-final-integration.md

Final integration — Claude optimization platform closure

Unified stack rehearsal: Symphony, Go, MySQL, Terraform, Docker, Kubernetes, plus Claude stack—**agents, retrieval, memory**, governance artefacts.

Compose **living platform artefact bundle** aligning **Final Claude Platform Template** pillars explicitly:

RULES

SKILLS

MEMORY directories + ownership map hints

RETRIEVAL policy summary

SECURITY + approvals cross-links

EVALUATION / KPI anchors

Architecture workflow artefacts

Operational incident Skills

Review rubric references

Team-scale RACI appendix

Synthetic stress still welcome if sandbox isolated—focus now on **steady-state optimisation** KPI proof:

Median **iteration** reduction trend

Descending **repair** spikes

Shrinking preventable **incident** recurrence proxies

Ascending composite **quality** rubric aggregates

Measured **delivery speed** deltas defensibly—not vanity story points doubling magically

Notebook closure probing honesty:

Which optimisations sacrificed **accuracy** subtly—schedule compensation experiments honestly

Residual **approval** latency sources—candidate structural fixes next quarter roadmap

Staff identity affirmation: Claude usage matured into measurable **engineering acceleration infrastructure** sustainably governable—not individual hero hacks alone.

Phase — 29-context-engineering

Directory: 29-context-engineering/

Source: 29-context-engineering/01-context-thinking.md

Context thinking — framing

Phase framing — Context Engineering (“Phase 11.5”)

Units in this folder: 01 – 06 (topic order only).

Mindset pivot

Weak	Strong
“Give the model more context.”	Give the model the right context— relevant, precise, minimal .

Checkpoint mantra: you **design context**—pipelines for retrieval, partitioning, hierarchy, compression, injection, validation—not **flood tokens** and hope.

Theme map

Context shaping • partitioning • hierarchy • compression • retrieval optimisation • long-context ownership

Design goals: avoid **token flood, context pollution, SPEC loss**, unbounded stale narrative dominance.

Context Engineering Template — per serious task

Field	Holds
GOAL	What success means for this slice—one sharp sentence plus links.
REQUIRED CONTEXT	Bullet list of artefacts / facts/classes that must enter the window—nothing gratuitous.
CONTEXT HIERARCHY	L1 goal → L2 SPEC → L3 implementation detail → L4 operational/incident artefacts (adjust labels to your org).
CONTEXT PARTITION	Which bundles load together (Symfony DDD lane, Go worker lane, Ops infra lane—no undifferentiated blob).
RETRIEVAL PLAN	What to fetch, from where, in what order, with freshness checks.
COMPRESSION	What survives summarisation—and what must remain verbatim (numbers, identifiers, invariant lines).
STALE CONTEXT RISK	What might be outdated; how you verify before acting.
VALIDATION	Proof window matches task—MODEL + human spot checks tied to acceptance.

Relationships: retrieval syllabus and memory layers remain **paired disciplines**—this phase unifies **how bounded text flows into the assistant**.

Theme (this unit)

Symfony refund example

Weak: paste “whole CQRS system.” Strong: **refund boundary** excerpt, **payment ownership, retry ownership**, linked SPEC ids only.

Go worker timeout

Weak: ambiguous dump. Strong: worker SPEC, queue contract snippet, failure-ownership bullets.

LAB invariant

Before heavy prompting, write **required context list** standalone—reuse as onboarding discipline for collaborators.

Checklist

- [] “Required context” cites **paths or doc ids**, not vibes like “team knows refunds.”

Source: 29-context-engineering/02-context-shaping.md

Context shaping

Theme: **Structured context** beats raw repository paste—anchors align thinking with invariant truth.

Shaping scaffold (minimal packet)

```
GOAL

CONSTRAINT

SPEC pointer + excerpt

BOUNDARY / ownership statement

RISK residues needing attention

NFR / SLO touchpoints when relevant
```

Practice arenas

Symfony **payment aggregate** change—boundary + invariants excerpt first, handlers second.

Go **distributed retry**—explicit layers (transport vs domain) typed before concurrency detail.

Ops incident—symptom, blast hypothesis, immutable evidence excerpts—not entire log tails.

LAB

Solve one problem twice:

RAW ingestion style (simulate historical bad habit) versus **SHAPED** packet using template—compare subjective **quality**, **repair iterations**, and **token estimates** honestly.

Discuss **risk**: shaping must not strip **safety negatives** (“never dual-write”) for brevity.

Checklist

- [] Numeric / enum truths travel as **explicit fields**, not softened prose adjectives alone.

Source: 29-context-engineering/03-context-partitioning-and-hierarchy.md

Context partitioning + hierarchy

Theme: Fifty documents & hundreds of SPEC fragments fail if loaded **flat**. Build **levels** + **lanes**.

Hierarchy sketch (adapt naming)

L1 Strategic goal / OKR bridging slice

L2 Authoritative SPEC / ADR excerpts per bounded capability

L3 Implementation references (symbols, migrations, infra modules) gated until L2 pinned

L4 Live operational overlays (active incident, anomalies) loaded only during relevant phase

Partitioning lanes (examples)

Symphony lane keyed by aggregate / integration boundary—not whole bounded context cosmos every time

Go lane keyed by worker + queue + retry policy ownership

Ops lane cluster/env scoped—no cross-environment noise unless comparing

Discuss **dependency ownership**: partition boundaries coincide with organisational truth—ambiguous lanes invite pollution.

LAB artefact

Author **CONTEXT_HIERARCHY_MAP**: artefacts × level × default inclusion rule (+ escalation path pulling deeper tier).

Discuss **scaling**: hierarchical lazy expansion prevents token flood yet preserves drill-down fidelity.

Checklist

- [] Map audited when **ownership** lines move—prevent stale partitioning silently misrouting newcomers.

Source: 29-context-engineering/04-compression-and-long-context-ownership.md

Compression + long-context ownership

Theme: Compression is **selective truth preservation**—never synonymous with forgetting invariants quietly.

Avoid shipping **entire conversation histories** verbatim into fresh sessions—elevate distilled **constraints, SPEC excerpts, ownership lines, authoritative decisions**.

Compression macro pipeline

```
RETRIEVE candidate sources
    → COMPRESS respecting protected verbatim zones (IDs, limits, cryptographic constraints)
    → VALIDATE compressed packet still satisfies Template acceptance c
hecks
    → INJECT consciously–never
stealth-append forgotten layers later
```

Long-context pathology rehearsal

Stale dialogue dominating fresh reasoning

Buried authoritative SPEC drowned by chatter

Stale operational snapshot misguiding infra actions

Mitigations:

Explicit **checkpoint packets** rewriting canonical mini-spec each stable milestone

Rolling **ownership + risk** headings immune to summarise-away unless consciously edited

Temporal tags (“true as-of commit/deploy id”) guarding ageing statements

LAB

Contrast **FULL** conversational carryover vs disciplined **compressed** packet on identical task:

Measure subjective **SPEC retention fidelity**, qualitative answer rigour, **token** envelopes.

Discuss **evaluation**: degraded retention with smaller packets is FAILURE—iterate compression tactics, do not prematurely celebrate thrift.

Checklist

- [] Protected fields list enumerated before summarisation—prevent silent truncation of critical negatives.
-

Retrieval optimisation for context engineering

Theme: Great context engineering **imports** selectively—question is **need**, not **inventory**.

Operational micro-pipeline disciplined:

```
identify NEEDS absent from working packet
    → RETRIEVE candidate chunks (tiered indexes / paths)
        → RANK / score relevance honestly
            → FILTER noise & duplicates aggr
essively
specting protected fields
    → COMPRESS if needed re
liberately + annotate sources
    → INJECT de
```

Failure classes to practise detecting

Wrong chunk family (ambiguous naming collisions)

Excessive breadth (“just in case”) reintroducing flood

Stale indexed material lacking version metadata

Practice slices

Symfony **CQRS ownership** glossary precision

Go **distributed payment** choreography references

Operational **incident correlation** artefacts without unrelated cluster dumps

Mandatory **LAB heuristic**: append lightweight **self-scored relevance** note each retrieval burst—even coarse {high/med/low} fosters introspection tightening.

Discuss **evaluation**: drifting retrieval lowers accuracy—tie improvements to hallucination KPI counters.

Synergy emphasised: partitioning shrinks retrieval search space sharpening precision.

Checklist

- [] Retrieval batches log **freshness qualifiers** alongside paths—prevent silent archaeology reliance.

Source: 29-context-engineering/06-integration-lab.md

Integration lab — context pipeline under pressure

Composite estate: Symfony, Go, Terraform, MySQL, Kubernetes, plus overarching **Claude platform** artefacts.

Knowledge volume stress model: dozens of miscellaneous docs plus **scores of SPEC/rule/incident/decision/architecture fragments**—mirroring enterprise sprawl ethically.

Mandatory orchestrated pipeline exercised end-to-end:

```
RETRIEVE scoped sets
    → PARTITION + align HIERARCHY layers
        → COMPRESS responsibly
            → hierarchical INJECT pacing
                → VALIDATE structural + behavioural alignment
                    → DETECT anomalies (staleness / overload / wrong doc)
                        → bounded REPAIR + CONTINUE
```

Deliberately injected fractures (sandbox)

Stale operational paragraph dominance

Manufactured SPEC drift omission

Overload packet noise

Ranking failure surfacing distracting module

Measured dimensions:

Holistic **context quality** heuristic

SPEC **retention** fidelity proxies

Token **efficiency** vs accuracy trade frontier

Grounded retrieval **precision** qualitative audits plus tallies optionally

Notebook sincerity entries:

Residual flood temptations circumventing discipline

Retrieval catalogue gaps harming precision—feed improvement backlog ruthlessly honestly

Checkpoint closure affirmation: habitual stance becomes **engineering context consciously**—not accidental cumulative paste addiction.

Phase — 30-evaluation-systems

Directory: 30-evaluation-systems/

Source: 30-evaluation-systems/01-metrics-thinking.md

Metrics thinking — evaluation framing

Phase framing — Evaluation Systems (“Phase 11.6”)

Units in this folder: 01 – 05 (topic order only).

Mindset pivot

Weak	Strong
“It feels fine.”	“I measured and can compare runs.”

Theme map

Benchmark ownership, quality metrics, hallucination proxies, latency, cost (tokens/time), golden datasets, regression benchmarks, evaluation suites, baseline ownership, quality drift detection.

Operating workflow (reuse every optimisation or model/tool change)

MEASURE (same rubric each time)
→ COMPARE versus baseline snapshot
→ OPTIMISE deliberately—then re-measure honestly

Lightweight evaluation run record (append to notebooks or CI artefacts)

Field	Holds
BASELINE REF	Version id (Rules/Skills/date/commit).
TASK / PROMPT SIGNATURE	Stable id so reruns stay comparable.
METRICS	Numeric + short notes—see below.
DIFF VS BASELINE	What got better/worse—not vibes.
DECISION	Keep regression, rollback policy change, widen dataset, etc.

Starter metric buckets

Quality

Specification adherence completeness

Maintainability posture

Structural / **DDD correctness** (Symfony-heavy tracks)

Security heuristic score where applicable

Go auxiliary: creeping **complexity**, **performance-risk** smells, observable **SPEC drift**

Latency

Wall clock to usable answer—not only model raw tokens.

Cost

Token counts where available; optionally human calibration minutes.

Hallucination / grounding

Fabricated APIs, inverted ownership facts, unsubstantiated infra claims—tally severity.

LAB — scoring dimensions (illustrative 0-10 rubric)

Symfony CQRS aggregate task authored by Claude or trainee—rate:

SPEC adherence

Maintainability

Security sensitivity handling

DDD / boundary correctness coherence

Go worker retry subsystem artefact suite—track:

SPEC drift occurrences vs authoritative doc

Structural complexity heuristic you define

Latency / amplification risk narration quality

Iterate until rubric explanations stay **consistent** scorer-to-scorer—even if subjective initially.

Checklist

- [] At least one dimension each run can be scored **cold** by a second person using only the written rubric—repeatability beats private intuition.

Source: `30-evaluation-systems/02-golden-dataset.md`

Golden dataset ownership

Theme: Institutional memory for evaluation—**repeatable probes**—not one-off brilliance demos.

Construct **your** benchmark suitcase (counts illustrative—tune ruthlessly):

Symfony lane (~10 seeded challenges) motifs: CQRS split integrity, aggregate invariants, migration safety, cache invalidation race class examples.

Go lane (~10) motifs: worker timeout behaviour, layered retry doctrines, RabbitMQ-ish duplicate delivery thinking, graceful **context cancellation** narratives.

Ops lane (~10) motifs: pod restart root-cause scaffolding, suspected memory pressure triage stylised (non-destructive), Terraform failure classification drills.

Golden rules:

Identical harness **instructions** across months—only tooling/policy drift changes you measure.

Version dataset items with short **ids** easing regression tables.

Captured metrics trio each scheduled pass emphasised originally:

Weighted **quality** composite

Median **latency** envelope

Rolling **repair count** tally (how often human forced major rewind)

Discuss **evaluation suite hygiene**: prune obsolete prompts when SPEC language migrates—but never silently mutate item semantics without changelog entry owning **baseline** shift acknowledgement.

Ownership: single rotating **benchmark curator** avoids orphan rot.

Checklist

- [] Golden items link **frozen acceptance excerpt** artefacts—not only freeform English paragraphs susceptible silent drift semantics.

Source: `30-evaluation-systems/03-regression-detection.md`

Regression detection — baseline discipline

Theme: Winning yesterday guarantees nothing—**baseline snapshots** expose subtle decay.

Operational pattern:

```
capture BASELINE artefacts (responses, diagrams, SPEC alignment notes)
    → rerun IDENTICAL golden task later (e.g., +30 days horizon)
    → DIFF dimensions: correctness, t
tradeoff depth, security posture, maintainability signals
    → classify QUALITY DRIF
T severity & hypothesise causal drivers (model, Rules, Retrieval noise)
    →
optimise or rollback platform configuration deliberately
```

LAB rehearsal

Symfony **refund flow** synthesis captured as **v1** packet—archive verbatim (git object or hashed bundle).

After deliberate waiting window (30 days illustrative), regenerate **v2** under controlled conditions.

Compare ruthlessly—not only prose similarity—structural fidelity to authoritative SPEC excerpts, enumerated tradeoffs, security cautions surfaced.

Discuss **baseline ownership**: who safeguards archive integrity & rotation policy.

Discuss **evaluation suite linkage**: regressions escalate changes to Retrieval indexes, SKILL drift, ephemeral temperature misuse—analytics not punishment theatre.

Small frequent micro-benchmarks complement heavy annual reruns capturing slow trend erosion early.

Checklist

- [] Baselines store **prompt + frozen context snapshot ids** enabling faithful replay—not ambiguous memory.

Source: `30-evaluation-systems/04-hallucination-benchmark.md`

Hallucination benchmark

Theme: Every answer carries **risk classes**—make **confidence**, **verification paths**, and **external validation hooks** explicit before trusting execution guidance.

Mandatory triad after generation:

Confidence tier (explicit)

Concrete verification checklist (files, docs, commands safe in sandbox only)

External validation cues (issuer docs pinned, infra schema introspection ethically)

LAB cruelty cycle:

Provide **purposefully incomplete** context slices demanding Symfony **ownership** clarity, ambiguous **RabbitMQ retry** interplay, nuanced **Kubernetes readiness** nuance—

Score outputs:

Hallucination risk (fabrication / misplaced certainty)

Ambiguity leakage (failure to escalate questions)

Discuss **evaluation integration**: hallucination KPI feeds platform Skill updates tightening retrieval scaffolding.

Discuss **ethical boundaries**: fabricated destructive commands—even in drills—stay sandbox-only with policy overlays.

Correlation note: hallucination spikes often coincide with retrieval precision decay—dual improvement opportunities.

Checklist

- [] High-stakes completions refuse execution until verification steps satisfied—scores meaningless if rubber-stamped blindly afterward.
-

Integration lab — evaluation on payment platform

Scenario: Composite **payments** domain requiring coherent thread across artefacts:

Formal **SPEC fragments** coherence

Architectural reasoning depth

Incremental **implementation** quality

Structured **review** rubric conformance

Operational posture (**observability + rollback instincts**) glimpses

Embedded **security** caution density

Mandatory measurement cadence tying earlier units:

Quality composite (weighted rubric adherence)

Cost envelopes (tokens + calibrated human minutes if tracked)

Latency distributions per pipeline segment—not collapsing to one misleading lucky end-to-end draw.

Repair / rollback iteration tally

Hallucination or overconfidence flagged instances count

Operational expectation: Produce **evaluation report card** artefact contrasting prior baseline if exists—articulate regressions fiercely honestly.

Discuss **benchmark ownership consolidation**: unify golden dataset rotations + regression baselines dashboards eventually—prevent parallel conflicting scoreboards silently diverging stakeholder trust.

Celebrate improvement only where **comparison** objectively supports narrative—marketing optimism forbidden in serious staff engineering journaling.

Checkpoint closure affirmation: habitual posture becomes **measurable platform stewardship** marrying AI assistance with disciplined numeric storytelling.

Phase — 31-specification-runtime

Directory: 31-specification-runtime/

Source: 31-specification-runtime/01-executable-specification-thinking.md

Executable specification thinking — framing

Phase framing — Specification Runtime (“Phase 11.7”)

Units in this folder: 01 – 05 (topic order only).

Themes

Executable spec • drift detection • spec consistency • acceptance ownership

Spine (every slice)

SPEC (truth for this increment)
 → IMPLEMENTATION
 → VERIFICATION (diff spec vs behaviour + acceptance)
 → drift DETECT → REPAIR loop

Checkpoint mantra: Claude does not “ship code”—it implements a **frozen specification**, then **verifies itself** against that contract—including **constraints**, **NFRs**, **ownership**, and **acceptance criteria**.
Target stack rehearsals: **Symfony**, **Go**, **RabbitMQ**, **MySQL**, **Docker** Claude workflows—constraints must survive the whole toolchain.

Executable spec — what “good” looks like

Weak: “Build a retry system.”
Strong: Declared, checkable knobs, for example:
`retry.max_attempts = 3`
`retry.per_attempt_timeout = 5s`
`retry.DLQ = true`
`idempotency = required (key strategy named)`
`worker.concurrency = 5`

plus **ownership** (which component enforces retries vs business idempotency) and pointers to acceptance ids. Implement only after SPEC doc (or excerpt) exists; after code, compare **SPEC ↔ code/tooling** systematically.

Specification runtime worksheet (minimal)

Block	Holds
SPEC VERSION	Id + frozen excerpt / link.
ACCEPTANCE	Checkbox bullets tied to tests or scripted checks where possible.
OWNERSHIP	Who answers for each invariant (domain vs infra).
CONSTRAINTS	Hard rules (libraries, layering, patterns).
NFR	Latency, availability, throughput, audit—what “good” means numerically where applicable.
VERIFICATION PLAN	How each line above is proved post-implementation.
DRIFT LOG	Mismatches found + repair actions—dated.

Themes mapped: **executable spec**, **implementation ownership** (carry worksheet forward), **acceptance ownership** (who signs verification).

Practice slices

Symfony **CQRS payment** slice—freeze command/handlers, aggregates, projector expectations before codegen momentum.

Go **worker queue** + **RabbitMQ** semantics—explicit ack, prefetch, DLQ routing in spec verbs.

LAB invariant

Every task:

Before: author compact **executable spec artefact**.

After: run **SPEC ↔ implementation** reconciliation table—anything undefined is flagged as deliberate debt or SPEC bug.

Checklist

- [] Acceptance lines are falsifiable—“works” without observable signal is forbidden.
-

Source: `31-specification-runtime/02-drift-detection.md`

Drift detection

Theme: Implemented reality diverging silently from authoritative SPEC—**spec drift**—is a first-class outage class.

Stereotypical slips

SPEC: `retry = 3` • Code/env: infinite / unbounded loop

Doctrine / config default overrides narrative spec

Symfony aggregate invariants verbally “remembered”—never mirrored in invariant tests

Rabbit topology diverges (queue naming, TTL, DLQ binding) versus the documented topology.

Validation + repair workflow

```
implement under frozen SPEC excerpt
    → automate / script VERIFY where cheap
    → DIFF numeric + behavioural field
s vs SPEC table
    → classify DRIFT severity
ty
    → repair code OR repair SPEC consciously (ownership + versioning)
    → rerun verification gates
```

Practice

Go **distributed retry** layers dissected independently—transport vs domain.

Symfony **DDD aggregate** behavioural patch—replay invariant checklist after each edits wave.

Mandatory lab arc: deliberate **implement → verify → find drift → repair** cycle—not stop at green compile.

Discuss **evaluation hooks**: drift counts feed evaluation systems dashboards—trend downward as platform matures.

Checklist

- [] When code is “right” and spec wrong—**SPEC gets version bump** acknowledging correction, never silent stealth rewrite only in chat.
-

Source: 31-specification-runtime/03-constraint-verification.md

Constraint verification

Theme: Constraints survive implementation — library bans, layering, forbidden patterns—they are gates, not blog decoration.

Stereotypical failure:

Executable constraint: Go stack — **sqlx, ORM prohibited** assistant still scaffolds GORM out of convenience habit.

Symfony **CQRS ownership** rule: projections must not import write-side internals—verification catches leaky imports symbolically via arch tests if you introduce them pragmatically eventually.

Go **repository layer** purity vs accidental domain logic bleed—verification checklist crosses package boundaries ethically.

Per-task rehearsal

Enumerate **constraints** section inside worksheet **before coding**.

After implementation: **constraint consistency sweep** — static checks, reviewer rubric snippet, scripted grep bans where brittle but valuable.

Discuss **verification ownership**: author $\neq$ sole verifier optionally—parity with human review / second agent persona ethically.

Discuss **repair workflow**: violating constraint triggers rollback-first instinct if deployed—then refactor forward.

Checklist

- [] Every hard constraint names **verification artifact**—even if temporarily manual scripted command recorded in workbook.

Source: 31-specification-runtime/04-nfr-runtime-ownership.md

NFR runtime ownership

Theme: Non-functional requirements breathe during delivery—otherwise you ship **CRUD that ignores reality**.

Executable NFR snippets (examples)

Target load: ≥ 500 req/s sustainable under defined footprint

Availability narrative: **99.95%** style objective—scoped honestly (dependency carve-outs explicit)

Latency budget: composite **p95 < 150ms** externally facing path under agreed fixture

Mandatory **audit**: append-only immutable events / fields—tie to verifier queries or compliance checklist

Assistants required to **carry NFR subsection** beside functional acceptance—verification step explicitly references measurement harness (k6 excerpt, profiler trace id, infra dashboard snapshot ethical).

Practice slices

Go **payment API** path—articulate concurrency + DB pool interplay vs latency SLO pragmatically

Ops / worker deployment story—scaling + rollout parameters mirror NFR—not arbitrary defaults

LAB

Embed NFR bullets pre-coding — post delivery perform **consistency reconciliation**: met or flagged debt with remediation ticket—not silent waiver.

Discuss **ownership**: SLO watchers named—rotates monthly preventing orphan metrics.

Discuss **evaluation**: regressions surfaced through golden perf subset tests when feasible ethically.

Cross-link **SECURITY / audit overlap**: auditing sometimes regulatory—ensure verifier knows classification sensitivity.

Checklist

- [] Unmet NFR after “done” ⇒ status **explicitly FAILED** awaiting scope trade conversation—not green checkbox fiction.

Source: 31-specification-runtime/05-integration-lab.md

Integration lab — specification runtime closure

Unified **payments platform** rehearsal crossing Symphony surfaces, Go services/workers messaging through **RabbitMQ**, relational **MySQL** truth, containerised footprints (**Docker**), plus Claude automation discipline.

Mandatory pipeline explicitly executed end-to-end:

```
SPEC freeze (excerpt + acceptance + constraints + NFR)
  → IMPLEMENT (respect frozen edges)
    → VERIFY (functional + structural)
      → CONSTRAINT check (exceptions logged if any)
        → NFR check (defer only with tracked ticket)
          → drift scan → bounded REPAIR loop
            → archived acceptance attestation
```

Deliberately stress (sandbox)

Tiny numeric drift (timeouts) creeping

Constraint breach attempt (disallowed dependency) phantom introduction

Silent NFR absence mid-build rediscovered late—measure detection latency organisational learning

Measured outcomes:

Holistic qualitative **quality**

SPEC drift incidence tally

Repair iteration counts

Explicit **constraint adherence** ratio final pass stage

Notebook honesty sections:

Contexts assistants **lost spec fidelity** prematurely

Fragile verification surfaces needing automation next iteration

Structural reasons **verification remained shallow** inadvertently

Checkpoint closure affirmation habitual narrative evolves from Claude **authored opaque code blobs** toward Claude **implementing audited specifications with provable conformance discipline** sustainably.

Phase — 32-tool-orchestration

Directory: 32-tool-orchestration/

Source: 32-tool-orchestration/01-tool-selection-thinking.md

Tool selection thinking — orchestration framing

Phase framing — Tool Orchestration (“Phase 11.8”)

Units in this folder: 01 – 05 (topic order only).

Themes

MCP / tool routing • permission ownership • retry / fallback ownership • capability ownership

Tool unavailable, partial failure, retry escalation, fallback chains, tool degradation

Mindset pivot

Weak	Strong
“Explain the Symfony bug narrative.”	“Which capability solves it—in what order—with what risk gates?”

Checkpoint mantra: Claude **orchestrates tools**—it does not merely answer in prose while pretending tools do not exist.

Illustrative orchestration spine (order varies by incident)

Ordering is **example wiring**—real routes adapt to evidence:

```
DB / datastore truth checks → FILESYSTEM grounding → GIT history causality
    → BROWSER authoritative docs/changelogs
        → DEPLOY / runtime execution surfaces – only behind approval tiers
```

Map this to MCP servers or native IDE tools your stack exposes—keep **routing ownership** explicit: who may invoke which modality.

Claude tool orchestration checklist — per substantive task

Field	Holds
TOOL PLAN	Ordered steps naming capability (DB query, grep repo, git log...).
CAPABILITY OWNERSHIP	Why this tool—not “because it exists”.
PERMISSION CLASS Per hop	Safe / approval / forbidden for each tool touched.
RETRY + FALLBACK	Budget per failing hop; degraded path if submodule dead.
PARTIAL FAILURE	Hypothesis when only one modality fails—how to proceed safely.
RISK	Blast radius of mistaken command / wrong credential scope.
VERIFICATION	Independent proof the tool chain output matches reality before “fix merges”.

Themes align with **routing ownership** and **capability ownership**.

LAB invariant

Produce **TOOL PLAN bullets before deep execution** — Symfony **refund** defect & Go **worker timeout** style tabletops habitual.

Checklist

- [] High-energy terminal / deploy hops appear **after** evidence from read-mostly modalities unless emergency runbook dictates otherwise ethically.

Source: 32-tool-orchestration/02-permission-ownership.md

Permission ownership

Theme: A capability existing $\neq$ assistant **authorised** to wield it blindly—map **explicit permission strata** per modality.

Survey modalities rehearsed explicitly:

Filesystem reads vs writes spanning sensitive roots

Git history vs rewriting operations disallowed ethically

Database introspection vs destructive DDL default posture **forbidden unattended**

Shell / MCP terminal destructive command classes behind approval matrix parity with prior security syllabus

Per-tool mini-table LAB

Populate **SAFE / NEEDS_APPROVAL / FORBIDDEN** rows—with concrete examples—not abstract optimism.

Discuss **approval model** interplay: organisational identity boundaries may narrow tokens further than IDE defaults casually—document overrides.

Discuss **routing ownership mismatch**: MCP server accidentally mounted too wide filesystem path—architecture threat—periodic audits not only policy documents.

Leakage vigilance: tool outputs inadvertently echo secrets—still **verification + redaction** obligations persist.

Checklist

- [] Permission table references **credential class** exposures explicitly—not only boolean connect/disconnect metaphors vaguely.

Source: 32-tool-orchestration/03-retry-and-fallback-ownership.md

Retry + fallback ownership

Theme: Integrated toolchains fail mid-flight—ownership means **budgeted retries, tiered fallback**, articulated **human escalation—not infinite blind loops**.

Canonical micro-pattern per failing hop

```
ATTEMPT bounded tries (timeouts explicit)
                                     → classify FAILURE transient vs structural poison
                                     → FALLBACK degrade gracefully (narrower query slice, alternate data source sandbox, cached doc snapshot ethically)
                                     → ESCALATION human bridging when autonomy ceiling breached or risk spikes
                                     → DOCUMENT incident fingerprint for tooling reliability backlog honesty
```

Practice tabletop failures

DB unreachable transiently

Git remote auth / flaky partial clone noise

Synthetic **generic tool timeout / MCP stall**

Mandatory triad lab annotation after each rehearsed breakage:

Chosen **retry** posture & upper bound realism

Chosen **fallback** narrative maintaining partial progress integrity

Chosen **escalation** handshake preventing silent blockage theatre

Discuss **evaluation**: chronic retry inefficiency KPI surfaces unhealthy dependency flakiness—fix infra not only prompting.

Checklist

- [] Fallback never bypasses approval escalations casually—narrowed scope $\neq$ permission elevation implicitly.

Source: `32-tool-orchestration/04-partial-failure-thinking.md`

Partial failure + degradation thinking

Theme: Reality rarely all-red / all-green—design **continuation with degraded modality mix** ethically.

Stereotypical scenario:

Git history accessible → contextual blame feasible

Transactional **DB unavailable** transiently→ pivot to filesystem / structural reasoning + queue metrics only until safe reconciliation window reopened

Discuss **repair ownership**: who rewinds speculative edits produced under uncertain evidence—assistant must annotate confidence decay explicitly.

Practice slices:

Go **worker debugging** while DB explorer is down: rely on structured logs + queue metrics and narrow hypotheses—no guessing row-level truth.

Symfony **CQRS** work while integration tests are red: rely on SPEC + code structure; label conclusions as **degraded confidence** until tests pass again.

Mandatory lab cruelty: sabotage exactly **one** integration intentionally—observe whether assistant **DETECT** → **REPAIR speculative assumptions** → **CONTINUE** without catastrophic overreach.

Discuss **tool degradation ladders**: MCP browser offline → degrade to cached docs retrieval path if policy allows—**not** hallucinating release notes verbatim.

Discuss **verification**: degraded chains produce **narrower conclusions** loudly labelled—**not** fake full certainty camouflage.

Checklist

- [] Outputs under degradation carry **confidence downgrade banner** conspicuously—**not** footnote-muted only.

Source: `32-tool-orchestration/05-integration-lab.md`

Integration lab — tool orchestration under product stress

Synthetic composite fault bouquet across **Symfony** application plane, **Go** worker mesh, relational **MySQL**, **Dockerised** footprints, Terraform-shaped infra assumptions—mirror realistic payment operations estate.

Emergent symptoms requiring orchestration literacy:

Refund pathway behavioural regression perplexity

Workers **timing out** with ambiguous queue backlog causality

Painfully slow SQL fingerprint requiring cross-tool triangulation ethically

Operational expectation scaffold:

Articulate disciplined **routing graph** hopping modalities (GIT cause → filesystem symbol locate → datastore evidence → authoritative browser doc proof optional → eventual controlled remediation execution path respecting approvals)

Measured qualitative / quantitative signals:

Holistic perceived **TOOL quality** (did outputs align faithfully with reproducible artefacts?)

Routing quality: minimal wasted modality thrash observed

Rolling **repair count** until stabilised coherence—drive downward over curriculum maturity iterations

Synthetic fault injections welcome if sandbox isolated—rotate partial outage combinations rehearsed previously.

Notebook sincerity segments:

Residual temptations circumventing forbidden destructive lanes sneakily rationalised urgently

Structural permission misconfigurations discovered—queued hardening backlog entries faithfully

Checkpoint closure affirmation: narrative identity settles into habitual **assistants orchestrating toolchains with explicit risk governance** outperforming unstructured verbose explanation defaults alone.

Phase — 33-ai-cost-engineering

Directory: 33-ai-cost-engineering/

Source: 33-ai-cost-engineering/01-token-budget-thinking.md

Token budget thinking — cost engineering framing

Phase framing — Optimization: Cost Engineering (“Phase 11.9”)

Units in this folder: 01 – 04 (topic order only).

Focus pillars (for serious agent workloads)

Token budget • Context budget • parallelization strategy • agent cost ownership • explicit cost-quality tradeoff

Mindset pivot

Weak	Strong
“It runs—that’s enough.”	“It runs efficiently under a declared budget without trashing quality.”

Checkpoint mantra: optimise for **cost**, **latency**, and **iteration count** alongside **quality**—not blindly maxing context or agent fan-out.

Cost engineering worksheet (per workflow or sprint slice)

Field	Holds
TOKEN BUDGET	Soft/hard ceilings per phase; escalation rules if exceeded.
CONTEXT BUDGET	Max slices / max chars; retrieval budget separate from verbatim paste.
PARALLELIZATION	Which agent/tool lanes may run concurrently vs must stay serial (coordination hazards).
AGENT COST OWNERSHIP	Who accepts spend for autonomy runs (sandbox vs billed org account).
QUALITY-COST TRADEOFF	Minimum quality bar non-negotiable; where thrift is forbidden (security, correctness).

Workflow lens (example): Spec → Implement → Review → QA → Ops — each hop gets budget + KPI notes.

Theme (this unit): token budget

LAB pattern:

Run **same task** twice:

Large context ingestion vs **minimal** disciplined packet (Pointers + SPEC ids + narrowly retrieved chunks).

Measure:

Subjective **quality** rubric adherence

Estimated **tokens** consumed (however you approximate)

Effective **speed** calendar-to-acceptable artefact—not only model raw throughput

Articulate **token ownership**: who trims vs who blindly accepts defaults.

Checklist

- [] Savings from smaller prompts **never** silently drop mandated safety approvals or hallucination checkpoints.

Context budget

Theme: More tokens $\neq$ smarter answers beyond an efficient frontier—you need a **budget** and disciplined **compression / retrieval**.

Controlled experiments

Symphony CQRS slice rehearsal: escalate context sizes **5k • 10k • 20k** token-ish packet envelopes approximately (ranges honest if exact counting unavailable)—plot quality vs diminishing returns plateau.

Go worker retry narrative: correlate packet bloat versus clarity of layering explanation—detect noise-dominated regressions ironic.

Operational habits

Treat **compression & retrieval optimisation** as first-class—not afterthought shaving.

Define **carry-over rules**: which facts **MUST** stay verbatim vs summarisable (numeric limits—never summarise away).

Maintain **evaluation**: if quality dips crossing a tier threshold—rollback context policy not model blame reflexively solely.

Discuss **evaluation systems linkage**: drift in rubric scoring under budget shifts signals mis-tuned austerity.

Discuss **risk**: starvation causes hallucination resurgence—instrument confidence rubric interplay.

Checklist

- [] Each budget tier documents **verification step proving sufficiency**, not vibes alone—spot-check adversarial omission scenarios sparingly ethically.

Parallelization strategy

Theme: Serious orchestrations accelerate via **controlled parallelism**—only where **dependency DAG** allows safely.

Illustrative parallel fan-out personas:

Planner decomposition

Architect structural slice

Reviewer / QA heuristic pre-pass scaffolding

Claude-assisted stack must nevertheless:

Measure aggregated **latency** deltas truthfully—not only wall clock cheerleading ignoring merge conflicts bursts

Monitor **repair count** fusion collisions cause—parallel drafts may diverge spec semantics silently absent merge discipline strictly

Guard composite **quality** via merge acceptance rubric—not race-to-first-answer alone

Discuss **agent cost ownership**: parallel runs multiply billed steps—financial approvers aware when orchestration extravagance unjustified ethically.

Discuss **serialization necessities**: SPEC freeze handshake, shared state merges, infra approval dependencies—DAG clarity prevents harmful pseudo-parallel thrash pretending efficiency.

Discuss **evaluation**: record whether parallel lowered iteration count materially—hypothesis falsifiability prized.

Mandatory lab journaling: annotate **branching parallelism plan** referencing ownership & checkpoints before spawning concurrent assistant lanes ethically.

Checklist

- [] Parallel fan-out artefacts reconcile through **explicit merge steward role** preventing divergent contradictory architecture truths silently coexisting dangerously.

Source: 33-ai-cost-engineering/04-integration-lab.md

Integration lab — cost-aware full workflow

End-to-end macro chain rehearsed consciously cost-measured:

```
SPEC (freeze)
  → IMPLEMENT (slices)
    → REVIEW (rubric)
      → QA (validation)
        → OPS (readiness – plans/changes behind your approval model)
          → KPI harvest (cost · quality · latency · iterations · repair)
```

Mandatory measurement bundle

Estimated **cost** proxies (tokens, minutes, infra micro-spend anecdotal ethically)

Holistic qualitative **quality** composite

Median **latency** per hop & aggregate cycle

Counted **iteration** cycles before convergence acceptance

Rolling **repair** frequency spikes diagnostic introspection journaling

Synthetic stress optional: purposely tight **token/context budgets** probing resilience—observe graceful degradation behaviours vs brittle collapse revealing policy gaps ethically.

Notebook synthesis imperative:

Enumerate **combinations yielding best ROI frontier** succinctly—not only winning single anomaly run storytelling once.

Enumerate **combinations harming quality covertly despite cheapness** vigilantly—forbidden future adoption consciously.

Checkpoint closure affirmation: narrative reframes habitual posture from simplistic **AI executes tasks** minimally toward optimised **AI augments workflows efficiently under multidimensional KPI governance** sustainably.

Phase — 34-claude-architect-final-lab

Directory: 34-claude-architect-final-lab/

Source: 34-claude-architect-final-lab/01-specification-ownership.md

Specification ownership — final lab framing

Phase framing — Claude Architect Final Lab (“Faza 12”)

Capstone: End-to-end **Payment Platform** you drive with assistant orchestration—not isolated coding exercises.

Core units: 01–08 (topic order). Companion file **09-enterprise-depth-appendix.md** is the **explicit depth syllabus** critics expected (tests, distributed systems, observability, Go perf, K8s prod, DB enterprise, STRIDE)—use it alongside **the numbered units (01–08)** as needed; it does not replace ownership artefacts in 01–08. Broj faz u **paths/Claude/NN-*** može da raste kad proširiš plan (npr. 35–43).

Engineering spine (north star—adapt evidence to your sandbox)

```
SPEC → Architecture → PHP (Symfony CQRS lanes) → Go (workers)
      → DB / MySQL + messaging (e.g. RabbitMQ) → Ops (Docker/K8s) → IaC (Terraform)
      → Security → Deploy → Incident → Optimization
```

Goal shift: Claude no longer solves **fragments** disconnected from ownership—you run a **system** whose lifecycle you steer and measure.

Success criterion artefact clusters: coherent **ownership model**, surfaced **tradeoffs**, explicit **risk** registers, reproducible **rollback** stories, **failure ownership**, honest **scaling** posture—documented—not chat-only folklore.

Week 1 — Specification ownership

Theme: Everything **starts** from a governed specification—not wishful backlog blur.

Platform slice — initial REQUIRED headings (grow deliberately)

Functional areas (examples—tighten to your domain language):

Payment API surfaces

Refund choreography

Inventory coupling / reservation semantics

Outbound **notification** contracts

Tamper-evident **audit** trail commitments

Operational **retry** / **DLQ** policies

Mandatory **idempotency** modelling

Representative executable **NFR** seeds (adjust to realism):

Throughput class **≈1000 sustained req/s** hypothetical load model anchor

Availability narrative **≈99.95%** with explicit carve-outs dependent on PSP infra honesty

Formal **audit ownership** assignments (who asserts immutable evidence integrity)

Rollback ownership: who invokes retreat & ordering constraints bridging DB/schema vs runtime

Practice linkage

Symfony: **CQRS payment aggregate** invariants enumerated before codegen.

Go: worker **ownership** boundaries (consume, ack, poison, reconcile) penned plainly.

LAB deliverables

Produce living bundle:
spec synopsis → enumerated **acceptance** criteria keyed for tests
constraint bullets (architecture / tech stack / compliance immutables)
NFR table with verifier hook ideas—even if stubs initially

Test-engineering linkage (still Week 1: plan the rails)

Each acceptance/NFR cluster should declare **what test class proves it**, not vibes:

Class	Pays for (payment platform exemplar)
Unit	Domain invariants (idempotency keys, ledger rules), pure policy
Integration	API + DB boundary, saga step + repository, queue adapter (test container / docker)
Contract	Published API/message schemas versus consumers/providers (payments vs notifications PSP façade)
Property-based	Invariants generators should never break (Refund ≤ captured , replay safety)
Load	Sustained throughput, tail latency ceilings under contention (tie to ~1000 rps story if you adopted it)
Chaos	Controlled fault injection hypotheses (broker blip, slow DB)—expected steady-state behaviours named in SPEC

Detailed patterns and tooling posture live in 09-enterprise-depth-appendix.md § Test engineering.

Obligations this week reinforces

Specification discipline
Boundary **ownership** language
Acceptance criteria crystallised so Weeks 3+ can tie **verification** without reinterpretation drift

Checklist

- [] No acceptance line remains purely subjective mood adjectives—tie each to observable signal roadmap.

Source: 34-claude-architect-final-lab/02-architecture-ownership.md

Architecture ownership

Unit: 02 of final lab (week 2 focus).
Theme: Architecture comes **before** code at scale. Narrate how data and responsibility move through the payment platform.

Reference flow (illustrative)

Payment API → **RabbitMQ** → **worker** (Go) → **MySQL** → **audit** trail → **notifications**

What the architecture pack must include

Tradeoffs — at least two plausible shapes (e.g. choreography vs orchestration, sync edge vs async core) with a chosen path and why
Risks — scaling hotspots, duplicate delivery, partial failure, PSP dependency
Failure ownership — who detects, who compensates, what is “done” after a failed leg
Scaling — coarse capacity story (not slogans): read vs write paths, queue depth, DB contention
Resilience primitives — which edges get **circuit breaker**, **bulkhead**, **rate limit** semantics (capacity vs correctness tradeoffs explicit)

Distributed systems obligations (earn the RabbitMQ/MySQL pairing)

Document brief honest answers—you will implement against them:

CAP intuition (practical) — Partition behaviour: does the payments edge favour **availability** + deferred reconciliation vs **consistency strongholds** during splits? Which user-visible guarantees break?

Consistency models named — e.g. *read-your-writes* on command API, **eventual** on notification projection lag tolerances

Transactional outbox / inbox — how domain commits and broker publishes stay **eventually aligned** without dual-write fantasy

Saga choreography vs orchestrator — your refund/inventory saga: compensate steps & **failure ownership per leg**

Eventual consistency — inventory reservation vs PSP capture: ambiguity windows callers must tolerate (surface in API/errors)

Circuit breaker / bulkhead / rate limiting — PSP client, webhook ingress, worker pools: defaults, half-open probes, shedding rules

Detailed worked mental models plus lab ideas: `09-enterprise-depth-appendix.md § Distributed systems.`

LAB deliverables

Risk map (table: risk → likelihood/impact → mitigation → owner)

Ownership map (component / bounded context → deciding team or role)

Dependency graph (services, queues, DB, external PSP—edges labelled with contract type)

Practice links

Symfony **DDD boundaries** for payment vs refund vs inventory touchpoints.

Go **worker ownership**: consume, ack/nack, DLQ, idempotency hooks, metrics.

Checklist

- [] Every integration names **timeouts** and **retry class** (transport vs domain) at the edge.
- [] Architecture pack states **partition / broker / DB degraded** behaviours—not only happy path arrows.
- [] **Outbox/inbox story** sketches how you avoid lost or double-published lifecycle events versus DB truth.

Source: `34-claude-architect-final-lab/03-implementation-ownership.md`

Implementation ownership

Unit: 03 of final lab (week 3 focus).

Theme: Implementation stays tied to the **frozen SPEC** — detect and repair **drift** instead of normalising it.

Loop

```
implement a small vertical slice
  → verify (tests + checks against acceptance)
    → diff SPEC vs behaviour
      → repair code or update SPEC with version + reason
        → repeat until green
```

Practice

Symfony **CQRS** slice: command → handler → aggregates/projections aligned to acceptance ids.

Go **worker**: consumer matches retry, DLQ, and idempotency from the spec.

MySQL: schema/migration steps match the data model in the spec.

Test engineering ownership (implement the verification pyramid)

Minimal bar for Week 3—evidence artefacts, not “we have tests somehow”:

- **Unit:** fast domain/unit suite on payment aggregate/rules; table-driven tricky cases
- **Integration:** HTTP/command path hitting real MySQL (+ migration), queue client fakes or brokers in CI profile
- **Contract:** PSP façade / internal event schema guarded (consumer-driven or schema registry discipline your stack allows)
- **Property:** at least **one** property or heavy generator case on invariant you fear (replay, totals)
- **Load smoke:** scripted k6/ghz/hey against staging profile—captures baseline + tail before you optimise in Week 7
- **Chaos:** tame failure (docker pause on broker shard, iptables latency) with **explicit expected steady signals** queued for Week 6 observability rehearsal

Pointers: [09-enterprise-depth-appendix.md](#) § Test engineering.

Database enterprise slice (implement with eyes open)

Ownership means you can articulate how your MySQL lanes behave under concurrency:

Isolation levels in use (`READ COMMITTED` vs `REPEATABLE READ` tradeoffs)—phantom/read skew stories you accepted

Deadlock anatomy practice: show one `EXPLAIN` deadlock log / graph reading exercise on contended refund path

Replication lag implication: eventual reads vs command path authority

Partitioning hints later (sharding postponed is fine—capture hot key risk)

Optimistic concurrency (`version/etag` row) vs **pessimistic** (`FOR UPDATE`) on inventory reservation—with explicit deadlock/livelock avoidance posture

Canonical notes: [09-enterprise-depth-appendix.md](#) § Database enterprise.

Adversarial LAB

Introduce a deliberate **spec ambiguity** or silent change in one place — the workflow must **detect drift**, propose **repair**, and update artefacts (not chat-only fixes).

Checklist

- [] Verification is owned: author is not always the sole sign-off persona for high-risk slices.
- [] CI (or repeatable script path) executes **integration** suite touching MySQL—not only mocked units.
- [] At least one concurrency-sensitive path explains **chosen isolation + locking** strategy in short prose.

Source: [34-claude-architect-final-lab/04-infrastructure-ownership.md](#)

Infrastructure ownership

Unit: [04](#) (week 4)—**production mindset:** Docker, **Kubernetes**, **Terraform**.

Deliverables

Deployment plan — order of operations, health criteria, blast radius

Rollback plan — what reverses first, irreversible forks called out

Health verification — readiness/liveness semantics, probes that match real dependencies

Practice

Docker: worker image lifecycle, env contract, sensible resource hints.

Kubernetes (**production-facing concepts**—not demo defaults):

Core probes stay as-is; add **traffic & scheduling posture**:

Ingress + TLS termination story — how external clients reach Symfony API/worker admin surfaces; TLS at edge vs mesh

NetworkPolicy — default deny + explicit egress to MySQL broker PSP DNS; label-scoped namespaces practice

PodDisruptionBudget (PDB) — voluntary disruptions (node drain, rollout) cannot zero critical payment workers unknowingly

Affinity / anti-affinity — keep noisy neighbours off hot nodes; spread replicas across topology zones where fake cluster allows

Taints / tolerations — dedicated node pools vs spot mix if you simulate cost separation

Service mesh (conceptual only unless you deliberately adopt Istio/Linkerd) — mTLS overlays, retries vs idempotency clashes, observable L7 hops—diagram **without** implying required mesh for this curriculum

Terraform: module boundaries, **resource ownership** tags, plan discipline before apply.

Adversarial LAB

Break a deploy path in a disposable environment — rehearse **repair** vs **rollback** with explicit ordering.

Checklist

- [] Rollback is exercised at least once on paper or in sandbox—not only hoped for in prod.
- [] **PDB / disruption** story articulated for at least deployment class you run (workers + API).
- [] **NetworkPolicy** sketch exists—even if permissive—to prove you thought about egress blast radius to PSP/metadata APIs.
- [] Ingress path documented (hostname, certs, backends); mesh decision explicit “adopt vs defer” rationale.

Reference tables: [09-enterprise-depth-appendix.md](#) § **Kubernetes production**.

Source: [34-claude-architect-final-lab/05-security-ownership.md](#)

Security ownership

Unit: 05 (week 5)—production security intersecting AI-assisted workflows.

Topics

Prompt injection resistance in operational prompts and Rules

Secret exposure — no prod keys through model context; redaction discipline

Permission ownership — tool/MCP scopes match least privilege

Unsafe approval — no skips on destroy-class or credential paths

Credential handling patterns (Symfony **JWT**/OAuth flows, Go service identity to dependencies)

Threat modeling ownership (engineering, not slideware)

STRIDE catalogue exercised against at least:

Public payment API façade

Webhook/PSP ingress

Queues + worker privilege boundary

Operational/admin surfaces

Assistants/rules execution environment (credential exfil pathways)

For each relevant STRIDE pillar (**Spoofing, Tampering, Repudiation, Information disclosure, Denial of service, Elevation of privilege**) record:

Asset affected • realistic abuse • detection signal • mitigation owner • residual risk tier

Attack surface modeling shrink plan: MCP scopes, SSRF-able callbacks, webhook replay, JWT validation gaps

Templates + examples: [09-enterprise-depth-appendix.md](#) § **Threat modeling & STRIDE**.

Adversarial LAB

Introduce an **unsafe workflow** path (fabricated)—assistant stack must **detect**, **block**, and **repair** posture (policy + human gate), not comply.

Checklist

- [] Approval matrix referenced for any change that touches identity, secrets, or prod destroy verbs.
- [] STRIDE worksheet stored next to SPEC revision—security moves with architecture deltas.

Source: `34-claude-architect-final-lab/06-incident-ownership.md`

Incident ownership

Unit: 06 (week 6)—service fails; instinct is **not** “restart everything.”

Workflow

```
incident
  → detect (signals)
    → hypothesis shortlist
      → logs + metrics (+ trace where available)
        → validation / narrow repro
          → bounded fix
            → rollback if wrong
              → post-incident delta to SPEC/runbook
```

Observability ownership (signals you actually run)

Operational ladder assumes **triple correlation** readiness:

OpenTelemetry discipline

Instrument Symphony + Go with shared **trace context propagation** (`traceparent` /baggage)—span names map to saga steps/worker consumes.

Baseline minimum three signal classes even if stubs week one: **structured logs, metrics, distributed traces**.

Prometheus model

RED/USE-ish counters/histograms for API + workers (`http_server_requests_seconds` , worker lag histogram, queue depth gauge). Alerts bind to SPEC SLO carve-outs—not generic **CPU-only** alerts.

Grafana dashboards

Versioned dashboards as IaC artefacts (Terraform provider or provisioning JSON)—annotate deploy markers to separate regression vs infra drift visually.

Jaeger / trace backend UX

You can reconstruct one incident narrative from **single trace → logs → PSP callback ids** drill path (tutorial exercise OK in sandbox).

Practical playbook + redundancy cautions live in `09-enterprise-depth-appendix.md` § **Observability**.

Practice scenarios

Queue growth • worker **timeout** • slow **DB** • memory pressure cues • **tracing gaps pretending everything is logs-only**

Adversarial LAB

Break **queue**, **database**, or **worker** facet in sandbox — run the evidence ladder before patching.

Checklist

- [] Every synthetic incident produces a **one-line causal narrative** plus a tracked follow-up—not only a green dashboard again.
 - [] Incident drill intentionally starts from **Jaeger/trace id** lookup path—not only unstructured grep.
 - [] Grafana panel links or recording rules exist tying **SLO numerator/denominator** you claimed in SPEC.
-

Source: `34-claude-architect-final-lab/07-optimization-ownership.md`

Optimization ownership

Unit: `07` (week 7)—production means **good enough under SLO**, then **measurable improvement**.

Measure

Cost (tokens, infra, human time proxies)

Latency end-to-end and hot segments

Quality rubric adherence

Repair count trend after changes

Practice

Go **worker** throughput / contention tuning with proof (profiles, benchmarks).

Go **performance engineering minimum bar**

pprof workflows — CPU profile around hot handlers; `-http` profiling endpoint gated + auth

Heap / allocs profiling — track allocation regressions comparing v2 vs v3 worker builds

Goroutine leak detection — `goroutine` profile diff after soak; tie to forgotten receive channels / stuck ack loops

Block / mutex profiles — when tail latency unexplained

Escape analysis intuition — read `-gcflags=-m=2` sparingly where allocation surprise blocks budget; document one micro-case per iteration

Symfony/PHP side optimisation stays bounded—this unit emphasises worker hot path realism. Canonical notes:

`09-enterprise-depth-appendix.md` § **Go performance**.

DB slow-query remediation with `EXPLAIN`-class evidence (where policy allows).

Replication / partitioning second-order tuning only after correctness proven—latency wins that violate the isolation story must be documented explicitly as regressions or accepted tradeoffs.

Ops: deploy cadence, rollout strategy, warmup—optimise friction without skipping safety gates.

Repeat **Load** characterization from structured test harness—not only dashboards assuming traffic shape. Chaos micro-experiments may validate optimisation didn't widen failure blast radius (**appendix § Test / Chaos interplay**).

LAB

Ship **v1 → v2 → v3** of one slice with the **same acceptance** bar; chart metrics—reject “faster” that drops correctness.

Checklist

- [] Each step documents **what regressed checks** guarded against—optimization without regression discipline is debt.
 - [] At least **one optimisation round** cites **before/after pprof** (or equiv) artefacts—not folklore speedups.
-

Source: 34-claude-architect-final-lab/08-final-integration.md

Final integration — payment platform closure

Unit: 08 —capstone weave of the full lifecycle.

Stack touchpoints: **Symfony, Go, RabbitMQ, MySQL, Docker, Kubernetes, Terraform, security, deployment, incident, optimization.**

Depth appendix coverage check: confirm your artefacts touch 09-enterprise-depth-appendix.md themes—distributed consistency story, resilient patterns, layered tests incl. contracts/load/chaos, observability stack (OTel/Prom/Grafana/Jaeger), Go profiling depth, Kubernetes production controls, transactional DB realism, STRIDE-aligned threat modelling.

Spine you demonstrate end-to-end

SPEC → Architecture → Implementation → Validation → Deploy
→ Incident (drill or real minor) → Repair → Optimization pass

Claude-guided work must preserve **ownership** across:

tradeoffs • risks • rollback • failure • security • cost

Final measurement bundle

Quality (rubric composite)

Latency (critical paths)

Repair / iteration counts

SPEC drift incidence

Rollback rehearsal **completeness** (documented ordering)

Retrospective notebook (honest)

Where **ownership** blurred

Where **verification** was thin

Where assistants **skipped ahead** without evidence

Which **workflow** gaps you will encode in Rules/Skills next cycle

Checkpoint — Faza 12

You stop treating Claude as something that merely **helps with isolated tickets**. You operate it as part of an **engineering system you control**: spec-led, observable, reversible, measurable, secure, improvable—with **ownership** that survives handoffs between people and tools.

Source: 34-claude-architect-final-lab/09-enterprise-depth-appendix.md

Enterprise depth appendix (Faza 12)

Companion to 01-08. Ownership still lives in weekly deliverables—this file is the **explicit syllabus critics wanted**: serious **test, distributed systems, observability, Go perf, Kubernetes production, database enterprise**, and **STRIDE** depth for the Payment Platform capstone.

How to use: skim once in Week 1; drill sections as each week assigns hooks (01.* cites this file). Maintain **living links** from your SPEC/architecture notebooks.

1. Test engineering (serious layering)

Goal: acceptance + NFR become **executable**—each line knows its proof class.

Layer	What it guards	Typical failure modes if skipped
Unit	Pure domain logic, deterministic branches	regressions disguised behind integration flake
Integration	ORM/repo + migrations + HTTP edges + broker client wiring	migrations break prod only
Contract	Stable consumer/provider schemas (REST/OpenAPI/async schema)	deploy order lottery
Property-based	Invariants (“no refunds exceed capture”, idempotent replays yield same outward effect class)	edge combinatorics missed
Load	Throughput/saturation envelopes, percentile latency	graceful degradation unknown
Chaos / fault	Behaviour under bounded real faults (paused broker TCP, flaky DNS)	folklore resilience

Contract testing posture

- Choose **producer-driven** (`schemathesis`, Pact provider) vs **consumer-driven** (classic Pact) deliberately. Payment edge often = **provider** strictness; PSP adapter might be consumer of external contract tests you don't fully control — document **verification gaps**.

Property testing intuition

Encode **generator** + **shrink** workflows for one scary invariant slice (replay storms, randomized partial saga completion). Prefer fewer properties with strong meaning over random fuzz noise.

Load testing realism

Define **steady state**, **stress**, **soak**. Track **p95/p99** + error budget—not only average. Correlate saturation with worker pool + DB conn limits.

Chaos testing guardrails

- Always **steady-state hypotheses** beforehand (“queue depth bounded”, “DLQ unchanged”, “0 money inconsistent states”). Abort when blast radius escapes sandbox.

Test × chaos interplay with optimisation

After a performance change, re-run a **small chaos or load slice** (same harness as Week 3/7) to ensure optimisations did not widen retry storms, shrink bulkheads dangerously, or mask saturation. Treat “faster but flakier under fault” as a failed iteration.

Cross-links: Weeks **1** (plan classes), **3** (implement suites), **6-7** (observe + regress under change).

2. Distributed systems

CAP (nuanced—not a vendor checkbox)

partitions happen. Document for **each critical path**:

- Availability vs consistency choice under partition **what user sees**. Example: PSP unreachable → degrade to **explicit failure** vs optimistic accept + reconciliation queue.

Consistency models (name them honestly)

Examples to tag in architecture:

Strong consistency inside single MySQL transactional boundary

Sequential / linearizable seldom across PSP + ledger—avoid claiming unless true

Causal / read-your-writes for UX after mutate commands

Eventual for analytics projections, notifications, non-money secondary reads—with **lag SLA** wording

Outbox pattern (transaction + message alignment)

Classic fix for dual-write hallucination:

1. Business mutation + enqueue **same DB transaction** (outbox row).
2. Publisher relay process ships to broker reliably.
3. Consumer idempotency + dedupe keys correlate with ledger events.

Discuss **polling vs log-tailing relay**, ordering guarantees, backlog pressure ownership.

Saga pattern

- **Choreography** (events bounce) vs **orchestration** (central saga brain). Ownership clarity often favours orchestrator for refunds/inventory choreography complexity—still document **every compensating action + timeouts**.

Circuit breaker • bulkhead • rate limiting

Circuit breaker — fail fast downstream; beware half-open churn; align with retries + idempotent verbs only.

Bulkhead — thread pools per dependency class (PSP vs internal query) isolation.

Rate limit — edge protection vs fairness; distinguish **capacity shedding** vs **abuse mitigation**.

Related Week 2 architecture + 3 implementation (client middleware).

3. Observability (OpenTelemetry + Prometheus + Grafana + Jaeger)

OpenTelemetry (OTel)

- Standard **traces + metrics + logs correlation** via context propagation.
- Symphony: HTTP middleware + DB spans; Go: consumer spans per message.
- Export to **OTLP** collector → fan-out to Prometheus/Jaeger stacks.

Prometheus

- Counter/histogram naming discipline; exemplars (where supported) linking latency buckets to trace ids.
- Alert on **golden signals adapted** — not “CPU high” orphan alerts.

Grafana

- Dashboard-as-code mindset; overlays for deploy annotations.
- Use **recording rules** sparingly—clarity beats cleverness early.

Jaeger / tracing practically

Practice drill:

1. Take **trace id** from slow payment.
2. Jump to PSP callback span boundaries.
3. Correlate log fields (`payment_id`)—assert single coherent story within SLO budgets.

Incident Week 6 demands this—not optional gloss.

4. Go performance engineering

Minimal production literacy:

Technique	Typical question answered
<code>go test -bench</code>	micro hot path regressions
<code>pprof CPU</code>	where time goes under load
<code>pprof heap/allocs</code>	unexpected allocation churn
<code>pprof goroutine</code>	leaks / stuck concurrency
<code>pprof block/mutex</code>	scheduling / lock contention
Escape analysis (<code>-gcflags=-m=2</code>)	surprise heap escapes from hot tiny helpers

Always pair profiles with **correctness tests**—faster wrong ledgers are worse.

5. Kubernetes production controls

Concern	Why it matters for payments
Ingress	stable external API + TLS story
NetworkPolicy	blast radius to metadata/PSP endpoints
Service mesh (conceptual)	uniform mTLS, traffic policy, retries vs idempotency hazards— optional but understand tradeoffs
PodDisruptionBudget	voluntary disruption safety during drain/rollout
Affinity / anti-affinity	AZ spread, noisy neighbour isolation
Taints / tolerations	dedicated pools (GPU unlikely here; maybe spot vs on-demand simulation)

Mesh adoption is **not required**—but you must write **one paragraph** why adopt or defer with risk acceptance.

6. Database enterprise (MySQL-flavoured)

Transaction isolation

Know differences at least between **READ COMMITTED** vs **REPEATABLE READ** (MySQL/InnoDB specifics: gap locks, phantom risks). Map chosen levels to **payment vs reporting** paths.

Deadlock analysis

Practice reading deadlock graph output: lock order inversion across refund + inventory updates. Mitigations: consistent lock ordering, smaller transactions, optimistic paths.

Replication

Async replica lag → stale read risks. Document which reads may hit replicas vs must stay on primary.

Partitioning / sharding (future you)

Capture **hot partition key** risk now even if unsharded—prevents surprise scale rewrite.

Locking strategies

Optimistic — version column / compare-and-swap; fewer long locks; requires conflict UX.

Pessimistic — `SELECT ... FOR UPDATE`; stronger immediate consistency; deadlock sensitivity.

7. Threat modeling — STRIDE & attack surface

STRIDE quick map

Class	Example payment concern
Spoofing	forged webhooks, stolen tokens
Tampering	altered callback payloads, message replays
Repudiation	missing audit correlation on money movement
Information disclosure	verbose errors leaking PAN metadata, log PII
Denial of service	webhook floods, queue stuffing
Elevation of privilege	worker creds over-scoped to DB rows

Attack surface modeling

Enumerate **entrypoints** (public API, admin, queue consumers, CI agents, assistant tool scopes). For each: trust boundary, data carried, blast radius, reduction options.

Store STRIDE tables **versioned** next to architecture—update when flows change.

8. Cross-week lab ideas (pick 2-3 total if time-constrained)

1. **Outbox relay failure** — pause relay; verify lag metrics + no money inconsistency; recovery path.
 2. **Contract break** — introduce incompatible event field; ensure pipeline surfaces before prod.
 3. **Jaeger-led incident** — remove logging clue; navigate via trace spans only until hypothesis.
 4. **Deadlock rehearsal** — contrived concurrent refund/inventory locking; document ordering fix + test.
 5. **NetworkPolicy accidental deny** — break egress; practise triage dashboards vs blind restart.
-

Appendix checklist (merge into Week 8 rubric selectively)

- [] STRIDE artefacts updated $\geq$ once after architectural change
 - [] Contract/schema tests enforced in CI for published edges
 - [] OTel propagation verified cross Symphony $\leftrightarrow$ Go boundaries
 - [] Load profile reproduced from scripts—not anecdotal dashboards
 - [] PDB + NetworkPolicy rationales documented with rollback paths
 - [] Known isolation/deadlock story for concurrency hot path written down
 - [] pprof before/after saved for at least one optimisation cycle
-

Phase — 35-observability-engineering

Directory: `35-observability-engineering/`

Source: `35-observability-engineering/01-sli-slo-error-budget-framing.md`

SLI • SLO • error budget — observability framing

Phase framing — Observability Engineering (35)

Units in this folder: 01 – 06 (topic order).

Why a dedicated phase exists

Incident workflow without **measurable reliability contracts** degenerates into heroics. Staff-level expectation: you **own observability discipline** tied to **product risk**, not only dashboards.

Core vocabulary (fluent, not ceremonial)

SLI — concrete ratio you trust (latency threshold, availability, error rate freshness).

SLO — target + window for that SLI (what «good» means monthly / quarterly).

Error budget — headroom remaining before SLO breach; governs velocity vs freeze decisions.

Alerting strategy — every alert maps to owner, runbook snippet, severity, consumer (not «CPU high» orphaned pages).

Golden Signals & families

Briefly reconcile **golden signals** (latency, traffic, errors, saturation) with **RED** (Rate, Errors, Duration) for request services and **USE** (Utilization, Saturation, Errors) for resources—but **adapt labels** to your stack (Symfony edge, Go workers, RabbitMQ depth, DB connections).

Cross-links

Amplifies `10-claude-ops-engineer` (incidents) and `06` final-lab appendix—now first-class spine.

Theme: Reliability choices **written down** → SLI selects → dashboards prove → paging policy matches cognitive load budgets.

Source: `35-observability-engineering/02-opentelemetry-distributed-tracing.md`

OpenTelemetry & distributed tracing ownership

Theme: One **propagation contract** across Symfony, Go workers, brokers, DB client spans where valuable.

Practice checklist

- **Context carriers** (`traceparent` , baggage) consistent at HTTP ingress and async hop boundaries.
- **Span naming** maps to domain steps (e.g. `payment.capture` , `worker.consume`) not only library defaults.
- **Sampling** policy explicit (head-based vs tail-based awareness)—debuggability vs cost.
- **Trace ↔ log correlation** fields (`trace_id` , `span_id` , business keys) standardised in structured logs.

Distributed tracing correlation drill

For one slow path, prove you can move **trace id** → **span graph** → **log lines** → **DB statement fingerprint** without guessing pod names.

LAB: introduce synthetic latency → narrate culprit class using **three signals** jointly.

Source: 35-observability-engineering/03-prometheus-grafana-jaeger.md

Prometheus • Grafana • Jaeger operating model

Theme: Scrapes, panels, traces—**artifact ownership**, not ephemeral UI clicks.

Prometheus

- Naming & label cardinality discipline (**avoid high-cardinality explosion**—user id as label is usually wrong class).
- Histogram buckets aligned to SLO thresholds (meaningful p95 / p99, not arbitrary).
- **Recording rules & alerts** versioned beside code/laC; change review like application logic.

Grafana

- Dashboards as **code** (Terraform / provisioning JSON)—environment parity.
- Annotations tying deploys/incidents to graph shifts (**causal hygiene**).

Jaeger (or compatible backend)

- Service map sanity vs reality of mesh/ingress.
- Compare **sync path** vs **async consumer** trace completeness—close blind spots deliberately.

Alerting strategy (reprise)

Each alert answers: **symptom, SLO link, owner, first triage step, expected false-positive rate class**.

Source: 35-observability-engineering/04-red-use-golden-signals.md

RED • USE • golden signals in practice

Theme: Pick **one primary model per component class** so on-call brains do not context-switch models every page.

Component class	Favoured lens	Example signals
HTTP / gRPC API	RED (+ saturation of thread pool)	RPS, 5xx ratio, latency histogram
Worker / queue consumer	Rate, age of oldest message, process time	lag, poison rate, retry storms
DB primary	USE + query latency	connections, lock waits, disk saturation
Cache	USE + hit ratio	eviction pressure, latency

LAB: Sketch **one dashboard row per service tier** tying RED/USE to your SLO numerator/denominator story.

Source: 35-observability-engineering/05-alerting-burn-rate.md

Alerting hygiene & burn rates

Theme: Alerts are **budget burn detectors**, not anxiety generators.

Multi-window / multi-burn awareness (conceptual)

High-severity pages should correlate with **fast budget consumption**—study Google SRE alerting patterns conceptually even if tooling differs.

Severity model alignment

Wire observability alerts to **39** incident severity vocabulary—avoid dual incompatible scales.

Checklist

- [] paging alert count **bounded**—team can review monthly without fatigue
 - [] non-page «info» channel alerts still owned and triaged within SLA
 - [] runbook **one-liner** exists for every page route
-

Source: `35-observability-engineering/06-integration-lab.md`

Integration lab — observability contract

Deliverable: One-page **Observability Contract** for a service you own (real or synthetic):

1. SLI definitions + SLO targets + error budget policy (what freezes when budget low).
2. Dashboard links / UIDs as code references.
3. Top 5 alerts with owners + runbook anchors.
4. Trace propagation map (which hops **must** create spans).
5. Known blind spots + acceptance rationale.

Pass bar: A peer can **onboard to on-call** from this artifact alone in <30 minutes for happy & degraded paths.

Phase — 36-production-database-engineering

Directory: `36-production-database-engineering/`

Source: `36-production-database-engineering/01-production-db-framing.md`

Production Database Engineering — framing

Phase framing — Production Database Ownership (36)

Units in this folder: 01 – 06 (topic order only).

Upgrade from «Indexes & migrations» mindset

Ownership here means answering **failure & consistency class questions** executives and peers actually fear—under load.

Threads this phase tightens

Connection pooling semantics & starvation

Isolation levels consciously chosen—not default cargo-cult

Deadlock graphs readability & remediation patterns

Replication lag surfaced to API contracts

Partitioning vs sharding trade space honest

Read replica routing correctness windows

Hot partition/key vigilance

Slow query ownership (who files the ticket—not who «ran explain once»)

Vacuum/maintenance posture awareness (concept applies to PostgreSQL analogue; translate for MySQL to **engine maintenance** equivalents you operate under—stats, fragmentation, archival).

Cross-links strengthen `12-database-enterprise-systems` and `34` final-lab appendix.

Checkpoint: Choose one **financially consequential** transactional flow and declare its **locking & read path** posture in writing.

Source: `36-production-database-engineering/02-connection-pooling-saturation.md`

Connection pooling & saturation ownership

Theme: Pools **hide** misconfiguration until blackout.

Artefacts you should be able to reason about

- Max connections vs pods vs thread pools—**inventory inequality** maths.
- Wait times & queue depths under burst (expose via metrics—not only pool counter).
- Statement timeout interplay with HTTP/worker timeouts.
- Credential rotation transient behaviour (sticky sessions vs pooled identity).

LAB

Simulate contention in safe env; chart **latency vs saturation** crossover—document escalation before collapse.

Source: `36-production-database-engineering/03-isolation-deadlocks-deep-dive.md`

Transaction isolation & deadlock investigation deep dive

Theme: Locks are **engineering facts**, not DBA folklore.

Isolation recap (actionable questions)

Which anomalies (dirty read / non-repeatable / phantom / skew) remain **accepted** vs **fatal** money risks for each query class?

Deadlock anatomy

Practice reading deadlock graph excerpts: ordering inversions vs single hot row vs lock escalation class confusions—then choose **ordering discipline**, shorter transactions, or **optimistic** shift.

Deliverable snapshot: annotate one historical (or fabricated) deadlock with **preventive test** strategy.

Source: `36-production-database-engineering/04-replication-read-replicas-lag.md`

Replication lag & read replica routing consistency

Theme: Replica reads are eventually consistent **unless proven otherwise**.

Contracts

Declare per endpoint: **PRIMARY only** • **eventually OK with lag SLA** • **monotonic read hacks** where needed (`read-after-write` bridging strategies).

Expose **seconds behind** (& appropriate equivalent)—not only dashboards but **routing policy**.

Incident class

Stale balance / stale authorization cache style bugs—tie to **verification pattern** (`37` JWT/session overlap awareness).

Source: `36-production-database-engineering/05-partitioning-sharding-hot-keys.md`

Partitioning, sharding tradeoffs & hot partitions

Partition strategy heuristics

Range vs hash vs list—**replay** / **bulk export** ergonomics—not only write spread.

Sharding realism

Operational tax: migrations, resharding dramas, transactional boundaries—**explicit non-goals** when deferring shards.

Hot partition detection

Key skew metrics, monitoring tail latency per shard hypothetical—document **safeguard dashboards** you'd add before admitting multi-tenant growth.

Query plan ownership, slow queries & maintenance posture

Slow query ownership (process)

Who **classifies**, who **prioritizes**, regression guard after rollout—pair with **11-go-performance** & Symphony query audit habits.

EXPLAIN discipline

Cardinality estimates lies you accept; **histogram freshness** caveat; intentional index addition vs workload rewrite ROI.

Maintenance / housekeeping

Operational calendar: backups proof restore, archiving, compaction analogues relevant to engine—tie to **risk register** (07 workflows).

LAB: Maintain a **Monthly DB Health Snapshot** artefact template (bullet page: top slow, top contested tables, anomalies).

Phase — 37-security-engineering-operations

Directory: 37-security-engineering-operations/

Source: 37-security-engineering-operations/01-security-operations-framing.md

Security Engineering Operations — framing

Phase framing — SecOps Reality (37)

Units: 01 – 05 (topic order only).

Separation from adjacent tracks

Track	Focus	This phase fills
22-ai-security-engineering	Model prompt/tool injection class	Operational identity & infra hardening cadence
23-threat-modeling-engineering	STRIDE & attack surfaces	Lifecycle controls (secrets, certs, scanners) tying mitigations

Staff expectation: articulate **lifecycle** mechanics (rotation, revocation, blast radius)—not solely «we use JWT.»

Themes: **OAuth2/OIDC**, **JWT lifecycle**, **Secrets rotation**, **KMS/Vault conceptual ownership**, **mTLS/cert lifecycle**, **rate limiting**, **WAF framing**, **OWASP ASVS as checklist lens**, **SBOM/supply chain**, **SAST/DAST positioning**, **container & image scanning**, **runtime security**, **K8s RBAC & NetworkPolicies** reinforcing **13-production-kubernetes** & **05** appendix.

Source: 37-security-engineering-operations/02-oauth-oidc-jwt-lifecycle.md

OAuth2 • OIDC • JWT lifecycle ownership

OAuth2 / OIDC differentiation (operational viewpoint)

Flows you permit (auth code + PKCE class patterns for SPAs/mobile); **opaque vs JWT access tokens strategy** consciously chosen.

JWT lifecycle realities

Issuer trust, JWKS rotation, **kid rollover playbook**, revocation stories (often partial—document holes), skewed clocks defensive coding, **aud**, **exp**, **nbfi** rigor—not optional decoration.

Symfony / Go vignettes: middleware validation chokepoints consistent.

LAB

Produce **JWT Failure Modes Appendix** tied to rollout & dependency upgrade windows.

Source: 37-security-engineering-operations/03-secrets-kms-vault-lifecycle.md

Secrets rotation • KMS • Vault patterns

Principles

Separate **bootstrap** secrets from workload secrets—rotation cadence proportional to blast radius classification.

KMS framing

Envelope encryption mental model—even if cloud-managed, know **who can decrypt** implicitly.

Dynamic credentials (where applicable)

Short-lived DB creds patterns—benefits vs connection pool friction.

Deliverable

Secrets Rotation RACI miniature table (roles not names)—who approves outage window class changes.

Source: [37-security-engineering-operations/04-mtls-certs-rate-waf-asvs.md](#)

mTLS • certificate lifecycle • rate limiting • WAF • OWASP ASVS

Certificate lifecycle maturity

Acquisition automation, issuance renewals observability (35 alerting tie-in), breakage drills before expiry blackout.

mTLS selectively

East-west workload identity uplift vs cognitive tax—articulate adoption boundary.

Rate limiting layering

Edge vs service vs per-tenant fairness—coordinate with **09-distributed** resiliency mental models.

WAF framing (thinking, not vendor religion)

Effective rule classes (bot, injection surface reduction) vs bypass realism—paired logging.

ASVS leveraged as gap list

Pick **baseline profile** realistically—not ASVS-Level-3 Cargo Cult day one inventory.

Source: [37-security-engineering-operations/05-supply-chain-scans-kubernetes-hardening-lab.md](#)

SBOM • supply chain • container & scan pipelines • K8s RBAC • NetworkPolicy

SBOM stance

Generation points (build), ingestion into registry/policy gate (conceptual)—**risk acceptance** documented when tooling partial.

Supply chain layering

Signed images / provenance class awareness—even if phased adoption.

SAST • DAST balance

Early signal vs noisy triage backlog—coverage targets vs diminishing returns realism.

Image scanning runtime security

Baseline hardening checklist (readonly rootfs class mitigations)—contextual allowances.

K8s RBAC & NetworkPolicy crosslink

Minimal privilege service accounts tied to workloads; egress policy guardrails aligning **13** deepening.

Integration lab

Map one critical deploy path: **pipeline stages** → **scanners** → **rollback gate** annotated.

Phase — 38-platform-engineering

Directory: 38-platform-engineering/

Source: 38-platform-engineering/01-platform-engineering-framing.md

Platform Engineering — framing

Phase framing — Internal Platforms (38)

Units: 01 – 04 (topic order only).

Mindset pivot

Infrastructure modules exist ⇒ **golden paths** crystallize **how correct work stays easy**.

Core motifs

Internal Developer Platform (IDP) responsibilities—not feature factory vs enablement contradiction resolved consciously.

Golden Path templating (**Symfony service**, **Go worker**, **Kafka/Rabbit scaffold**, previews, CI)—reduce «snowflake divergence tax».

Self-service infra guarded rails (quota, policy bundles, TTL sandboxes).

Developer Experience (DX) ownership metrics: lead time excerpts, flaky pipeline rates, scaffold adoption ratio.

Platform KPI dashboard hypothetical (enablement SLA class—not vanity DORA meme replication only).

Backstage conceptual lens catalog & scaffolding discipline—even if OSS stack deferred.

GitOps posture (**ArgoCD / Flux**) progressive alignment with drift detection vs imperative emergency bypass discipline.

Progressive Delivery (canary/layered flags awareness) cooperating with **21-agent-reliability** release safety.

Cross-links amplify **04 IaC lanes**, **13 Kubernetes**, **28-claude-optimization** platform-thinking without duplicating every deployment recipe.

Source: 38-platform-engineering/02-golden-path-self-service-idp.md

Golden paths & self-service infrastructure

Golden path decomposition

Starter kit components: scaffolding, lint/test baseline, observability instrumentation hooks (35), secrets bootstrap (37).

Declare **variation budget** («supported extension points»)—where customization must route through platform backlog.

Self-service safeguards

Namespaces / clusters / quotas with **policy-as-code gates** preventing silent cost & security divergence.

Deliverable blueprint

One golden path README («from idea to observable deployment»)—even if hypothetical.

Source: 38-platform-engineering/03-gitops-argocd-flux-progressive-delivery.md

GitOps • ArgoCD • Flux • progressive delivery

GitOps invariants

Declared state ⇒ reconciled observable diff; rollback is **rollback commit** choreography—coordinate with 34 rollback narrative.

Argo CD vs Flux (conceptual)

Sync waves, drift detection ergonomics vs CRD density tax—pick lens matching org maturity—not dual pilot chaos.

Progressive delivery interplay

Canary metrics tie to **SLO error budget burns** (35)—automatic promotion rules vs human gate realism.

Source: 38-platform-engineering/04-platform-kpi-dx-integration-lab.md

Platform KPIs • DX telemetry • Integration lab

KPI candidates (adapt ruthlessly)

Scaffold reuse rate • median CI wall clock • flaky test recurrence • MTTR infra incident class • infra cost anomalies (41)

LAB

Produce **Quarterly Platform Narrative** (one page synthetic):

- Adoption metrics trend line(s) plausible
 - Three friction themes removed / three upcoming bets
 - One risk accepted explicitly (snowflake exception count)
-

Phase — 39-incident-leadership

Directory: `39-incident-leadership/`

Source: `39-incident-leadership/01-incident-commander-framing.md`

Incident Commander & Production Leadership — framing

Phase framing (39)

Units: 01 – 05 .

Expansion beyond tactical triage

You already practise evidence ladders (10). Leadership layer adds:

Incident Commander (IC) role clarity & succession

Stakeholder communication pacing (internal & external optics)

Status page ownership factual minimalism discipline

Blameless postmortem culture & corrective action backlog hygiene

Severity model mapping symptom → SLA class → escalation

Oncall culture sustainable load budgeting / shadow / handoff artefacts

Runbook ownership lifecycle versioning next to 25 **rules/skills maturity**

Success signal

Incident narrative quality improves when **outside observer** trusts timeline—not only when root cause patched.

Source: `39-incident-leadership/02-severity-matrix-ic-structure.md`

Severity model • IC structure • escalation paths

Define **severity matrix** aligning customer impact × data integrity × unresolved uncertainty windows.

Practice IC rotation micro-drill tabletop: overloaded specialist scenario—who absorbs comms slice vs technical slice?

Escalate criteria explicit: **ambiguous blast radius triggers automatic IC assignment**.

Source: `39-incident-leadership/03-stakeholder-comms-status-page.md`

Stakeholder communication & status page ownership

Cadence messaging: factual deltas, ETA honesty policy («unknown» permissible), jargon translation for leadership.

Status page realism

Audience segmentation (internal engineers vs tenants vs regulators class scenarios).

Template

60 / 180 / tail update script scaffolding for hypothetical outage.

Source: 39-incident-leadership/04-postmortem-blameless-ownership.md

Blameless postmortems & corrective action backlog

Separate **immediate tactical patch** backlog from systemic **risk reduction** backlog items.

Five-whys sparingly replaced with **contributing factor graph awareness** fatigue caution.

Deliverable snippet: hypothetical anonymized retro summary excerpt showing learning vs blame tone.

Source: 39-incident-leadership/05-oncall-runbooks-integration-lab.md

Oncall culture • runbooks • integration lab

Sustainable pacing: rotations length, workload shadowing onboarding success metrics.

Runbook fields: precondition checks, escalation contacts, degraded-mode toggles (**pair with 34 rollback** narratives).

LAB: Simulate complex incident tabletop—assume **lost domain expert**—you must stabilize using runbook + observable ladder only (35 overlays).

Phase — 40-retrieval-engineering-evaluation

Directory: `40-retrieval-engineering-evaluation/`

Source: `40-retrieval-engineering-evaluation/01-retrieval-evaluation-framing.md`

Retrieval Evaluation Engineering — framing

Phase framing (40)

Units: 01 – 06 .

Bridges prior retrieval (17) upward

Operational RAG stacks fail quietly—this phase encodes **measurement conscience**.

Themes: **embedding strategy tradeoffs**, **chunking granularity ownership**, **reranking tiers**, **hybrid retrieval architectures**, **vector DB operational ownership**, **retrieval offline eval discipline**, **recall vs precision tradeoffs consciously chosen**, synergy with hallucination benchmarking (30), **groundedness & citation verification** habits.

Source: `40-retrieval-engineering-evaluation/02-embeddings-chunking-hybrid.md`

Embeddings • chunking • hybrid retrieval architectures

Contrast dense vs sparse vs hybrid merges for **Symfony + infra doc** heterogeneous corpora scenarios.

Chunking heuristics: structural vs fixed token windows vs semantic boundaries **cost realism**.

Operational tradeoff taxonomy: ingestion cadence staleness vs query freshness SLA.

Source: `40-retrieval-engineering-evaluation/03-reranking-vector-ownership.md`

Reranking • vector store ownership discipline

Operational layers: lexical prefilter → semantic recall → rerank refinement—where compute budget caps live.

Vector DB vigilance topics: compaction, duplication drift, deletion correctness (GDPR analogue awareness), latency tail sensitivity.

Source: `40-retrieval-engineering-evaluation/04-offline-eval-harness.md`

Retrieval offline evaluation harness

Synthetic query sets seeded from canonical docs + adversarial ambiguity prompts.

Grade rubric scaffolding: graded relevance tiers **not naive binary**.

Report at minimum: Recall@k class metrics, contamination rate (junk chunks), staleness sentinel hits.

Source: 40-retrieval-engineering-evaluation/05-groundedness-citations.md

Groundedness • citations • hallucination benchmark linkage

Theme: Responses must reference **inspectable anchors** proportionally risky domain areas.
Citation verification heuristic patterns (quote span alignment, contradictory doc detection).
Glue to 30-evaluation-systems golden dataset evolution when AI answers touch internal truth.

Source: 40-retrieval-engineering-evaluation/06-integration-lab-scoreboard.md

Integration lab — RAG QA scoreboard

Maintain living **spreadsheet-lite artefact**:

Scenario	Retrieval config id	Recall proxy	Precision proxy	groundedness verdict	escalation
Payment rule lookup	baseline-dense-v3	↑ / → / ↓	...	PASS / EDGE / FAIL	human review queue

Iterate one regression each sprint synthetic—prove closure loop awareness.

Phase — 41-finops-cloud-cost-ownership

Directory: `41-finops-cloud-cost-ownership/`

Source: `41-finops-cloud-cost-ownership/01-finops-cloud-framing.md`

FinOps & cloud cost ownership — framing

Phase framing (41)

Units: 01 – 03 .

Relation to `33-ai-cost-engineering`

That track focuses **LLM/context spend**. This expands to **infra & SaaS/cloud economic ownership** aligning Staff expectations.

Themes: **cloud cost attribution discipline, anomaly detection & investigation, budgeting & forecasting maturity, capacity planning loops, rightsizing rhythms, storage lifecycle realities, FinOps KPIs, performance per euro framing** tying `35` SLO ethos.

Source: `41-finops-cloud-cost-ownership/02-allocation-anomalies-budget-capacity.md`

Cost allocation • anomalies • budgeting • capacity forecasting

Practice labelling cost dimensions (environment / squad / SKU class) realistically—not infinite tag explosion.

Weekly anomaly heuristic pass: unexplained deltas → hypothesis classes (traffic vs misconfig vs orphaned resources).

Rolling budget checkpoints synchronized with roadmap intake decisions.

Source: `41-finops-cloud-cost-ownership/03-rightsizing-lifecycle-kpi-integration-lab.md`

Rightsizing • lifecycle storage • KPI scoreboard integration lab

Iterate candidate rightsizing backlog with **risk class** tagging (cheap experiment vs nuanced perf regression).

Cold storage tiers & ephemeral env TTL linkage to platform (`38`).

KPI mini-board

Monthly capture: infra cost trajectory vs **released value proxy** realism (avoid vanity normalisation fantasies—but force explicit assumption).

Phase — 42-organizational-production-realities

Directory: `42-organizational-production-realities/`

Source: `42-organizational-production-realities/01-org-constraints-framing.md`

Organizational production realities — framing

Phase framing (42)

Units: 01 – 04 .

Enterprise delivery includes **constraints not reducible to YAML**.

Themes: **legacy ownership psychology, political realities & stakeholder conflict navigation, partial migration choreography, strangler fig pattern realism, technical debt backlog transparency, buy vs build decision rigor, influence lacking formal authority archetypes, RFC ownership discipline, ADR evolution & supersession etiquette**.

Amplifies `07-claude-architect-workflow` + `24-approval-workflow` with social layer explicitness—not replacing them.

Source: `42-organizational-production-realities/02-rfc-adr-staff-influence.md`

RFC • ADR • influence without authority

RFC anatomy

Problem narrative • options with explicit kill criteria • phased adoption • escalation path—not slide deck substitute.

Lightweight vs heavyweight selection matrix.

ADR chain: supersede explicitly; staleness auditing quarterly synthetic.

Practice artefact

Rewrite a **rejected proposal** into a neutral learning capture—not a trash-bin shame artifact.

Source: `42-organizational-production-realities/03-legacy-strangler-stakeholders.md`

Legacy strangler • migrations • stakeholder conflict choreography

Incremental boundaries—keep old world stable while carving seams.

Debt register surfacing translating **pain felt** vs **pain priced**.

Conflict defusal patterns: aligning incentives via metrics both sides trust (35 observability artefacts as occasional neutral evidence).

Source: `42-organizational-production-realities/04-integration-lab-brief.md`

Integration lab — portfolio narrative rehearsal

Produce **Quarterly Architectural Reality Brief** (hypothetical):

- Three bets actively funded.

- Two deferred with explicit rationale (political / capacity constraints named honestly).
 - One debt payoff scheduled with a measurable payoff metric.
-

Phase — 43-staff-ai-evaluation-harness

Directory: `43-staff-ai-evaluation-harness/`

Source: `43-staff-ai-evaluation-harness/01-evaluation-harness-framing.md`

Staff AI evaluation harness — framing

Phase framing (43)

Units: 01 – 06 .

Unified lens across:

Benchmark dataset discipline (30 extension—not duplicate theory)

Golden dataset versioning & lineage

Regression harness choreography tying prompt / agent config / retrieval config toggles (31 **drift detection** interplay)

Quality baselines immutable history snapshot policy

Agent benchmark scenarios repeatable scripts / harness folders

Approval quality KPIs aligning 24 confidence governance

Dedicated **retrieval benchmark lane** cooperating with 40

Source: `43-staff-ai-evaluation-harness/02-golden-datasets-versioning.md`

Golden datasets • benchmarks • versioning hygiene

Treat golden sets like **migration artifacts**: schema version field, ingestion provenance snapshots.

Negative & edge adversarial augmentation discipline—not only happy corpus.

Linkage to CI analogue (even conceptual) gated merges when regression deltas exceed tolerated bands.

Source: `43-staff-ai-evaluation-harness/03-regression-matrix-prompts-agents-tools.md`

Regression suites • prompts • agents • tools

Matrix thinking: altering tool permissions must re-run 32 partial failure regression scenarios.

Partition suites: ultra-fast gates vs nightly deep hunts.

Source: `43-staff-ai-evaluation-harness/04-quality-baselines-thresholds.md`

Quality baselines • acceptance thresholds

Define rollback triggers not only for infra—signal **assistant quality slump** freezes or narrowed autonomy posture.

Source: `43-staff-ai-evaluation-harness/05-approval-quality-kpis.md`

Approval quality KPI linkage

Mis-approval taxonomy: false reject slowing teams vs harmful accept blast radius estimation.

Tune approval friction using incident / audit feedback loops—even when early proxies imperfect.

Source: `43-staff-ai-evaluation-harness/06-integration-eval-playbook-lab.md`

Integration lab — consolidated evaluation playbook

Produce **EVAL_PLAYBOOK.md** synthetic combining:

Sections:

- 1 Dataset registry snapshot fields
- 2 Suite execution matrix
- 3 Metric dashboard sketch
- 4 Incident tie-back loop for eval regression discovery

Ship **dry-run tabletop**: pretend failed golden suite on Friday evening—timeline you follow before Monday blame begins.
